

FUNDAMENTAL PROCESSES IN INITIATED CHEMICAL VAPOR
DEPOSITION

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Simon Shindler

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FUNDAMENTAL PROCESSES IN INITIATED CHEMICAL VAPOR DEPOSITION

Simon Shindler, Ph. D.

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Polymer chemical vapor deposition (CVD) offers superior conformality and substrate compatibility compared to solution-based thin film fabrication. Unique among polymer CVD processes, initiated CVD (iCVD) decouples radical generating reactions from polymerization reactions, enabling use of mild conditions, and diverse polymer chemistries. Most iCVD research falls into one of three categories: (i) implementation, (ii) process innovations, and (iii) post-deposition modifications. This dissertation targets critical knowledge gaps in these areas by investigating the fundamental processes relevant to them. Beginning with implementation, Chapter 2 addresses the gap in understanding of defects which might pose a risk to use of iCVD in industries requiring smooth, uniform films, such as optical coatings and soft electronics. Films fabricated using iCVD at sample temperatures below $\sim 30^{\circ}\text{C}$ are shown to be susceptible to defects originating from build-up of oligomeric species ($n < 10$) and prevention methods are developed. Chapters 3 and 4 address the understudied iCVD vapor phase. Even though innovations like batch iCVD and condensed droplet polymerization were built on understanding of the vapor phase, no models of the iCVD vapor phase have been validated experimentally. Chapter 3 first investigates heat transfer and mixing in the iCVD reactor, introducing the Peclet and Knudsen numbers to quantify these effects. Chapter 4 then thoroughly characterizes

the reactive vapor phase, developing a robust model of initiator (di-*tert*-butyl peroxide) decomposition at a range of flowrates, pressures, temperatures and monomer (cyclohexyl methacrylate) compositions. Moving to post-deposition modification, Chapter 5 investigates post-deposition base catalyzed hydrolysis of two fluoroacrylic polymers. The high pH conditions required to hydrolyze the passivating fluoroacrylate moiety pushes the limits of post-deposition modification. Quantification of hydrolysis kinetics shows that both crosslinking and covalently grafting the film to the substrate are necessary. The eventual degradation of the grafting agent indicates a need for improved grafting procedures.

BIOGRAPHICAL SKETCH

Simon Shindler was born in Port Angeles, Washington where he gained an appreciation for the outdoors. He graduated from Port Angeles High School in 2015, and studied Chemical Engineering at the University of Idaho where he was first exposed to surface science working in the lab of Dr. Prof. Eric Aston on thermal degradation reactions of hydrophobic core-shell structure electro-spun poly(vinyl pyrrolidone) non-woven mats. After graduating in 2019 with his Bachelors of Science in Chemical Engineering, with a minor in Mathematics, Simon began work in the Yang Lab at Cornell University pursuing his PhD.

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TABLE OF CONTENTS

Biographical Sketch	iii
Acknowledgments	iv
Table of Contents	v
List of Figures	ix
List of Tables	xiii
List of Abbreviations	xiv
List of Symbols	xvi
CHAPTER 1. INTRODUCTION	1
1.1 Background	2
1.2 Reactor Configuration	2
1.3 Scope of Thesis	5
1.4 References	8
 CHAPTER 2. PREVENTION AND CHARACTERIZATION OF THIN FILM DEFECTS INDUCED BY CONTAMINANT AGGREGATES IN INITIATED CHEMICAL VAPOR DEPOSITION	 10
Abstract	11
2.1 Introduction	12
2.2 Materials and methods	16
<i>2.2.1 Materials</i>	<i>16</i>
<i>2.2.2 Film Synthesis</i>	<i>17</i>
<i>2.2.3 Native Aggregate Synthesis</i>	<i>20</i>
<i>2.2.4 Polymer Distillation</i>	<i>20</i>
<i>2.2.5 Conditioning Film Method</i>	<i>21</i>
<i>2.2.6 Characterization and Data Analysis</i>	<i>21</i>
2.3 Results and Discussion	23
<i>2.3.1 Defect Characterization and Atomic Force Microscopy</i>	<i>23</i>
<i>2.3.2 XRD and Anisotropy</i>	<i>30</i>

2.3.3 Surface Characterization of Contaminant Material	32
2.3.4 pPFDA Polymer Distillation	36
2.3.5 Defect Prevention by the Conditioning Film Method	38
2.4 Conclusions	41
2.5 References	43
CHAPTER 3. HEAT TRANSFER AND MIXING IN INITIATED CHEMICAL VAPOR DEPOSITION	48
Abstract	49
3.1 Introduction	50
3.2 Materials and Methods	54
3.2.1 Mixing simulations	54
3.2.2 Flow Calibration	55
3.2.3 Vapor phase composition measurements using a Pirani gauge	56
3.2.4 Residence time distribution measurement	56
3.2.5 Effective temperature measurement	57
3.2.6 Heat transfer in Argon and IPA vapor	58
3.3 Results and Discussion	59
3.3.1 Dimensional analysis	59
3.3.2 Mixing simulations	62
3.3.3 Step response curves	66
3.3.4 Heat transfer	70
3.3.5 Reactor Effective temperature	78
3.4 Conclusions	88
3.5 References	91
CHAPTER 4. CHARACTERIZATION OF THE VAPOR PHASE DURING ICVD POLYMERIZATION OF CYCLOHEXYL METHACRYLATE	96
Abstract	97
4.1 Introduction	98
4.2 Materials and Methods	103

4.2.1 Materials	103
4.2.2 Filament temperature calibration	104
4.2.3 Vapor phase composition measurements using a Pirani gauge	104
4.2.4 Batch mode TBPO decomposition	105
4.2.5 Continuous flow-through TBPO decomposition	105
4.2.6 iCVD depositions	106
4.2.7 Thin film characterization	109
4.3 Theory	110
4.3.1 Temperature Calibration	110
4.3.2 Kinetic model of batch decomposition	112
4.3.3 Kinetic model for continuous decomposition	117
4.4 Results and Discussion	122
4.4.1 Batch mode decomposition of pure TBPO	122
4.4.2 Flow-through decomposition of pure TBPO	129
4.4.3 Relationship between Pirani gauge and composition	133
4.4.4 Flow-through decomposition of CHMA / TBPO mixtures	138
4.4.5 Deposition Kinetics	144
4.5. Conclusions	155
4.6. References	158
 CHAPTER 5. HYDROLYSIS OF POLY(FLUOROACRYLATE) THIN FILMS SYNTHESIZED FROM THE VAPOR PHASE	 164
Abstract	165
5.1 Introduction	166
5.2 Materials and Methods	171
5.2.1 Materials	171
5.2.2 iCVD deposition	172
5.2.3. Grafting	173
5.2.4. Hydrolysis	173
5.2.5 Characterization	174

5.3 Theory	177
5.3.1 <i>Derivation of Fractional conversion from XPS data</i>	177
5.3.2 <i>Equations for determination of reaction order</i>	178
5.4 Results and Discussion	183
5.4.1. <i>iCVD synthesis and FTIR analysis</i>	183
5.4.2 <i>Composition analysis using FTIR with ellipsometry</i>	186
5.4.3 <i>Role of Grafting and Crosslinking in Film Stability</i>	188
5.4.4. <i>Hydrolysis Kinetics</i>	191
5.4.5. <i>Dynamic contact angle of hydrolyzed films</i>	201
5.5 Conclusions	204
5.6 References	206
 CHAPTER 6. CONCLUSIONS AND SUMMARY	 214
6.1. Conclusions	215
6.2 Future Work	218
 APPENDIX A	 220

LIST OF FIGURES

Figure 1.1. Reactor schematic showing (a) an oblique view, (b) a side view and (c) a heat-map surrounding the filament array (pg. 4)

Figure 2.1. Schematic showing a mono-lamellar polymer crystallite on a surface (pg. 14)

Figure 2.2. Optical micrographs of polymer thin film defects (a-g) and diffusion limited aggregation phase diagram (h) (pg. 24)

Figure 2.3. AFM data showing emergence of defects (pg.29)

Figure 2.4. Powder XRD of contaminated p(AA-co-DEGDVE) film (pg. 31)

Figure 2.5. XPS survey scans for samples placed in the reactor. (pg. 32)

Figure 2.6. FTIR Vari-GATR scans for samples placed in the reactor. (pg. 33)

Figure 2.7. SEM images depicting remnants of melting or sublimating contaminant aggregates prepared using the procedures described in Section 2.2.3 (pg. 35)

Figure 2.8. pPFDA distilled samples showing defects caused by pPFDA distillation and blank samples showing baseline reactor contaminant levels. (pg. 37)

Figure 2.9. Experimental setup showing polymer distillation of pPFDA from a wafer placed on a hot TEC at 45oC to the cold wafer placed on a TEC cooled to 0°C (pg. 37)

Figure 2.10. Optical micrographs showing defects resolved by pAA using a stage temperature of 0oC, defects resolved by pAA using a stage temperature of 13oC, defect-free pAA film using a stage temperature of 50oC and a film made using the conditioning film method (see Section 2.2.5), showing the absence of defects. (pg. 40)

Figure 3.1. Schematic of the two iCVD reactor geometries used in simulations (a) single chamber design with inlets and outlet labeled (b) double chamber design with inlets, outlet, side port and back port labeled. This double chamber design matches the design used in experiments. (pg. 55)

Figure 3.2. Binary gas mixing for a range of Peclet numbers over the transitional range of Peclet number in two-chamber and one-chamber reactor configurations. (pg. 63)

Figure 3.3. Step response curves for two chamber reactor configuration showing ideal mixing, measurement at the back-port of the reactor, measurement at the sideport and measurement at the reactor outlet. (a) Simulation at a Peclet number of 5.3 (b) Simulation at a Peclet number of 1.3 (c) Simulation at Peclet number of 0.3 (pg. 65)

Figure 3.4. (a) Pirani gauge ratio standard curve for IPA and Ar (b) composition versus time calculated from the IPA/Ar standard curve for three repeats (c) derivative of composition versus time fit to ideal CSTR curve (pg. 69)

Figure 3.5 (a) Schematic showing the coordinate system used in modeling of

temperature at the filament array (b) Plot showing equation (3.2) for three adjacent filament wires as a visual aid (pg. 71)

Figure 3.6. Temperature profile in the Knudsen layer, showing relevant quantities (pg. 73)

Figure 3.7 Data fitting for pressure dependent power versus filament temperature (pg. 77)

Figure 3.8 Ratio of effective temperature to reference temperature versus filament temperature for argon at nominal pressures of (a) 50 mtorr (b) 100 mtorr (c) 200 mtorr and (d) 300 mtorr (pg. 86)

Figure 3.9 Ratio of effective temperature to reference temperature versus filament temperature for IPA at nominal pressures of (a) 50 mtorr (b) 100 mtorr (c) 200 mtorr and (d) 300 mtorr (pg. 87)

Figure 4.1. Calibration of temperature example showing flowthrough TBPO at 100mtorr (pg. 111)

Figure 4.2. (a) Fractional conversion calculated from equation (4.14) plotted on linearized axes against time for batch-mode TBPO decomposition. (b) Arrhenius plot of k_1^{eff} from the slopes of fits in (a) (c) Mole fraction of TBPO during the thermal decomposition of pure TBPO in batch mode calculated from Equation (4.21) plotted against the adjusted Pirani gauge ratio $c-C_{products}$. (pg. 126)

Figure 4.3. (a) Arrhenius plot for thermal decomposition of TBPO in flow-through model showing pressure independence. (b) Plot of calculated rate coefficient k_1/F_{in} versus flow of TBPO into the reactor, showing behavior consistent with CSTR model. (pg. 131)

Figure 4.4. (a) Pirani gauge ratio versus mole fraction of TBPO for a range of total reactor pressures (b) Pirani gauge ratio versus inlet TBPO mole fraction for unreacted TBPO and fully reacted TBPO (pg. 135)

Figure 4.5. (a) Predicted Pirani gauge ratio versus actual Pirani gauge ratio for thermal decomposition of TBPO in CHMA at stage temperature of 50°C assuming no deposition or gas phase reactions occur (b) Predicted Pirani gauge ratio versus actual Pirani gauge ratio for thermal decomposition of TBPO in CHMA at stage temperature of 30°C assuming no gas phase reactions or deposition occur (c) Measured Pirani gauge ratio versus filament temperature for data taken at a stage temperature of 50°C including model prediction which assumes no gas phase reactions and deposition does not affect vapor (d) Measured Pirani gauge ratio versus filament temperature for data taken at a stage temperature of 50°C, including model prediction assuming no gas phase reactions and deposition does not effect vapor. (pg. 140)

Figure 4.6. Plot showing the mole fraction of CHMA predicted by the feed composition alone, accounting for vapor phase processes, but not deposition, and accounting for deposition quantified by deviation from prediction in figure 4.5 (b) and (d). (pg. 143)

Figure 4.7. FTIR absorbance spectra for CHMA monomer and CHMA polymer

showing disappearance of the vinyl peak at 1700 cm^{-1} consistent with polymerization. (pg. 145)

Figure 4.8. (a) Mole fraction of TBPO measured at the Pirani gauge sensor versus inlet mole fraction of TBPO aligns well with predicted value from figure 4.4. deviation from the model is expected to be due to insufficient time to equilibrate before measurement (b) Dimensionless FTIR absorbance versus thickness measured by ellipsometer showing a clear non-linear relationship, indicative of density variation with thickness, high MSE correlates with deviation. FTIR absorbance is used as a proxy for quantity of film instead of thickness. (c) Arrhenius plot for deposition rate constant based on equation (4.24) (pg. 147)

Figure 4.9. Stage temperature dependence fit from data of Chan and Gleason. (pg. 151)

Figure 4.10. (a) Arrhenius plot of deposition rate data collected at y_{TBPO}^{in} of 0.2 corrected for effects of changing stage temperature (b) Arrhenius plot of deposition rate data from Xu and Gleason (pg. 154)

Figure 5.1. Reaction schema and chemical structures relevant to thin film hydrolysis. (a) Schematic showing the working principle of using surface reaction to react samples (b) reaction scheme showing TCVS grafting (c) reaction scheme showing polymerization followed by hydrolysis (d) chemical structure of DEGDVE (e) chemical structure of PFOA (f) chemical structure of HFBA (pg. 170)

Figure 5.2. FTIR spectrum of pHFBA with and without baseline correction (pg. 174)

Figure 5.3. FTIR spectra of monomer and polymers. FTIR spectra of DEGDVE monomer, p(PFOA-co-DEGDVE), pPFOA, PFOA monomer, p(HFBA-co-DEGDVE), pHFBA and HFBA monomer. Polymer spectra were collected in transmission mode and were normalized by their film thickness. Monomer spectra were collected using FTIR-ATR, and were normalized to be of the same general scale as they cannot be used for quantification. (pg. 184)

Figure 5.4. Optical micrographs of iCVD films treated with aqueous solutions of NaOH (pg. 189)

Figure 5.5. XPS data for unreacted p(HFBA-co-DEGDVE) and p(PFOA-co-DEGDVE) thin films and films reacted for 300 min in 1M NaOH. Spectra corresponding to other time points are included here. (pg. 192)

Figure 5.6. XPS scan of pHFBA film showing Si peaks from film degradation after reacting with 0.5M NaOH for 30 minutes. (pg. 193)

Figure 5.7. XPS data for p(HFBA-co-DEGDVE) in the concentration series. Where NaOH concentration was varied from 0.05-2M, but reacting time and temperature were kept constant at 50 min and 30°C (pg. 193)

Figure 5.8. XPS data for p(HFBA-co-DEGDVE) time series. Where the reaction time was varied from 0-300min, and concentration and temperature was kept constant at 1M and 30°C . (pg. 194)

Figure 5.9. XPS data for p(PFOA-*co*-DEGDVE) concentration series. Where NaOH concentration was varied from 0.2-2M, but reacting time and temperature were kept constant at 180 min and 30°C (pg. 194)

Figure 5.10. XPS data for p(PFOA-*co*-DEGDVE) time series. Where the reaction time was varied from 0-300min, and concentration and temperature was kept constant at 1M and 30°C. (pg. 195)

Figure 5.11. Angle resolved XPS data from p(HFBA-*co*-DEGDVE) reacted for 200min in 1M NaOH (pg. 195)

Figure 5.12. Data to determine reaction kinetics. (pg. 198)

Figure 5.13. Dynamic contact angle data for hydrolyzed p(HFBA-*co*-DEGDVE) and p(PFOA-*co*-DEGDVE). (pg. 203)

LIST OF TABLES

- Table 1.1.** Categories of applied iCVD research (pg. 6)
- Table 1.2** Summary of chapters (pg. 7)
- Table 2.1.** Deposition conditions used for film syntheses (pg. 19)
- Table 2.2.** XPS mole fraction (pg. 33)
- Table 2.3.** Ellipsometry thickness (pg. 33)
- Table 2.4.** Defect coverage data for pPFDA distillation. (pg. 38)
- Table 3.1** Dimensionless numbers relevant to mass transfer in iCVD (pg. 60)
- Table 3.2.** Table of dimensionless numbers relevant to heat transfer in the iCVD system (pg. 61)
- Table 3.3** Parameters used to fit equation (3.11) (pg. 75)
- Table 3.4** Dimensionless interfacial heat transfer resistance for Ar and IPA fit from effective temperature (pg. 85)
- Table 4.1** Deposition conditions (pg. 108)
- Table 5.1.** Deposition conditions (pg. 176)
- Table 5.2.** Compositional and density data for homopolymer and crosslinked films (pg. 187)
- Table 5.3.** AR-XPS mole fractions for p(HFBA-*co*-DEGDVE) sample reacted for 200 minutes in 1 M NaOH. (pg. 199)

LIST OF ABBREVIATIONS

4VP	4-vinyl pyridine
AA	Acrylic acid
AC	Alternating current
AFM	Atomic force microscopy
Ar	Argon
AR-XPS	Angle resolved x-ray photoelectron spectroscopy
CD	Compact Dendrite
CS	Compact seaweed
CVD	Chemical Vapor Deposition
DC	Direct current
DEGDVE	di(ethylene glycol) divinyl ether
DVB	Divinyl benzene
FD	Fractal dendrite
FS	Fractal Seaweed
FTIR	Fourier transform infrared spectroscopy
FTIR-ATR	Fourier transform infrared spectroscopy with attenuated total reflectance
HFBA	2,2,3,4,4,4-hexafluorobutyl acrylate
iCVD	Initiated Chemical Vapor Deposition
IMFP	Inelastic Mean Free Path
IPA	Isopropyl alcohol
MW	Molecular Weight
NaOH	Sodium Hydroxide
PAN	Polyacrylonitrile
PFA	perfluorinated acid

PFDA	1H,1H,2H,2H-perfluorodecyl acrylate
PFOA	1H,1H,2H,2H-perfluorooctyl acrylate
PVC	Polyvinyl chloride
PVD	Physical vapor deposition
SEM	Scanning electron microscopy
S _N 2	2 nd order nucleophilic substitution
TBPO	di- <i>tert</i> -butyl peroxide
TCVS	Trichlorovinyl silane
TEC	Thermoelectric cooler
VariGATR	Variable angle grazing attenuated total reflectance
XPS	X-ray photoelectron spectroscopy
XRD	X-ray diffractometry

LIST OF SYMBOLS

A	Absorbance or Arrhenius pre-exponential factor
A_1	Area of the reactor stage
b	Pathlength
c	Concentration, Pirani gauge ratio or speed of sound
c_p	Constant pressure heat capacity
D	Diffusion coefficient, fitting coefficient for mixed monomer and TBPO
d	Collision diameter
d_0	Isotropic capillary length
Da	Dahmköler number
E_A	Activation energy
F	Molar flow
f	Area fraction of surface or volume fraction of reactor made up by the filament array
g	Acceleration due to gravity
h_{eff}	Geometric factor for determining effective temperature
h	Dimensionless heat transfer coefficient
H	Distance between the filament and the lid
H_{vap}	Enthalpy of vaporization
I	Electrical current
j_q	Heat flux
K	Curvature
k	Rate constant
k_B	Boltzmann's Constant
Kn	Knudsen number
L	Separation between the filaments

l	Characteristic length
L_{fil}	Length of the filament wire
M	Formula weight
m	Mass of a molecule
Ma	Mach number
M_r	Molar refractivity
n	Number of atoms, volumetric number density, or surface number density
$\mathbf{n}$	Unit normal vector
n_D	Refractive index
N_R	Number of times a reaction has happened
Pe	Peclet number
p	Pressure
q_{fil}	Heat generated per unit length from the filament array
R	Resistance
r	Rate of reaction
R_{fil}	radius of the filament array
R_G	Ideal gas constant ($8.314 \text{ J mol}^{-1}\text{K}^{-1}$)
Re	Reynold's number
T	Temperature
t	Time
U	Dimensionless chemical potential
V_C	Molar volume at the critical point
V_R	Reactor volume
V_1	Volume of the bottom chamber
V_2	Volume of the filament array
V_3	Volume of the edges of the top chamber
v	Velocity

x	Polymer mole fraction
$\hat{x}$	Characteristic length in Dahmkoler number
y	Vapor phase mole fraction
α	Arbitrary reaction order
α_0	Accommodation coefficient
β	Arbitrary reaction order
Γ	Equilibrium molar surface excess
γ	Isotropic surface energy
Δ	Dimensionless degree of supercooling
ϵ	Molar extinction coefficient
ϵ_{fil}	Emissivity of the filament wire
ε	Strength of anisotropy
η	Temperature adjustment coefficient in Arrhenius kinetics
θ	Angle in radians
κ	Thermal conductivity
κ'	Temperature independent thermal conductivity
λ	Analysis depth or mean free path
μ	Chemical potential or viscosity
π_m	Number of fluorine atoms in the fluoroacrylate monomer
ρ	Density
σ	Stephan Boltzmann constant
ϕ	Heat transfer potential
ϕ_m	Number of carbon and fluorine atoms lost when a single fluoroacrylate moiety is hydrolyzed
χ	Fractional conversion
ζ_T	Temperature jump coefficient

Superscripts

0	Refers to the unreacted state
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<i>eff</i>	Effective (average) quantity over the volume of the reactor
<i>f</i>	Refers to the fully reacted state
<i>i</i>	Refers to initial reactor conditions
<i>in</i>	Refers to the inlet or feed to the reactor

Subscripts

1	Refers to primary decomposition
<i>A</i>	Refers to amorphous adsorbed polymer
<i>av</i>	Average of the lid and stage
<i>bp</i>	At the boiling point
β	Refers to beta scission
<i>C</i>	Refers to the crystalline aggregate
<i>c</i>	Refers to copolymer
<i>CAL</i>	Refers to calibration data
<i>fil</i>	Applying to the reactor filament
<i>h</i>	Refers to homopolymer
<i>i</i>	Refers to initiation reaction
<i>int</i>	Refers to quantity at the interface of a growing aggregate
<i>m</i>	Refers to monomer
<i>meas</i>	Refers to measured data
<i>n</i>	Refers to the direction of the unit normal vector
<i>p</i>	Refers to polymerization or propagation
<i>Pirani</i>	Refers to data taken by the Pirani gauge
<i>r</i>	Refers to recombination of methyl radicals
<i>ref</i>	Reference condition
<i>sat</i>	At saturation
<i>stage</i>	Referring to the reactor stage
<i>tot</i>	Total quantity

<i>t</i>	Refers to termination reaction
<i>wall</i>	Applying to the wall of the reactor

CHAPTER 1:

INTRODUCTION

1.1 Background

Polymer thin films are widely used in applications ranging from organic electronics¹ to membranes.² While solution phase techniques like spin coating and spin casting are the most popular synthesis methods, the applications of polymer films have been further broadened by polymer chemical vapor deposition (CVD) processes.³ Polymer CVD is solvent free, synthesizes conformal films,^{4–8} doesn't suffer from surface tension induced defects like de-wetting, and often enables precise thickness control even for ultrathin films.⁹

Unfortunately, many polymer CVD processes require harsh processing conditions (like plasma enhanced CVD),^{10,11} or have a limited library of polymers which can be synthesized (like parylene CVD).¹² Initiated Chemical Vapor Deposition (iCVD) avoids these issues by decoupling the initiation and propagation steps, such that an initiator molecule is decomposed at a hot filament, while polymerization occurs on a cooled stage.¹³ Because separate molecules can be used for the initiator and monomer, initiators can be used which require milder conditions to successfully initiate polymerization. This separation also broadens the range of compatible monomers, enabling the synthesis of acrylate and methacrylate films, which are widely used in solution-based polymerization.

1.2 Reactor Configuration

Establishing a clear physical understanding of the iCVD reactor is helpful for readers unfamiliar with iCVD. We first discuss the basics of the polymerization process, since they underpin the reactor design principles. In iCVD, polymerization and film formation occur simultaneously during a process known as deposition. The fundamental reactions occurring during deposition mirror free-radical polymerization in the solution phase. Therefore, the surface concentrations of monomer and initiating radicals primarily determines how deposition occurs.

The surface concentration of monomer can be increased by either increasing the partial pressure of monomer or decreasing the temperature of the surface. To account for both these effects, the dimensionless ratio of monomer partial pressure to monomer saturation pressure ($p_m/p_{m,sat}$) serves as a useful proxy for monomer surface concentration. If monomer partial pressure exceeds the saturation pressure ($p_m/p_{m,sat} > 1$), then there is a risk of condensation, which can result in low film quality, and non-uniformity. If monomer concentration is too low ($p_m/p_{m,sat} < 0.05$), then deposition can become prohibitively slow. Determining the partial pressure of initiating radicals is difficult, so $p_m/p_{m,sat}$ is the simplest way of modulating the deposition rate.

To ensure deposition occurs on the sample being coated, but not the other interior surfaces of the reactor, these surfaces are heated to at least 60°C. The reactor stage, where samples are placed, is usually cooled to a temperature of 10-30°C using a re-circulating chiller. In many reactor configurations, including the one used for fabrication in this thesis, the deposition process occurs in a top chamber, while a bottom chamber houses cooling lines. Figure 1.1(a) and (b) provide a schematic of the reactor, labeling the cooled stage, top and bottom chambers, reactor inlets and outlet, and the heated filament array.

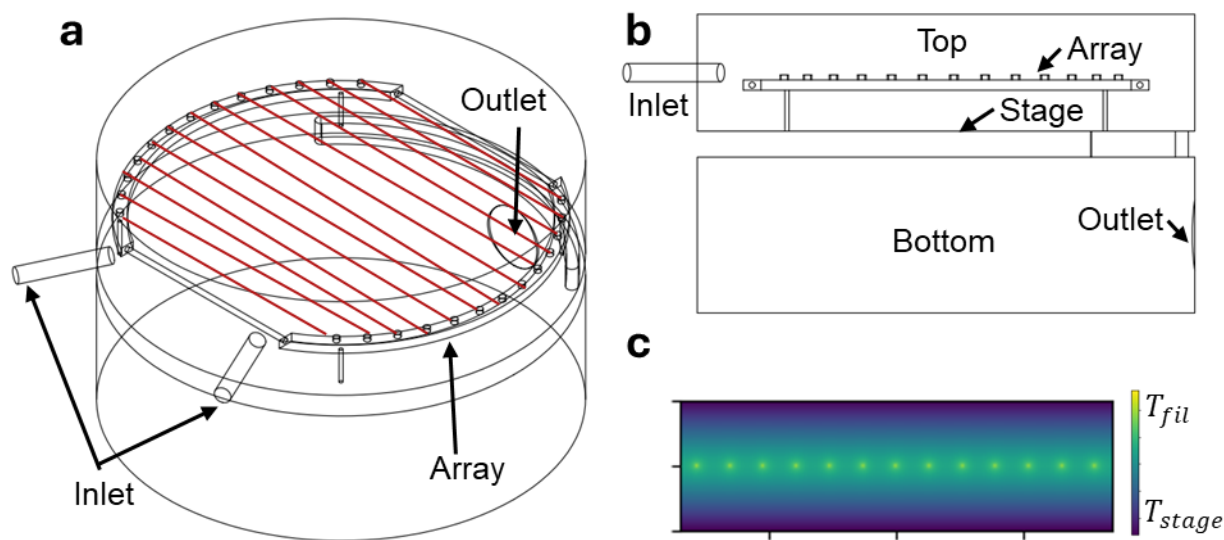


Figure 1.1. (a) Reactor schematic showing the inlets, outlet and the filament array (b) Side-view of the reactor showing the top and bottom chambers of the reactor, where the cooled stage is located, the array, reactor inlets and the reactor outlet. (c) Temperature distribution surrounding the filament array to show how the reactor is heated, where the stage and filament temperatures are labeled.

We now turn to consideration of the initiation process. The element of iCVD which distinguishes it from other polymer CVD processes is the use of an initiator species (often di-*tert*-butyl peroxide or TBPO) separate from the monomer which undergoes thermal decomposition into free radicals capable of initiating polymerization. By using an initiator like TBPO, the temperature required to generate radicals can be reduced from 500°C (pyrolytic CVD) to 200°C. Use of relatively mild conditions expands the range of potential chemistries that can be used in iCVD. The drawback of using a separate initiator species however is that the radicals tend to recombine and must be generated close to the surface. This is enabled by a resistively heated filament array consisting of nichrome wire stretched across a frame (Figure 1.1 (a) and (b)). Figure 1.1 (c) shows the approximate temperature distribution in the reactor, showing that the filament is at elevated

temperature.

All process gases are delivered to the vacuum chamber at flow-rates of between 0.1 and 10 standard cubic centimeters per second (sccm), which corresponds to 7.45×10^{-8} to 7.45×10^{-6} mol s⁻¹. To generate flow of monomer, liquid monomer is heated and flow is controlled using either a mass flow controller, or a needle valve.

The final element which enables successful operation of an iCVD reactor is the reactor control systems, and process monitoring. Temperatures of the stage, filament and reactor body are all monitored and controlled to maintain constant values throughout the deposition. Overall pressure in the reactor is also controlled at a constant value, generally below 1 torr (0.0013 atm). The thickness of the growing film is monitored using a laser interferometer.

1.3 Scope of thesis

The majority of iCVD research is applied research. Applied research in iCVD is either devoted to implementing iCVD in new ways or innovations to fabricate new materials. New fabrication methods generally apply either to the deposition process, or to the post-deposition modification of polymer films. These three categories are summarized in Table 1.1. It should be noted that work may fall into multiple of these categories. For instance, one can imagine research aimed at producing an anti-fouling surface (implementation) by performing depositions using novel chemistries (iCVD innovation) which are modified following deposition using surface reactions (post-deposition modification). In the field of iCVD, most research which does not clearly fall into the categories in Table 1.1 is fundamental in nature.

Table 1.1 Categories of applied iCVD research

	Example of research
Implementation	Teflon thin films
Innovation <i>iCVD</i>	iCVD at high monomer pressure
<i>Post-deposition</i>	Chemical etching

Fundamental research is devoted to improving understanding as opposed to doing something new. While applied research tends to make up the bulk of iCVD literature, it is often built upon understanding of underlying processes gained in more fundamental work. For example, batch iCVD, and condensed droplet polymerization were both built upon fundamental understanding of the iCVD vapor phase developed in earlier kinetic studies. Fundamental research can generally be placed into the categories in Table 1.1 based on the type of innovations it is likely to support. Since fundamental research can be more difficult to justify than applied research, categorizing it using Table 1.1 is helpful to illustrate its contribution to the field.

This thesis contains four fundamental research studies (Chapters 2-5) which are explicitly focused on supporting future applied work in the three categories in Table 1.1. These fundamental investigations focus on areas of research which have attracted significant attention but where a lack of fundamental understanding has created a bottleneck to progress. This is summarized in Table 1.2.

Table 1.2 Summary of chapters

		Section
Implementation		Chapter 2. Defect formation and prevention
Innovation	<i>Fabrication</i>	Chapter 3. Heat transfer and mixing in iCVD Chapter 4. Vapor phase reaction kinetics in iCVD
	<i>Post-fabrication</i>	Chapter 5. Poly(fluoroacrylate) hydrolysis

In Chapter 2, we address the presence of defects in iCVD. Though iCVD is not susceptible to surface tension induced defects which are common in solution processed thin films, defects are nevertheless possible. Furthermore, the high-quality films produced in iCVD have motivated implementation in numerous precision applications, such as optical thin films where a defect can cause faulty device performance. Defects have largely been unstudied in iCVD and as such the discovery of a defect mechanism is relevant to implementation of the process. In this section we show a defect type which has not yet been studied, demonstrate how it is caused, and provide a procedure for preventing it.

Chapters 3 and 4 deal with the development of a model of the gas phase in iCVD. Existing efforts to model the vapor phase have been limited in their scope, and do not provide generalizable methods. Chapter 3 deals with heat transfer and mass transfer in the vapor, while Chapter 4 deals with the kinetics of TBPO decomposition in the vapor, its potential interactions with monomer, and the effects on deposition. These investigations are made possible by the Pirani pressure gauge, which measures heat transfer from a fine wire as a proxy for pressure. Because this heat transfer is composition dependent, it can be used as a precise and effective composition sensor.

Chapter 5 investigates the base-catalyzed hydrolysis kinetics of two fluoroacrylic

polymers. This reaction has potential applications in surface energy patterning due to the marked contrast between the fluoroacrylic surface energy and the carboxylic acid reaction product. The system is further interesting due to the passivating properties of the fluoroacrylate surface, which push the limits of capability of the hydrolysis reaction. This study shows that grafting and crosslinking of the film are necessary to achieve successful hydrolysis, and characterizes the kinetics, showing it to be an S_N2 reaction.

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CHAPTER 2:

PREVENTION AND CHARACTERIZATION OF THIN FILM DEFECTS INDUCED BY CONTAMINANT AGGREGATES IN INITIATED CHEMICAL VAPOR DEPOSITION

Abstract

As initiated Chemical Vapor Deposition (iCVD) finds increasing application in precision industries like electronics and optics, defect prevention will become critical. While studies of non-ideal morphology exist in the iCVD literature, no studies investigate the role of defects. To address this knowledge gap, we show that buildup of short-chain polymer or oligomer during normal operation of an iCVD reactor can lead to defects which compromise film integrity. We used atomic force microscopy to show that oligomer aggregates selectively prevented film growth, causing these hole-like defects. X-ray diffraction and optical microscopy demonstrated crystallinity of the aggregates, pointing to a flat-on lamellar or mono-lamellar structure. To understand the origin of the aggregates, spectroscopic ellipsometry showed that samples exposed to the reactor consistently accrued low-volatility contaminants. X-ray photoelectron spectroscopy revealed material derived from polymerization in the contamination, while scanning electron microscopy showed the presence of defect-causing aggregates. We directly linked oligomeric/polymeric contamination with defect formation by showing an increased defect rate when a contaminant polymer was heated alongside the sample. Most importantly, we showed that starting a deposition at a high sample temperature (e.g. 50°C) before reducing it to the desired setpoint (e.g. 9°C) unilaterally prevented defects, providing a simple method to prevent defects with minimal impact on operations.

2.1. Introduction

Over the last two decades, initiated Chemical Vapor Deposition (iCVD) has emerged as a promising polymer thin film synthesis technique. Because iCVD is a single step process which produces low roughness, conformal, high purity films¹, it is being leveraged to produce better antifouling surfaces², porous polymer films³, passivating coatings⁴ and adhesives⁵ (for a comprehensive review, see Gleason¹). In particular however, the high quality of iCVD films has inspired numerous potential commercial applications in industries like optical⁶ and electronic materials^{7,8}, where uniform films with low roughness are required^{9,10}. In these industries, defects can reduce the product quality, prevent miniaturization, and decrease manufacturing yield (directly reducing revenue). As a result, for well-established manufacturing techniques like inorganic CVD and PVD, a robust body of literature exists to address defect prevention¹¹. Despite its emerging application in precision devices however, the polymer CVD literature has done little to discuss defect formation and prevention.

Taking iCVD as an example, while the literature contains investigations of roughness and island growth caused by low surface energy materials¹² and substrates¹³, discussion of defects has historically been limited to the observation that iCVD is unsusceptible to the types of surface tension-induced defects which affect techniques like spin coating¹⁴. The ability to create high-quality, defect-free films is instrumental for precision applications of iCVD, such as soft electronics, where iCVD has been successfully incorporated into devices^{15–17}. As iCVD continues to mature as a technology and a wider range of deposition conditions and monomer chemistries are implemented, the importance of filling the current gap in our understanding of defect prevention will only grow. In the context of this paper, the term ‘defect’ describes the optically observable structure caused by the contaminant in the polymer film, and the term ‘aggregate’

describes the optically invisible contaminant that caused the defect to form.

Herein, we demonstrate the existence of a previously unexplored defect type arising from crystalline aggregates with an in-plane fractal shape that is on the micrometer to millimeter scale. Our data indicates the aggregates are made up of oligomers that accumulate in the reactor during regular depositions. The morphology of such defects resembles surface-stabilized polymer crystals, which are known to form readily on solid surfaces and exhibit fractal structures (Figure 2.1)^{18–23}. Morphology of the fractal structures is dictated by diffusion-limited crystallization (Figure 2.1 (a)), and the anisotropic surface tension of the crystal's prism facet (Figure 2.1 (c))^{24,25}. Subtle differences in crystal anisotropy and supercooling can give rise to a variety of distinct in-plane fractal morphologies^{26,27}. In particular, the complex in-plane fractal morphologies are understood via 2D diffusion-limited crystallization theories¹⁸, while the out-of-plane morphology is governed by the semicrystalline lamellar structure characteristic of polymer crystals (Figure 2.1 (b))²⁸. In ultra-thin crystalline polymer films (<100nm), the polymer lamellae lay flat on the substrate surface¹⁸. Mono-lamellar and multi-lamellar polymer crystals can be differentiated using scanning electron microscopy (SEM) and atomic force microscopy (AFM)²¹ since multi-lamellar crystals exhibit stepwise thickness changes in integer multiples of the lamellar thickness¹⁸.

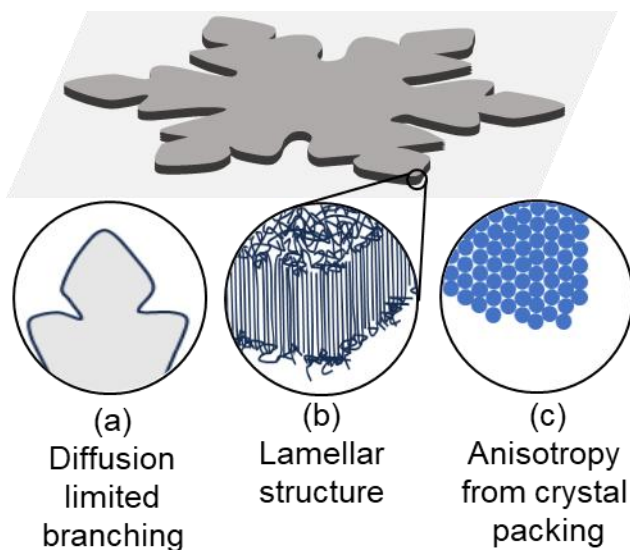


Figure 2.1. Schematic showing a mono-lamellar polymer crystallite on a surface, highlighting **(a)** Branching morphology driven by diffusion limited aggregation, **(b)** flat-on mono-lamellar structure of ultrathin crystalline polymer films, and **(c)** horizontal cross section of (b), showing crystalline packing of chains. Anisotropy and symmetry of the aggregate arises because new chains add to the rough tip more easily than they add to the flat prism facet. This phenomenon is also referred to as surface tension anisotropy of the prism facet (see Section A2.2).

So far, detailed reports of crystallinity in iCVD are limited to the crystallization of atactic poly(1H,1H,2H,2H-perfluorodecyl acrylate) (pPFDA), which crystallizes due to the bulkiness of its large perfluorinated side-chain²⁹. In this case, the close-packing of the bulky side chains is the primary driver of crystallization, and the polymer backbone (including its stereochemistry) has little impact on the crystal structure^{30,31}. While the semicrystalline nature of some pPFDA films synthesized in iCVD appears to affect the surface morphology of growing films, this phenomenon is distinct from the post-synthesis crystallization of polymeric material discussed herein³². Besides pPFDA²⁹, reports of crystallinity in iCVD are exceedingly rare³³. Indeed, NMR studies have

confirmed the expected atactic³⁴ nature of iCVD polymers³⁵, which are considered less prone to crystallization than their tactic counterparts³⁴. While some of the defects we observed are associated with fluorinated polymers, others emerge in deposition environments that have never been exposed to fluorinated species, suggesting that such defects arise from atactic oligomer/polymer aggregates, and crystallize by a different mechanism than atactic pPFDA.

Albeit uncommon, atactic polymers without bulky side chains have also been reported to crystallize – generally when factors are at play that allow locally tactic regions to find each other more easily. Furthermore, factors which enable greater chain extension can help overcome the energy barrier caused by chain entanglement³⁶, as in the case of atactic poly(methyl methacrylate) crystallization in the presence of chain stretching agents³⁷. Crystallization of poly(acrylonitrile) (PAN) and poly(vinyl chloride) (PVC) are both famous cases of atactic polymers which maintain relatively high crystallinity via enhanced interaction and packing among tactic segments that arise randomly within an atactic chain^{38,39}. Crystallization of atactic oligomeric polystyrene has been observed by Chai *et al.*, and is justified by considering that Bernoulli statistics would generate a sufficiently high concentration of stereoregular material to form crystallites²³. In these three examples, chain stretching agents, conformational flexibility, and low degree of polymerization play similar roles in enabling short sections of tactic polymer generated at random to aggregate and crystallize. Notably, free radical polymerization in iCVD is known to lead to a high dispersity⁴⁰ and often low molecular weight polymer⁴¹. This type of polymerization would be expected to give rise to an oligomer presence which resembles the conditions used by Chai *et al.*, and may explain the defects observed herein.

In this study, optical and AFM microscopy were used to elucidate the process of defect formation from the oligomeric material accumulated in the reactor over the course of regular

depositions. XRD was performed on heavily defect-laden samples to better understand the crystal structure. To elucidate the origins and nature of reactor contamination, spectroscopic ellipsometry clearly showed presence of low volatility contaminants, while XPS and SEM of the contamination showed material derived from polymerization and aggregate formation respectively. By heating a pPFDA sample alongside a cooled, clean wafer, we induced defect formation intentionally, illustrating that iCVD thin films can be a source of oligomeric contaminants. Finally, we show that defect-free films can be produced (even under conditions that would normally cause defects) when the deposition is begun at a high stage temperature to form a defect-free conditioning film, and stage temperature is subsequently dropped to the desired deposition setpoint. For consistency, we have chosen to use poly(acrylic acid) (pAA) to resolve defects unless otherwise specified (such as samples in Figure 2.2, which illustrate the presence of defects in a range of polymer types). We suspect that defects have been avoided historically by using relatively high stage temperature deposition conditions, explaining why defect formation has not been a problem for precision applications of iCVD to date¹. As we show, defects can be seen clearly with optical microscopy because the film thickness variation they cause manifests vividly as coloration, however optical microscopy has been rare in the iCVD literature compared with characterization by techniques like ellipsometry, SEM and FTIR, which do not show defects clearly. By addressing the overlooked aspect of defect formation in iCVD, our research aims to contribute fundamental understanding and provide practical prevention measures to current and future industries that may rely on iCVD.

2.2. Materials and Methods

2.2.1 Materials

4-vinyl pyridine (4VP) (95%, from Sigma Aldrich), divinyl benzene (DVB) (80%, from Sigma Aldrich), di(ethylene glycol) divinyl ether (DEGDVE) (99%, from Aldrich), 1H,1H,2H,2H-

perfluorodecyl acrylate (PFDA) (97%, from Sigma Aldrich), acrylic acid (AA) (99%, from Sigma Aldrich) were used as monomers during depositions. Di-*tert*-butyl peroxide (TBPO) (98%, from Aldrich) was used as an initiator. All monomers and the initiator were used without further purification. All depositions were performed on polished p/boron-doped silicon wafers (from University Wafer). Argon was obtained from Airgas and was used as a patch gas to maintain consistent overall gas flowrate in some depositions. Thermal grease (Ceramique 2, Tri-linear ceramic thermal compound from Arctic Silver) was used to ensure thermal contact between the 1"x1" thermoelectric cooling device (TEC) (from TE Technology Inc.) and the silicon wafer.

2.2.2 Film Synthesis

Polymer thin film synthesis occurred in a custom-built reactor. Samples were cut from silicon wafers, setting aside a reference sample in each case as a control for characterizations. Once samples were loaded into the reactor, it was closed off to atmosphere and evacuated to base pressure (2-3 mtorr). During this time, an Accel 500LT recirculating chiller (Fisher Scientific) was used to cool the stage to its operating temperature (10°C-50°C). For depositions requiring a sample temperature from -5°C to 10°C, the chiller was set to 15°C and a TEC was used. After pumping down for one hour, the leak rate was recorded in sccm (standard cubic centimeters per minute). In all depositions, the leak rate was less than 0.05 sccm. At this stage, flow-rate of each monomer and the initiator were calibrated by closing the throttle valve and measuring the linear increase in pressure. In all depositions, TBPO was used as a thermal initiator, the flowrate of which was controlled using an MKS mass flow controller type 1159B. Monomer feed into the reactor was controlled using a needle valve.

After pump down, all flowrates and pressure were brought up to their desired setpoints and allowed to equilibrate for 5 minutes. Pressure in the reactor was monitored using an MKS Baratron

capacitance manometer model 626C and was controlled using an MKS throttle valve (model no. 253B) at the outlet of the reactor. Once equilibrated, the filament array was turned on and brought up to 215°C. The filament was resistively heated using a BK Precision DC power supply, and its temperature was monitored with a k-type thermocouple during the deposition. Thickness of films was monitored during deposition using a 633 nm in-situ laser interferometer with a JDSU He/Ne gas laser (model number 1508P-1). Once the desired thickness was reached, the deposition was stopped by turning off the filament array and evacuating the reactor. Refer to Table 2.1 for deposition conditions. All pPFDA depositions were performed using the same deposition conditions and pAA was used as a standard polymer in this study to resolve defects for consistency between experiments.

Table 2.1: Deposition conditions used for film syntheses.

Location in paper	TBPO (sccm)	M1	Flow (sccm)	M2	Flow (sccm)	Ar (sccm)	Pressure (mTorr)	T _{stage} (°C)	Historical use of fluorinated monomers in the reactor
Figure 2.2 (d)	0.5	4VP	4	DVB	0.3	0.75	400	25	N
Figure 2.2 (a)	1.86	AA	5.05	DEGDVE	0.02	-	100	10	Y
Figure 2.2 (b)	0.9	PFDA	0.08	-	-	0.5	200	20	Y
Figure 2.2 (c)	0.39	AA	0.81	-	-	-	100	15	Y
Figure 2.2 (f)	0.6	DVB	0.9	-	-	2	300	30	N
Figure 2.2 (e)	0.5	4VP	4.1	-	-	0.75	400	25	N
Figure 2.3 (d)(g)	0.6	AA	1.4	-	-	-	100	15	Y
Figure 2.4 (inset)	1.55	AA	5.42	DEGDVE	0.05	-	40	4	Y
Figure 2.10 (a) & Figure 2.2 (g)	0.21	AA	0.43	-	-	-	10	0	Y
Figure 2.10 (b)	1.63	AA	6.77	-	-	-	100	13	Y
Figure 2.10 (c)	0.89	AA	1.78	-	-	-	500	50	Y
Figure 2.10 (d)	1.03	AA	2.09	-	-	-	500/100/100	52.4/9/2	Y
Table 2.4 and Figure 2.9	0.275	AA	0.275	-	-	-	100	45/0	Y
pPFDA (not imaged)	0.9	PFDA	0.1	-	-	-	100	5	Y

2.2.3 Native Aggregate Synthesis

Native contaminant aggregates reported in Section 2.3.3 were prepared according to the same procedures as a normal deposition (Section 2.2.2) including leak testing, monomer and initiator flow-rate testing and pressure equilibration. The only differences from normal deposition were that the filament array was not turned on (no film was deposited) and because the goal of these experiments were to observe contamination, an additional 2 hours were given for cooled samples to collect contaminant.

2.2.4 Polymer Distillation

Polymer distillation experiments required two consecutive depositions on the same day. First, five roughly 2 cm x 4 cm rectangles were cut from a single silicon wafer, three for the control deposition and two for the polymer distillation itself. For the polymer distillation, an additional pPFDA coated wafer from a previous pPFDA deposition (conditions reported in Table 2.1) of the same size was also used. For both the control deposition and polymer distillation, one of the wafers is set aside as a blank. For the control deposition, two TEC plates are placed adjacent to each other in the reactor and both remaining blank samples are placed on the TEC plates as in Figure 2.9. TEC temperatures are monitored with k-type thermocouples affixed to samples with Kapton tape and thermal grease and are controlled manually using a high voltage power supply. The reactor is pumped to base pressure, flow-rates are calibrated, and the reactor is brought up to target conditions as normal. Before turning on the filament array and starting the deposition, the voltages are set on the TEC plates so that one heats to 45°C (sink wafer), while the other simultaneously cools to 0°C (source wafer). This heating step drives off any aggregates from the sink which may have formed during pump-down. The voltages are then switched so that the temperature of the sink sample drops to 0°C and the source sample rises to 45°C, stabilizing after roughly 30 seconds.

The sink sample is then allowed to sit for 30 minutes at target pressure with gases flowing and collect aggregates. After the collection period, the filament array is turned on and the deposition is performed to resolve any aggregates which formed on the sink sample.

After the control deposition is performed and samples are removed from the reactor, a blank wafer (sink) is secured to the cold TEC using thermal grease and the pPFDA coated wafer (source) is secured to the hot TEC using thermal grease. The procedures outlined above for the control deposition are then repeated so that the pPFDA sample is heated and the blank wafer is cooled. This enables transfer of pPFDA from the hot (source) wafer to the cold (sink) wafer. Statistical significance between the sample and controls were demonstrated with a paired t-test with a one-tailed t-distribution.

2.2.5 Conditioning Film Method

To ensure rapid sample temperature change, a TEC plate is used. The TEC plate and sample are placed in the reactor and good heat transfer is maintained with thermal grease. The normal deposition procedure is followed with the chiller set to a temperature of 20°C and the TEC plate set to a temperature of 50°C. Once the film thickness reaches 100 nm, temperature is lowered to 9°C by switching the voltage on the power supply and manually controlling until setpoint is reached. The rest of the deposition is performed as normal.

2.2.6 Characterization and Data Analysis

A VHX-970F digital microscope from Keyence collected optical micrographs of films with appropriate magnification used to capture the defect scale. The ImageJ particle analysis program was used to quantify defect surface coverage for polymer distilled samples and controls synthesized as described in Section 2.2.4. To show statistical significance between defect populations generated in Section 2.2.4, a one-tailed paired t-test was performed on surface

coverage values from particle analysis in imageJ.

An Asylum Research MFP-3D-BIO atomic force microscope (AFM) was used in AC tapping mode to analyze the morphology of surfaces containing crystalline structures. A soft tapping mode AFM probe (PPP-NCSTR, NANOSENSORS) was used to scan a 30 x 30 μm region at 0.5 Hz. A 10 x 10 μm region was used for better resolution of island growth in the scan of fully developed defects.

A Bruker D8 Advance ECO powder diffractometer collected powder x-ray diffraction (XRD) scans of samples with high defect loading. Scans were performed over an angle range from 2° to 16° from a 1kW Cu-K α x-ray source.

A JA Woolam alpha-SE spectroscopic ellipsometer collected spectra at 65°, 70° and 75° angle of incidence on aggregate samples and their respective blanks. Data was fitted using Complete EASE 6 software version 6.51 with a 3-layer model of a Cauchy film resting on a silicon thermal oxide on a semi-infinite silicon metal substrate. For contaminant samples synthesized according to Section 2.2.3, statistical significance between reference samples and contaminant samples was demonstrated using a one-tailed paired t-test.

A ScientaOmicron ESCA 2SR XPS collected X-ray photoelectron spectroscopy (XPS) survey scans from aggregate samples prepared using the procedures in Section 2.2.3. The X-ray source pass energy was 150eV, and the filament voltage was 10kV. Data was collected with a step size of 1eV. Data analysis and peak integration were performed using CasaXPS software.

A Zeiss Gemini 500 Scanning Electron Microscope (SEM) collected SEM micrographs of aggregate samples.

A Nicolet i A1.50 FTIR spectrophotometer with a variable angle grazing attenuated total reflectance (VariGATR) attachment set at 60 degrees and a liquid nitrogen cooled MCT detector

measured infrared spectra of aggregate samples prepared using the procedures in Section 2.2.3. Spectra were collected from 700 - 4000 cm^{-1} with a resolution of 4 cm^{-1} . Omnic software was used for data analysis and built-in atmospheric correction was used to eliminate spurious peaks where necessary. To avoid bias, baseline correction was not performed. (FTIR spectra were inconclusive and may be found in Figure 2.6)

2.3. Results and Discussion

2.3.1 Defect Characterization by Optical and Atomic Force Microscopy

A selection of optical micrographs depicting the thin film defects and their shapes under study are displayed in Figure 2.2. Adopting the language used in the field of diffusion-limited aggregation, aggregates which show anisotropy are called “dendrites” while those that don’t are called “seaweeds”. Similarly, aggregates which form branching structures are called “fractal”, while those that form islands with no branching are “compact”^{26,27}. For background on diffusion limited crystallization theory, we direct the reader to Chapter 6 of Saito²⁴. An excellent review by Liu and Chen describes the application of this theory to 2D polymer crystallization¹⁸.

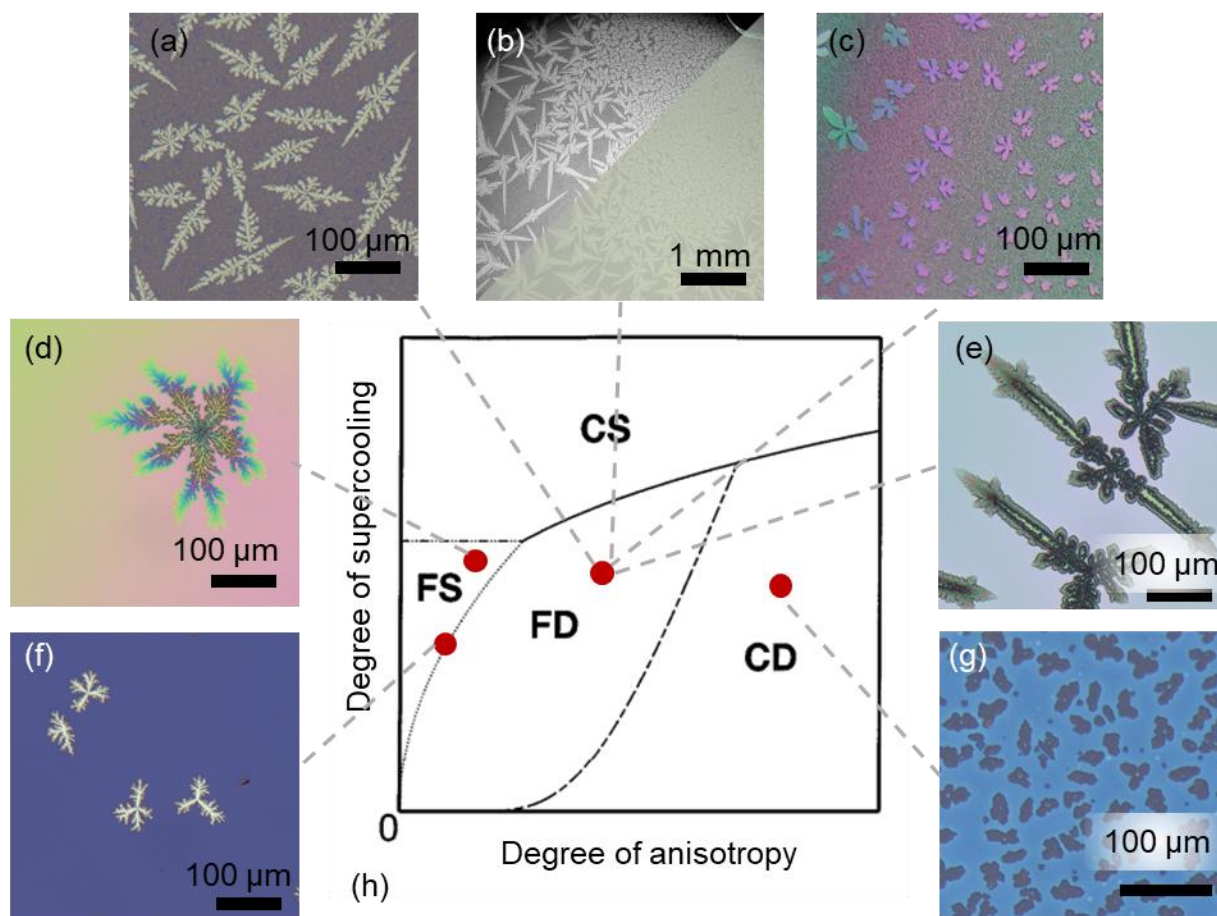


Figure 2.2. (a) Micrograph of a p(AA-co-DEGDVE) film exhibiting a defect with the fractal dendrite morphology (b) Micrograph of a pPFDA film exhibiting large fractal dendrite defects, indicating that defect formation occurs even with hydrophobic monomers like PFDA. Image was modified to clearly show defects (upper left), with the original image also provided (lower right) (c) Micrograph of defects formed on a pAA base-layer, resolved by a second layer of pAA film, showing that defects can form on a range of substrates, including silicon thermal oxide and polymer. (d) Optical micrograph of a p(4VP-co-DVB) film exhibiting a defect with the fractal seaweed morphology. (e) Micrograph of a p4VP film exhibiting fractal dendrite defects, showcasing the diversity of morphologies observed. (f) Micrograph of a pDVB film exhibiting defects with a transitional morphology between fractal dendrites and seaweeds. (g) Micrograph of a pAA film exhibiting a compact dendrite morphology (also in Figure 2.10) (h) An example of a

phase diagram taken with permission from Brenner et al.²⁷. The regions of the phase diagram describe relationships between morphology, labeled as compact seaweed (CS), compact dendrite (CD), fractal dendrite (FD), and fractal seaweed (FS), with the precise relationships depending on the system under study. Labels categorizing micrographs into these morphologies are an aid to the eye, and do not correspond to measured or calculated anisotropy and supercooling. The degree of anisotropy describes the anisotropy of surface tension on the prism facet of a growing crystal (e.g., a higher degree of anisotropy indicates a higher surface tension on the prism edges); degree of supercooling describes the driving force for crystallization, (e.g., a higher degree of supercooling represents a greater driving force to crystallize).

Defects shown in Figure 2.2 (a)-(c) and (e) are good examples of fractal dendrites (FD in the phase diagram in Figure 2.2 (h)). The fractal dendrite morphology indicates that contaminant aggregates formed at a moderate degree of anisotropy (caused by moderate surface tension anisotropy on the crystal's prism facet) and low supercooling (i.e., the driving force for crystallization)²⁷. Though these samples are all classified as fractal dendrites, their diversity indicates sensitivity to growth conditions or variability in aggregate composition. Figure 2.2 (d) shows an example of a fractal seaweed (FS in the phase diagram), expected to form under conditions of low anisotropy^{26,27}. Compact seaweeds (CS), which form under a high degree of supercooling, were not observed under our reactor conditions. Figure 2.2 (g) shows an example of a compact dendrite (CD in the phase diagram), with no branching. Compact dendrites form under a high degree of anisotropy. Figure 2.2 (f) exhibits a transitional morphology, indicating it was formed at conditions close to the coexistence between dendrites and seaweeds. Figure 2.2 (b) shows the largest single defects we have observed (which were resolved using a pPFDA film), showing numerous millimeter-scale defects (the upper left half of the micrograph was modified to

highlight defects, with the original image provided on the lower right).

It is important to note that the labels categorizing micrographs in Figure 2.2 (a-g) are purely based on morphology, and do not reflect estimates of supersaturation and anisotropy. The degree of supercooling and degree of anisotropy are well-defined quantities in a theoretical context, and are therefore instrumental in generating a phase diagram like that shown in Figure 2.2 (h)²⁷, however the conventional iCVD setup used to synthesize samples in Figure 2.2 is not capable of controlling aggregation conditions enough to estimate these parameters.

Degree of supercooling is a dimensionless metric of the driving force for aggregation and degree of anisotropy is a dimensionless metric of how strongly surface tension is affected by the crystal facets^{24,27}. The mathematical problem of diffusion limited aggregation which was used to generate the phase diagram shown in Figure 2.2 (h) is taken with modification from Brenner *et al.*²⁷ In Brenner *et al.* the problem is posed as a heat transfer limited aggregation problem, however, our system is a mass transfer limited aggregation problem. Because the governing equations (copied below) are identical, the same phase diagram can be used:

$$\frac{\partial U}{\partial t} = D \nabla^2 U \quad (2.1)$$

$$v_n = D \mathbf{n} \cdot (\nabla U_{C,int} - \nabla U_{A,int}) \quad (2.2)$$

$$U_{int} = \Delta - d(\theta)K \quad (2.3)$$

Here, subscripts ‘C’ and ‘A’ refer to the crystalline aggregate surface and the amorphous adsorbed polymer or oligomer while subscript ‘int’ refers to quantities at the interface of the growing aggregate and the amorphous phase. D is the diffusion coefficient, v_n is the local velocity of the

growing aggregate in plane with the surface, $\mathbf{n}$ is the unit normal vector to the growing aggregate in plane with the surface, $U = \frac{\mu - \mu_\infty}{\Gamma_C - \Gamma_A} \left(\frac{\partial \Gamma}{\partial \mu} \right)_{\mu = \mu_{ref}}$ is the dimensionless chemical potential, μ is the chemical potential, μ_∞ is the chemical potential far from the aggregate, Γ is the equilibrium molar surface excess at the given sample temperature (see Meier for background on surface thermodynamics)⁴², and $\left(\frac{\partial \Gamma}{\partial \mu} \right)_{\mu = \mu_{ref}}$ is the change in molar surface excess with chemical potential at a reference chemical potential close to that of the growing crystal (assuming this quantity varies slowly with chemical potential and can be treated as a constant). Δ is the degree of supercooling, defined by equation (2.4),

$$\Delta = \frac{\mu_m - \mu_\infty}{\Gamma_C - \Gamma_A} \left(\frac{\partial \Gamma}{\partial \mu} \right)_{\mu = \mu_{ref}} \quad (2.4)$$

where μ_m is the chemical potential at which the aggregate will begin melting at a given sample temperature (as molar surface excess varies). (equation 2.3) is a local equilibrium condition which accounts for the interface curvature, where K is the 2D curvature and $d(\theta)$ is the anisotropic surface tension (represented here for 6-fold symmetry as we observe in our system),

$$d(\theta) = d_o(1 - \varepsilon \cos(6\theta)) \quad (2.5)$$

where $d_o = \frac{\gamma \mu_m \left(\frac{\partial \Gamma}{\partial \mu} \right)_{\mu = \mu_{ref}}}{(\Gamma_C - \Gamma_A)^2}$ is the isotropic capillary length in terms of the isotropic surface energy γ . θ is the angle between $\mathbf{n}$ and a crystallographic direction (line of symmetry) and ε is the strength of anisotropy.

Beyond morphology, optical micrographs can provide some important insights into the nature of defects. Figure 2.2 (a)-(c) and (g) show samples synthesized in a reactor that was

frequently exposed to fluorinated monomers, whereas Figure 2.2 (d)-(f) show defects that were deposited in a reactor only exposed to chemistries which had been used without incident in other reactors and were not fluorinated. Observing defects in a fluorine-free reactor indicates that fluorinated material is not a prerequisite for defect formation, leaving built-up polymeric/oligomeric material from past depositions as a likely culprit. While all other micrographs show defects on a silicon wafer (with the top layer being an amorphous silicon thermal oxide), Figure 2.2 (c) shows defects on a (defect-free) pAA film (resolved by a second pAA deposition), demonstrating that defects can form on a range of surfaces.

The defect formation mechanism was illustrated using AFM scans of defects at subsequent stages of development (Figure 2.3). Defect-causing aggregates formed after samples were loaded into the iCVD reactor and during the pump-down step while stage and line temperatures stabilized. The AFM scans of the aggregates that formed after the pump-down step are shown in Figure 2.3 (a)-(c). After the reactor reached base pressure (1-3 mtorr), flowrates of the initiator and monomer were calibrated, the chamber pressure was set, and the deposition was started by turning on the filament array (see Section 2.2.2 and Table 2.1 for methodological details and deposition conditions). After 5 minutes of depositing a pAA film, growth only occurred on the pristine surface, and depressions began to form where the aggregates were located (Figure 2.3 (d)-(f)). By the end of the deposition, the aggregate surface remained film-free, but exhibited islands where polymer growth nucleated (Figure 2.3 (g)-(i)). These islands of polymer growth resembled those observed by Rumrill *et al.*¹³, where poly(hydroxyethyl methacrylate) islands were deposited on a poly(tetrafluoroethylene) film, implying a similar mechanism for island formation (i.e., driven by surface energy mismatch). Defects were visible in optical micrographs (Figure 2.2) because even small variations in thickness or roughness cause significant color changes. The “rainbow”

appearance of defects in Figure 2.2 (d) and (e) is also explained by this correspondence between color and thickness and the transformation of defects from holes to depressions in the film as islands merge. These results demonstrate clearly that hole formation is caused by the inhibition of deposition on the contaminant surface as opposed to the space-filling role played by immiscible small molecules in porous film formation from iCVD^{43–46}.

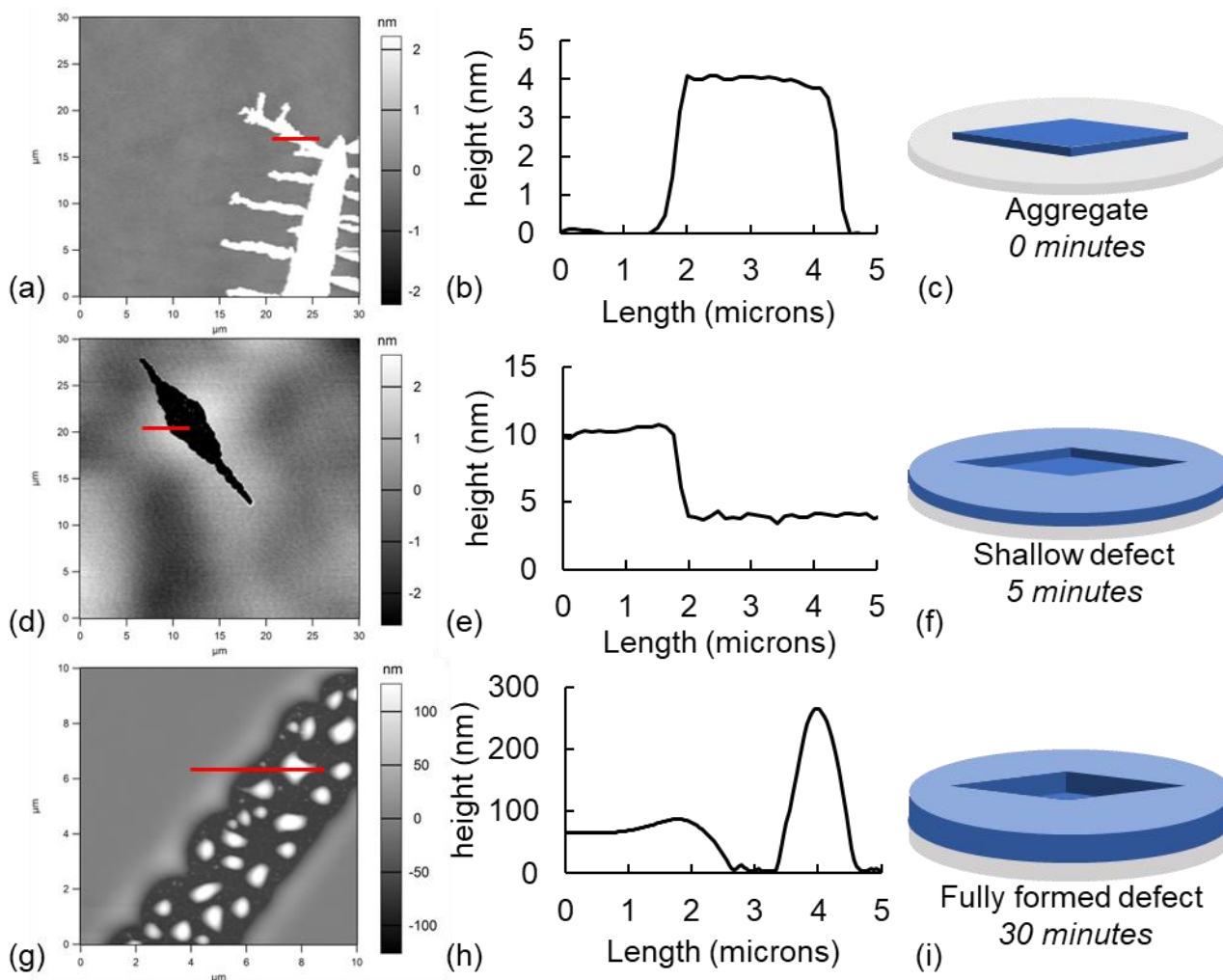


Figure 2.3. (a) AFM scan of a contaminant aggregate where the red bar indicates where the line scan in panel (b) was taken. (all line scans run left to right) (b) Line scan of a contaminant aggregate shown in panel (a). (c) Cartoon showing the contaminant aggregate represented in panel (a). (d) AFM scan of a shallow defect in a pAA film (deposition conditions in Table 2.1) where

the red bar indicates where the line scan in panel (e) was taken. **(e)** Line scan of the shallow defect shown in panel (d). Pristine film thickness was normalized to 10 nm based on spectroscopic ellipsometry measurements. **(f)** Cartoon showing the shallow defect represented in panel (d). **(g)** AFM scan of a fully developed defect in a pAA film (deposition conditions in Table 2.1) where the red bar indicates where the line scan in panel (h) was taken. **(h)** Line scan of the defect shown in panel (g) showing large islands of iCVD polymer on the defect surface. **(i)** Cartoon showing the fully developed defect represented in panel (g).

2.3.2 XRD and Anisotropy

To fully understand defect morphology, it is necessary to discuss the anisotropy intrinsic to our defects. Because the silicon substrate and polymer film are both amorphous, the anisotropy in fractal dendrites (Figure 2.2) is attributed to the crystallinity of the defect-causing aggregate. As such, we performed powder XRD on a sample that exhibited prodigious defect formation (Figure 2.4 top trace and inset). Because of the orientation of the crystal relative to the diffractometer, all d-spacings represent length scales normal to the surface. The observed d-spacing (3.75 nm) is consistent with the aggregate thickness observed in AFM traces (3.7-4 nm, Figure 2.3 (a)). Assuming the aggregates are oligomeric/polymeric in nature, this d-spacing corresponds well to the reported lamellar spacing¹⁸. Furthermore, the presence of diffraction in this sample as well as the remarkably high defect loading indicates a potentially multilamellar structure, in contrast to other samples analyzed with SEM and AFM (Figure 2.3 and Figure 2.7). Given the similarity of diffraction patterns, the crystallinity observed for iCVD poly(ethylene glycol dimethacrylate) and poly(isobornyl acrylate)³³ may also be due to lamellar structures.

Crystallinity in iCVD films has been observed most strongly for pPFDA, with the

crystallization explained by the packing of the bulky side chains ^{29,30}. Furthermore, fluorine was found in some contaminant material (Section 2.3.3), although not all (Figure 2.2 (d)-(f)). Therefore, we believe there is enough reason to assess the possibility that pPFDA may be the cause of defects in Figure 2.4. Based on the powder XRD data (Figure 2.4, top and bottom traces), there is no correspondence between the crystalline pPFDA film and that of the contaminant. Powder XRD patterns distinct from pPFDA have also been reported for poly(ethylene glycol dimethacrylate) and poly(isobornyl acrylate) synthesized by iCVD³³, though the origin of the crystallinity was not discussed.

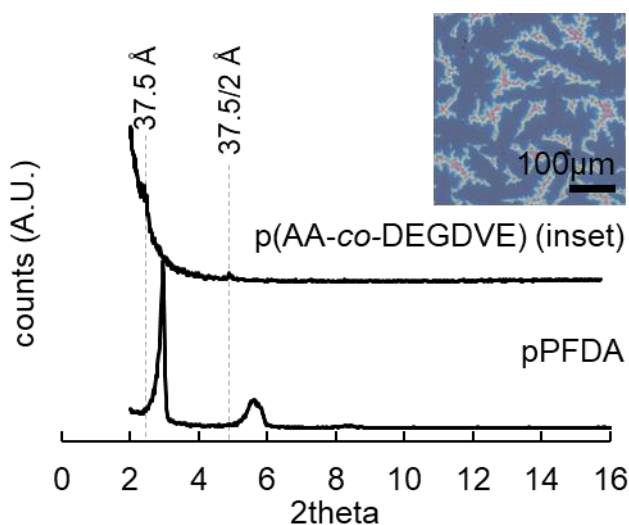


Figure 2.4. Powder XRD of contaminated p(AA-co-DEGDVE) film (top trace) shown in inset, crystalline pPFDA (micrograph not shown). The sample in the inset was produced from a pAA deposition, and shows crystallinity, with a d-spacing of 37.5 Å. Note that the powder diffraction pattern for the sample shown in the inset does not match pPFDA.

2.3.3 Surface Characterization of Contaminant Material

While optical and AFM micrographs show that defects were caused by contaminant aggregates, they provide few insights into the origins of contamination. To characterize contaminant material present in the iCVD reactor, contaminant was collected by placing three samples on the cooled stage for 3 hours. The next day, ellipsometry was conducted on samples exposed to contaminants in the reactor and blank samples not exposed to the reactor (Table 2.3). SEM and XPS were performed on samples after the exposure (Figures 2.5 - 2.7 and Table 2.2 and 2.3).

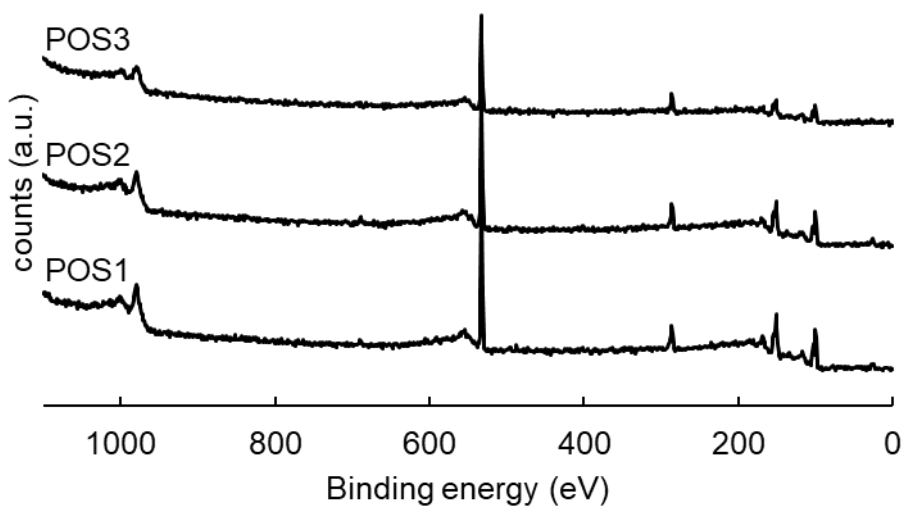


Figure 2.5. XPS survey scans for samples placed in the reactor.

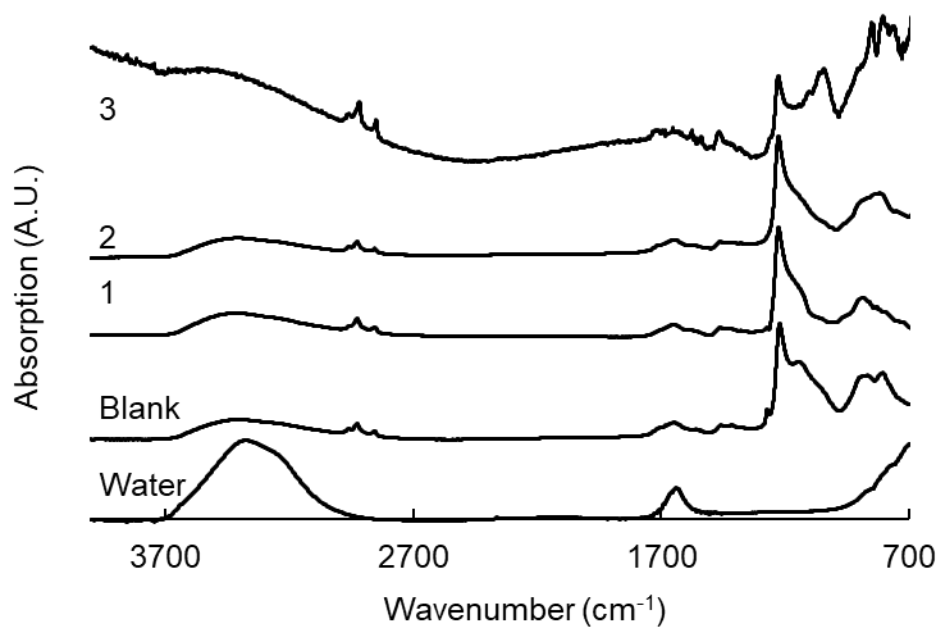


Figure 2.6. FTIR Vari-GATR scans for samples placed in the reactor.

Table 2.2. XPS mole fraction

	Run 1	Run 2	Run 3
F	0.01	0.01	0
C	0.19	0.24	0.25
O	0.43	0.40	0.44
Si	0.37	0.35	0.31

Table 2.3. Ellipsometry thickness

	Run 1	Run 2	Run 3
Thickness before (nm)	0.43	0.46	0.5
Thickness after (nm)	0.72	0.89	1
Age (days)	1	1	2

Spectroscopic ellipsometry data was fitted to a model representing a Cauchy film on silicon thermal oxide. The three Cauchy-film thicknesses for samples not exposed to the reactor averaged 0.46 ± 0.01 nm (mean $\pm$ standard error). This thickness value represents a layer of ambient contamination (e.g., adsorbed water) or model error and should be interpreted as a baseline as opposed to an actual “film”. Ellipsometry was also performed on the contamination-exposed samples (i.e., on wafers identical to the pre-exposure measurements). The average thickness was 0.87 ± 0.06 nm (mean $\pm$ standard error). The difference is statistically significant, with $p = 0.011$ from a 1-tailed paired t-test, showing that samples that spend time in the reactor picked up contaminant material. Because post-exposure samples were removed from the reactor a day before they were characterized, the contaminant material must be stable and non-volatile, thus likely macromolecular. The simplest source for such a contaminant is oligomeric material accumulated from past depositions.

SEM images of the post-exposure wafers revealed aggregate formation on two of them, with some aggregates appearing to be in the process of melting or sublimating (Figure 2.7 (a) and (b)). XPS survey scans performed on the same samples showed fluorine in 2 of the 3 samples (Figure 2.5 and Table 2.3). There was no notable correlation between samples with fluorine and observable aggregates in SEM. Observation of aggregates on samples never directly exposed to monomer vapor and showing stability under ambient conditions for hours to days eliminates the possibility that monomer crystals or small molecule crystals are involved in defect formation. Because synthesizing samples in a clean room setting was infeasible in this study, the obscuring presence of ambient contamination confounds a more detailed analysis of the XPS data. FTIR was also collected for these samples (Figure 2.6) but was inconclusive.

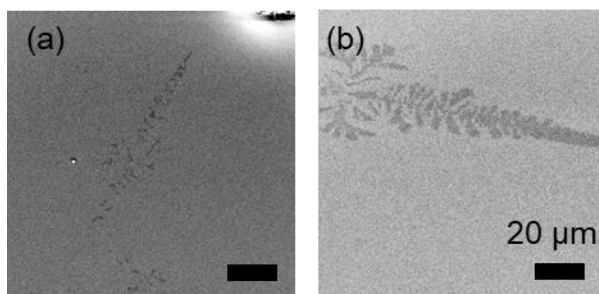


Figure 2.7. SEM images depicting remnants of melting or sublimating contaminant aggregates prepared using the procedures described in Section 2.2.3 **(a)** Almost fully melted or sublimated aggregate (corresponding to run 3 in Tables 2.2 and 2.3, and Figures 2.5 and 2.6). **(b)** Aggregate showing moderate melting or sublimation (corresponding to run 1 in Tables 2.2 and 2.3, and Figures 2.5 and 2.6).

We note that most surface analysis techniques are not sufficiently sensitive to characterize nanogram quantities of material (outside a cleanroom environment). This includes techniques like SEM with electron dispersive spectroscopy and polarized light microscopy. Similarly, bulk chemical analysis like nuclear magnetic resonance spectroscopy and matrix assisted laser desorption/ionization time of flight mass spectrometry require milligram quantities of material making such techniques infeasible. The reproducibility of these techniques is further jeopardized by non-defect-forming contamination (such as amorphous oligomer), contamination from the characterization technique itself (adventitious carbon), or damage to the aggregates by X-rays or electron beams. Considering these pitfalls, we can confidently draw two conclusions based on the surface analyses. First, samples that spend time in the reactor consistently and repeatably accrue non-volatile contaminants capable of remaining on the surface for at least 1-2 days. Second, fluorinated chemistry is often present in contamination. Because fluorine is only introduced to the reactor as a constituent of monomers, its prolonged presence in the reactor means that some

fraction of the contaminant material captured by ellipsometry is derived from polymerization and is most likely oligomeric or polymeric. In Section 2.3.4, an indirect method is discussed, which circumvents the limitations of surface analysis and enables a well-controlled study of defect formation.

2.3.4 pPFDA Polymer Distillation

While material(s) other than pPFDA can cause defects, as suggested by optical microscopy and XRD (Figures 2.2 and 2.4), in this section we demonstrate an example where defect formation was induced via exposure to pPFDA. The experimental setup used is shown in Figure 2.9, and detailed methods can be found in Section 2.2.4. Briefly, a pPFDA-coated “source wafer” was heated alongside a clean and uncoated “sink wafer” for 30 minutes, after which a pAA deposition was performed on both wafers. To obtain a baseline of defect formation in the reactor, the experiment was also performed with an uncoated source wafer (on the same day). Optical micrographs of the sink wafers for pPFDA and blank sources are provided in Figure 2.8. This experiment was performed three times, showing a statistically significant increase in surface coverage with $p = 0.027$ from a one-tailed t-test.

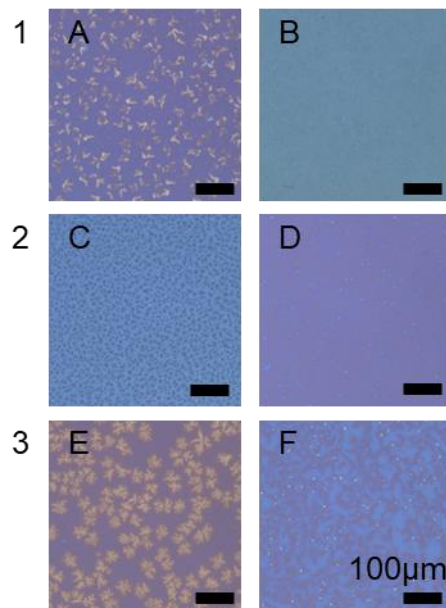


Figure 2.8. pPFDA distilled samples **(a)**, **(c)** and **(e)** for runs 1, 2 and 3 respectively showing defects caused by pPFDA distillation and blank samples **(b)**, **(d)**, **(f)** for runs 1, 2 and 3 respectively showing baseline reactor contaminant levels.

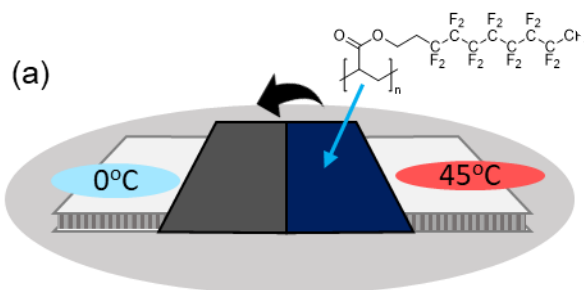


Figure 2.9. Experimental setup showing polymer distillation of pPFDA from a wafer placed on a hot TEC at 45°C to the cold wafer placed on a TEC cooled to 0°C

Table 2.4 shows the defect surface coverage for each distillation experiment and its control (quantified using particle analysis in imageJ). Having shown in this section that pPFDA can form defects and in Sections 2.3.1 and 2.3.2 that defects also occur without pPFDA (Figure 2.2 and 2.4), we have demonstrated that multiple chemical species are responsible for defect formation.

Furthermore, because the pPFDA-derived defect morphology observed in this distillation experiment matches defects that arise spontaneously (Figure 2.2 (d)), this experiment supports the conclusion that defects are caused by materials derived from polymerization – specifically a build-up of oligomeric material. More in situ distillation experiments and ex-situ investigation of oligomer crystallization²³ is necessary to understand better what properties make a given oligomer inclined to form defect-causing crystalline aggregates. However, that is outside the scope of this study.

Table 2.4. Defect coverage data for pPFDA distillation.

Run #	Control defect coverage	Sample defect coverage
1	0.6%	18.5%
2	7.3%	39.7%
3	7.8%	23.2%

2.3.5 Defect Prevention by the Conditioning Film Method

High-quality film synthesis using iCVD will require a procedure to avoid defect formation. Since crystals tend to have a negative entropy of formation, crystalline aggregates are expected to form more readily at lower temperatures. To confirm this for defects, pAA depositions were performed at a range of stage temperatures and were characterized with optical microscopy (see Table 2.1 for deposition conditions and Section 2.2.2 for methods). We found that, as the stage temperature increased from 0 °C to 13 °C, aggregate formation was suppressed (Figure 2.10 (a) and (b)). At 50 °C, no defects were observed (Figure 2.10 (c)). Because the stage temperature is often reserved as a key synthesis condition that regulates the kinetics of iCVD polymerization,

performing all depositions under such a high stage temperature may not be feasible to prevent defects.

Because the focus of this section is defect prevention during normal reactor operation, the samples shown in Figure 2.10 (a-c) were synthesized using normal deposition procedures, so their defect coverage necessarily depends on the unknown contaminant loading, which is subject to moderate variation over time. The trend shown in Figure 2.10 (a-c) is a qualitative verification of the well-established thermodynamic principle that crystal formation is inhibited at elevated temperature due to its negative entropy of formation. A quantitative study of the dependence of defect formation on temperature requires good controls which can be achieved using distillation procedures discussed in Section 2.3.4 but is nontrivial and is outside the scope of this study.

We demonstrate a generalizable approach to prevent defect formation, which consists of starting a deposition at a high stage temperature (e.g., 50°C) and, once the deposition starts (filament temperature reaches set-point), lowering the stage temperature without stopping the deposition (e.g., to 5°C), to obtain defect-free films at conditions which might usually be prone to defects (Figure 2.10 (d)). The thicknesses reported by AFM in Figure 2.3 coupled with consistent defect coloration within optical micrographs in Figure 2.2 (where light interference relates color and thickness) indicate consistent defect depth. This implies that all defects were formed at the start of a deposition, and defects do not form on actively growing films. Therefore, high stage temperatures only need to be used at the start of deposition and do not need to be imposed throughout. According to this logic, the defects studied here can always be prevented, provided the onset of a deposition utilizes a set of conditions that are not conducive to defect formation. We call this technique the conditioning film method, since it relies on establishing an actively growing defect free film which conditions the surface against defects.

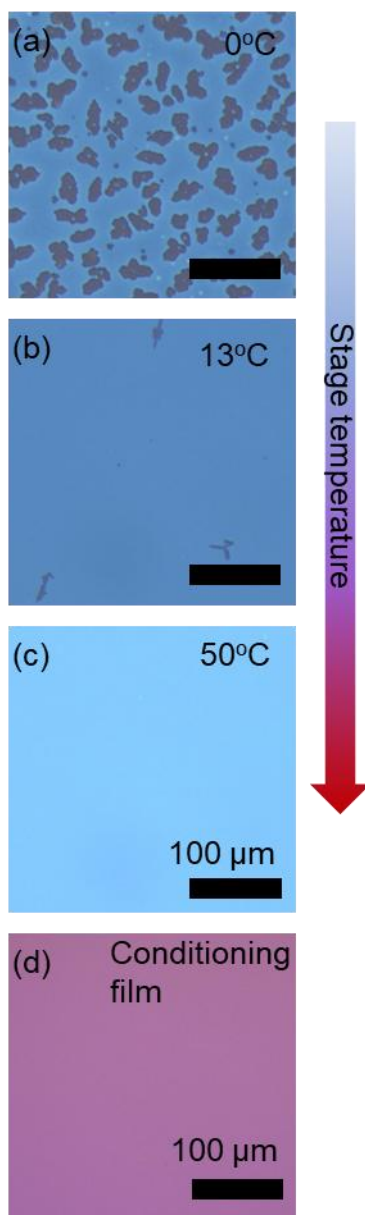


Figure 2.10. (a) Optical micrograph showing defects resolved by pAA using a stage temperature of 0°C. (b) Optical micrograph showing defects resolved by pAA using a stage temperature of 13°C. Noticeably fewer defects are present than panel (a). (c) Optical micrograph showing a defect-free pAA film using a stage temperature of 50°C. (d) Optical micrograph of a film made using the conditioning film method (see Section 2.2.5), showing the absence of defects.

2.4. Conclusions

We have introduced evidence for a new defect type that has the potential to impact low-stage temperature depositions ($<25^{\circ}\text{C}$) using the iCVD technique. AFM on samples obtained at various stages of defect development showed that contaminant aggregates form on the sample surface before deposition and cause holes in the final film by preventing film growth on the aggregate surface. Optical micrographs of defects match theoretical expectations for diffusion-limited aggregation²⁶. Observation of anisotropic defects on multiple types of amorphous surfaces shows that defect-causing contaminant aggregates are made of crystalline material and do not originate from flaws in the substrate.

XRD was performed on a highly defect-laden p(AA-co-DEGDVE) film to better understand the underlying crystal structure. The resulting XRD patterns showed peaks corresponding to a d-spacing of 3.75 nm, which matched the 3.7 - 4 nm lamellar thickness measured with AFM. Spectroscopic ellipsometry demonstrated that samples exposed to the reactor for 3 hours took up low volatility contaminants stable in air for at least 1-2 days. XPS survey scans detected fluorine in 2 of the 3 samples, indicating the presence of material derived from polymerization. SEM images confirmed the presence of aggregates on 2 of the 3 samples, notably showing melting or sublimating aggregates. By heating a source of polymer (pPFDA) alongside a cooled wafer and then performing a pAA deposition, we confirmed that polymeric material can cause defects. The simplest conclusion to draw from these data is defects are caused by a build-up of oligomeric material from regular iCVD depositions. This is supported further by crystallization of atactic oligomeric material under similar circumstances²³ and our expectation that iCVD polymerization will generate significant quantities of oligomeric material⁴⁰.

To prevent defects, we show that they occur less frequently at high sample temperatures,

and if a deposition is started at a high sample temperature, it will be defect-free regardless of whether the temperature is subsequently dropped to a much lower value throughout the rest of the deposition. We expect this procedure to be especially valuable for applications of iCVD, which require high-quality, defect-free films. It should be noted that this “conditioning” film method represents only one method by which defects can be prevented. The ability of contaminant to crystallize on the surface likely depends on the surface chemistry as well as temperature, so depending on the nature of the contaminant oligomer material, certain substrate surface chemistries may be effective at lowering the sample temperature at which defects proliferate. A similar method has been used to prevent roughness in iCVD at supersaturated conditions³. Furthermore, if cleaning procedures can be developed which prevent buildup of oligomeric material, this could also be a valid avenue to prevent defects.

In the future, it will be necessary to survey a range of potential contaminant materials and molecular weights to better understand the risk posed by this new defect type to precision application of iCVD in industry. We assume that some applications of iCVD will be unaffected by this defect type and would suggest future work to address the impact of such defects in the context of intended applications. Future experimental and theoretical work should be done to elucidate the relationships between monomer adsorption and contaminant aggregation as well as determine why aggregates inhibit film growth. Finally, we hope that the methods outlined here might enable a lower cost or higher throughput study of diffusion-limited aggregation by providing a simple way to observe large numbers of polymer aggregates, which might be difficult with AFM methods

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CHAPTER 3:

HEAT TRANSFER AND MIXING IN INITIATED CHEMICAL VAPOR DEPOSITION ENABLED BY IN-SITU GAS COMPOSITION SENSING

Abstract

Initiated Chemical Vapor Deposition (iCVD) stands out among polymer CVD technologies because it can produce a broad range of polymer thin film chemistries under mild conditions. Fundamental studies of iCVD kinetics often appeal to models of the iCVD vapor phase which have not been experimentally validated. This issue highlights an overlooked need for empirical studies of the vapor phase, specifically describing mixing, heat transfer and initiator decomposition. Since a rigorous understanding of heat transfer and mixing is necessary for any other vapor phase modeling, it is the subject of this work. We propose that an “ideal” iCVD reactor is well mixed with pressure-independent heat transfer. We show the system is well-mixed by measuring the residence time distribution for argon and isopropanol flowing at 1 sccm to show it matches a well-mixed ideal. The measurement of the residence time distribution was enabled by composition sensing based on the Pirani pressure gauge. We show that the Pirani gauge can serve as a low-cost, modular, and precise gas composition sensor, giving it practical benefits over methods like in-situ infrared spectroscopy and mass spectrometry. Using simple Multiphysics simulations of reactor mixing, we illustrate the impacts of non-ideal mixing and identify the critical Peclet number at which the reactor can no longer be assumed to be well-mixed. By measuring diffusive heat transfer from the filament at a range of pressures in argon and isopropanol, we quantified the effects of non-ideal heat transfer at low pressures and identified a critical Knudsen number below-which heat transfer is pressure-dependent. Our model is further verified by measurement of reactor effective temperature, showing strong agreement with heat transfer experiments. Our work provides tools to estimate the degree of mixing and temperature profile in the reactor for most cases relevant to iCVD, establishing a systematic method of determining the effect of reactor design choices and process variables like flow-rates and temperatures on these phenomena.

3.1 Introduction

Polymer thin films are used widely in applications such as organic electronics¹ and membranes². While solution phase techniques like spin coating and spin casting are the most popular synthesis techniques, polymer chemical vapor deposition (CVD) processes³ have further broadened the applications of polymer films. Polymer CVD is solvent-free, enables conformal film synthesis,⁴⁻⁸ doesn't suffer from surface tension-induced defects like de-wetting, and enables precise thickness control even for ultrathin films (<100 nm).⁹ Many polymer CVD processes require harsh conditions (like plasma-enhanced CVD)^{10,11} or have a limited library of compatible polymers (like parylene CVD).¹² Initiated Chemical Vapor Deposition (iCVD) overcomes these limitations by decoupling initiation and propagation. The initiator decomposes at a hot filament, while polymerization occurs on a cooled stage. As a result, iCVD operates under mild conditions and is compatible with a broad range of chemical functionalities¹³.

During iCVD, physical processes like heat transfer from the array and chemical processes like thermal decomposition of the initiator occur simultaneously. Elucidation of fundamental processes is necessary for material design, innovations in synthesis, and optimization for monomer yield¹⁴ and polymer quality. A common method of understanding complex systems like these is to perform Multiphysics simulations in which the governing partial differential equations for heat, momentum and mass transfer are solved. While we are aware of two studies like this in the iCVD literature,^{15,16} both speculate on novel reactor configurations instead of studying normal iCVD operation and, as a result, have not been incorporated into models of iCVD. Given the importance of Multiphysics simulations to inorganic CVD research,¹⁷ the comparative lack of computational study in iCVD highlights a potential for gaps in our understanding.

With few theoretical or computational studies to draw from, fundamental understanding of

iCVD comes primarily empirical. Most such work aims to model deposition rate, and final film properties (like molecular weight) based on process variables like the reactor pressure and filament temperature.¹³ Most of our current understanding of iCVD comes from this literature, but since the vapor phase is not directly modeled in these studies, little is known about the composition of the iCVD vapor phase under different conditions. While there is an emphasis on understanding the initiator decomposition and surface polymerization reactions occurring in iCVD, these reactions are almost always¹⁸ accompanied by mixing and heat transfer. Because mixing and heat transfer can be studied in isolation, it makes sense to establish a strong model of them before addressing reactive processes. While rare, existing efforts to directly measure heat transfer and mixing in the iCVD system warrant discussion.

Regarding heat transfer in iCVD, an excellent experimental investigation of the iCVD filament was performed by Bakker et al., precisely modeling the important radiative effects in heat transfer under variable pressure for vapors relevant to iCVD.^{15,19} The model of diffusive heat transfer proposed by this study is empirical, therefore it doesn't explain how system geometry or gas properties effect heat transfer. While the kinetic theory of gases is expected to explain these phenomena,²⁰ there have been no efforts to use this theory to develop a first-principles theory of diffusive heat transfer in the iCVD reactor.

The primary goal of mixing studies is often to determine the contacting pattern in the reactor, which generally involves comparing a measured residence time distribution (RTD) with an ideal well-mixed or plug-flow model.²¹ The only RTD measurement we are aware of in an iCVD reactor was obtained using the response of vapor composition at the outlet of the reactor to gas pulses measured using an in-situ FTIR.²² While this work concluded their system to be plug flow, their conclusion is suspect for two reasons. First, no mathematical analysis was provided,

making it difficult to tell what the judgement was based on. Second, plug flow in an iCVD reactor would almost certainly result in poor mixing and, thus, poor coating uniformity – contradicting ample evidence of good coating uniformity achieved by iCVD.³ Furthermore, the claim that iCVD can be classified as either plug flow or well mixed is incomplete, since the mixing state of the reactor is known to depend on process variables.²³

When a phenomenon like mixing depends on process variables it is generally described using dimensionless numbers which correlate with that phenomenon (Reynolds number or Peclet number for mixing). The magnitude of the dimensionless number allows a researcher to easily determine the mixing state of their system, and how it will change with variables like reactor dimensions or flowrate. While dimensionless numbers have been used in iCVD reactor scale-up,²⁴ there have been limited efforts to leverage them in the quantitative study of regime changes in iCVD like the onset of poor mixing, or onset of pressure-dependent heat transfer.²⁵

In iCVD, manipulating well-understood regime changes has been remarkably fruitful. For instance, deposition at or above the monomer saturation pressure has yielded insights into areas like polymerization kinetics,²⁶ nanoparticle fabrication,^{27,28} and porous film fabrication.²⁹ We should expect that less well understood regime changes in heat and mass transfer might enable similar innovations. For instance, poor precursor mixing in the vapor phase is well known to cause film non-uniformity.^{23,24} Therefore, a formalism to quantify mixing could be used to mitigate the need for batch iCVD,^{14,23,30} or be leveraged to produce gradient films (as has been done in parylene CVD).³¹ In the realm of heat transfer, it has been shown that heat transfer from the filament is strongly affected by the reactor pressure, but there are no methods of predicting the onset of this effect.¹⁹ The ability to do so might enable use of initiating systems which are only effective at low pressure. In this paper, we focus on developing models for both mixing and pressure-dependent

heat transfer. The end goal of our model is to allow a researcher to predict the temperature distribution and approximate mixing profile for their reactor using flow-rate, pressure and reactor dimensions.

To address the absence of low cost and precise real-time vapor phase composition sensing in iCVD we also introduce a method of detecting composition using a Pirani convection gauge. The technique of using the Pirani gauge to obtain real-time vapor-phase composition enables us to compare residence time distribution obtained from simulations to experiment. Pirani gauges operate by measuring heat transferred from a fine wire. At low pressures, the rate at which molecules collide with the wire limits heat-transfer so the rate of heat transfer is proportional to the pressure.³² Because larger molecules transfer more energy per collision than smaller molecules, the pressure reported depends on the vapor. In fact, since there are relatively few collisions between gas molecules the pressure reported by the Pirani gauge varies linearly with the mole fraction.³³

Vapor phase composition monitoring via in situ FTIR²² and in situ mass spectrometry³⁴ are expensive and have not gained wide-spread usage. Furthermore, they often need to be placed at the outlet of the reactor, potentially introducing a time delay in the measurement. Just as in-situ interferometry enables precise thickness control, an affordable, precise, and simple-to-implement composition sensor opens-up possibilities for low-cost composition-based feedback control during deposition. An excellent discussion of the potential benefits of this is contained in Schröder *et al.*³⁴ Because it enables process control, process monitoring like the Pirani gauge is not simply an incremental advancement in a process like iCVD, it can play a decisive role in the value it has in the industry.

We establish an ideal model of heat transfer and mixing in an iCVD reactor in which heat

transfer from the filament is pressure independent, and the reactor is well-mixed. Using the model, we show how deviation from the ideal state can be quantified using the Peclet number (mixing) and Knudsen number (heat transfer) using pressure-dependent heat transfer experiments for argon and isopropanol (IPA) gases and simulations of mixing. We demonstrate how this methodology can be easily generalized to different gases and reactor geometries. To characterize diffusive heat transfer, we establish simple first-principles models which can be generalized to most other reactor geometries to provide predictions of the temperature profile in the reactor. While mixing is more complex, we show that simple low-cost Multiphysics simulations can be used to evaluate the effect of design choices and estimate the composition throughout the reactor. We hope our work will provide iCVD researchers with tools to obtain accurate estimates of the mixing state of their reactor, radiative and diffusive heat transfer in their reactor (including the temperature profile), and how both these phenomena are effected by flow rates, gas composition, pressure, reactor dimensions and reactor temperatures.

3.2. Materials and Methods

3.2.1 Mixing simulations

A reactor model was built in Autodesk Inventor which roughly approximated a standard 2-chamber iCVD reactor (figure 3.1). A mixing study was then performed with two gases flowing into the reactor through two separate inlets at flowrates of 1 sccm respectively under steady state.

Transient simulations mirroring step-response experiments (Section 3.2.4) were also performed in which a steady 1 sccm flow of a primary gas through inlet 1 was established, then a 1 sccm flow of a secondary gas through inlet 2 was initiated while flow of the primary gas was ceased, such that total flow was constant. Videos of the changing composition were reported, and step-response curves were reported for composition at the outlet of the reactor, near the inlet in the

top chamber, and far from the inlet in the bottom chamber. Both simulations were performed with diffusion coefficients of 4 cm²/s, 16 cm²/s, 64 cm²/s and 678.5 cm²/s corresponding to Peclet numbers of 5.3, 1.3, 0.3 and 0.03 at the simulation reactor pressure of 0.1 torr and 298.15K.

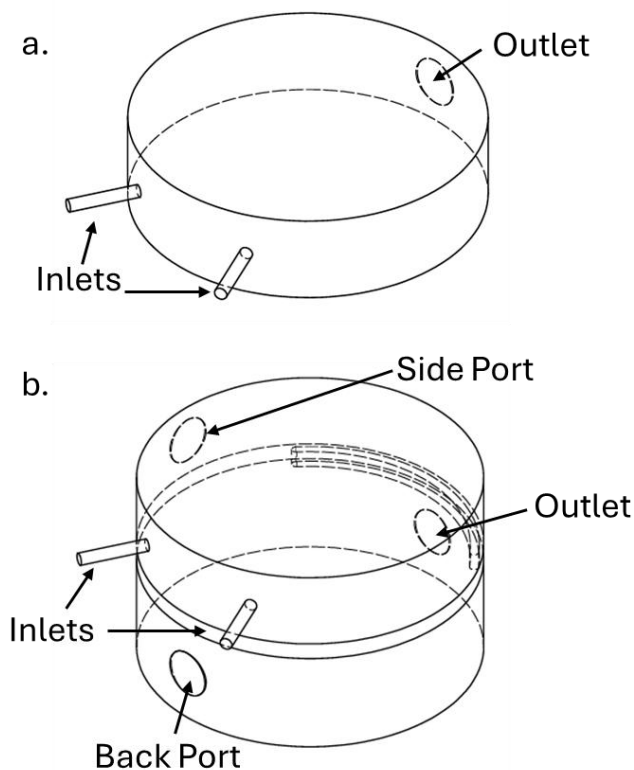


Figure 3.1. Schematic of the two iCVD reactor geometries used in simulations (a) single chamber design with inlets and outlet labeled (b) double chamber design with inlets, outlet, side port and back port labeled. This double chamber design matches the design used in experiments.

3.2.2 Flow Calibration

Argon gas, obtained from Airgas and 99.7% IPA obtained from Sigma Aldrich were used as model gases in experiments and were both used without further purification. Flow of argon gas was regulated by an MKS mass flow controller type 1159B while flow of IPA was controlled using a needle valve. Flow-rate measurements used in all experiments for IPA and argon were obtained

by flow calibration. First, the leak rate of the reactor was measured by closing the outlet to the pump at base pressure (2-5 mtorr). A gradual increase in pressure due to leak was then monitored using an MKS Baratron capacitance manometer model 626C by recording pressure every 10 seconds for 2 minutes and performing a linear regression. The slope was converted to a flow-rate in standard cubic centimeters per second (sccm) using the effective temperature (see Section 3.3.5). The flow-rate of gases were measured by setting a flow condition using either the mass flow controller (for argon) or the needle valve (for IPA). The outlet to the reactor was then closed and the increase in pressure was measured every 2 seconds for 20 seconds. A linear regression was performed on the pressure versus time, and the slope was converted to a molar flowrate in sccm using equation (3.15). Three replicates were performed for each flow.

3.2.3 Vapor phase composition measurements using a Pirani gauge

For binary gas mixtures in the iCVD reactor, the composition-sensitive Pirani gauge measurement is sufficient to determine composition. First a standard curve is created by varying the composition under a scenario where composition is known and both the Pirani gauge pressure and Baratron pressure, are measured. The ratio of the Pirani gauge pressure to the Baratron pressure is then plotted against the reactor composition to create a standard curve. In scenarios where the reactor composition is not known, but the Pirani gauge pressure and Baratron pressures can be read, the standard curve can then be used to determine the reactor composition.

3.2.4 Residence time distribution measurement

Step-response curves were generated by flowing argon into the reactor at a fixed flow-rate of 1 sccm, then rapidly shutting off the argon flow and opening up IPA flow into the reactor such that the total flow-rate remained constant. This caused a step change in composition of inlet vapor, which was registered by a Pirani gauge and composition was determined using a standard curve

(Section 3.2.3) matched to IPA/Ar mixtures to determine composition (figure 3.4 (a)).

3.2.5 Effective temperature measurement

The following procedures were used to estimate the effective temperature in the reactor over a range of conditions. The reactor was pressurized with either IPA or argon and was then allowed to sit for 60 seconds while pressure, filament temperature and stage temperature were measured. A linear regression on the pressure change over time enabled calculation of leak rate and precise determination of the initial pressure and initial filament temperature (taken as the average filament temperature over this period). The filament was then turned on in a stepwise manner by setting it to a constant value and then allowing time for the reactor pressure and temperature to stabilize, and after 60 seconds, stepping it up to the next condition. The experiment was repeated for pressures of 50 mtorr, 100 mtorr, 200 mtorr and 300 mtorr for argon gas and IPA vapor.

To determine effective temperature, we iterate through the pressure versus time data and using a window size of 6 (corresponding to 30 seconds). If the difference between the minimum and maximum temperature over the range is less than 2.7 K, we perform a linear regression on the baratron pressure over time to calculate a rate of change in pressure as a leak rate. These leak-rates are then interpolated, summed together, and subtracted from the Baratron pressure to give a leak-corrected baseline pressure which was used to determine the effective temperature for the data series using equation (3.15), where a Reimann sum is used to calculate the integrated leak-rate.

To fit the effective temperature to equation (3.27), we guess that the fraction $f = \frac{V_2}{V_R}$ has a value of 0.3 based on reactor parameters, the reference temperature T_{ref} is taken as 40°C and the effective temperature described by equation (3.27) was fit to data by minimizing the mean squared error.

3.2.6 Heat transfer in Argon and IPA vapor

Heat transfer from the filament array to the reactor body can be determined by performing an energy balance on the filament array following from the work of Bakker et al. (2009)¹⁹. Heat is generated resistively in the filament and is radiated and conducted away from the filament. At low pressures, radiative heat transfer dominates and a radiation-based model which depends only on the filament temperature and stage temperature can be generated. At higher pressures, this radiative heat transfer is subtracted from the total dissipated energy to determine the heat transfer by conduction to the gas. Heat transfer measurements are performed in argon and IPA gas flowing through the reactor at a range of pressures and filament temperatures and at constant flowrate of 1 sccm. The resulting data is subsequently fit to a simple model of the Knudsen jump²⁰.

Data was processed according to the following procedures. First, the filament temperature and the power to the filament taken at base pressure in the reactor (less than 5 mtorr), was fitted to equation (3.1), assuming q_{fil} to be zero. Measured values of the filament wire length (2.7 m) and filament diameter (0.4 mm) were used. Length was determined using total resistance across the array, and the known resistance of the wire, while diameter was determined using an instant readout digital caliper from Electron Microscopy Sciences. For incrementally higher pressures (up to 1 torr), the effective thermal conductivity was determined from the constant of proportionality relating the thermal driving force for heat transfer in gas $T_{fil}^{\frac{3}{2}} - T_{stage}^{\frac{3}{2}}$ to the calculated conductive heat loss. These were then fitted to equation (3.11), assuming that the mean free path was independent of temperature to obtain approximate guess values for fitting coefficients. Using these guess values, all the data was fitted to equation (3.11), now using the correct dependence of mean free path on temperature.

3.3. Results and Discussion

3.3.1 Dimensional analysis

The use of dimensionless numbers in engineering is well established, however their application to the iCVD system has been limited to discussion of reactor scaleup²⁴. Further elaboration on the role of dimensionless numbers in iCVD is needed to concisely describe physical processes, such as heat transfer and mixing.

The dimensionless numbers governing mass transfer in the iCVD system are the Peclet number relating advection to diffusion, Reynolds number relating inertial to viscous stresses, Mach number relating characteristic velocity to the speed of sound, and the Knudsen number, which relates the characteristic length scale of the system to the scale of molecular interactions²⁴. Viscous effects dominate inertial effects in all cases for iCVD, so the Reynolds number is small, and flow is laminar²⁴. The speed of sound in the gas is generally on the order of hundreds of meters per second, while the characteristic velocity of flow in iCVD is generally much less than 1 meter per second, so the Mach number is small, and compressibility effects can be neglected. Because the Mach number and the Reynolds number are sufficiently small, we can expect that flow through the reactor is laminar and incompressible.

Both the Peclet number and the Knudsen number are of a transitional scale, meaning that while they can ideally be neglected, they are often non-negligible. In fact, allowing for variations in how we define these two numbers, they fully define the mass transfer phenomena in an iCVD system. When the Peclet number is small (at low flowrates and for binary mixtures of high diffusivity), the well-mixed assumption is valid. We propose that this should be considered the ideal state of an iCVD reactor. For larger Peclet numbers, deviation from perfect mixing is quantified by the value of the Peclet number. When the Knudsen number is small (normal pressures

and in parts of the reactor with a large length scale), the flow is ‘viscous’ and can be described fully by continuum fluid mechanics. However, when the Knudsen number is not large (at low pressures and in parts of the reactor which have small length scales), flow in the reactor is no longer described by continuum fluid mechanics, and is called Knudsen flow, a transitional flow between viscous and molecular flow. Knudsen flow can generally be ignored in modeling iCVD, though it’s important for reactor design, since Knudsen flows are responsible for long pump down times required to outgas tubing connections to the reactor.

Table 3.1 Dimensionless numbers relevant to mass transfer in iCVD

Name	Equation	Description	Expected Range	Transitional Range
Peclet (mass)	$\frac{lv}{D}$	Advective versus diffusive mass transport	10^{-6} -10	10^{-1} -10
Knudsen	$\frac{\lambda}{l}$	Length scale of molecular collision to the scale of the system	10^{-5} - 10^{-1}	10^{-2} -1
Reynolds	$\frac{\rho vl}{\mu}$	Inertial versus viscous forces	10^{-8} -1	2300-4000
Mach	$\frac{v}{c}$	Characteristic velocity to the speed of sound	10^{-6} - 10^{-2}	>0.3

Dimensionless numbers relevant to the transfer of heat include the Peclet number for thermal diffusion relating advective heat transfer to diffusive heat transfer, the Grashoff number relating buoyant to viscous forces, and the Knudsen number. We neglect dimensionless numbers related to radiation in this discussion, and leave them to future work. Though there has been speculation about the presence of natural convection¹⁶, this has not been considered in our investigation as it would require inversion of the reactor.

Because large conductive heat fluxes occur close to the filament array (over a small length scale), the Peclet number for heat transfer from the filament is small, and heat transfer is purely

conductive. The Knudsen number describes the effect of low pressure on heat transfer near the filament.

The Peclet number is generally small but may describe some thermal gradients or non-ideality at high flow-rates. Because the conductive heat flux across most surfaces in the reactor is small, the Knudsen number is small. As such, the description of heat transfer from the walls of the reactor can be described primarily by Fourier conduction with non-ideality at high flow-rates quantified by the Peclet number.

Table 3.2. Table of dimensionless numbers relevant to heat transfer in the iCVD system

Name	Equation	Description	Expected Range		Transitional range
			(filament)	(reactor body)	
Grashof	$\frac{g\rho^2\left(\frac{T_s}{T}-1\right)l^3}{\mu^2}$	buoyant vs viscous force	10^{-6} - 10^2		$>10^2$
Péclet (thermal)	$\frac{lv\rho c_p}{\kappa}$	Conduction versus total heat transfer	10^{-6} -1	10^{-8} - 10^{-3}	10^{-1} -10
Knudsen	$\frac{\lambda}{l}$	Length scale of molecular collisions to system scale	10^{-2} -5	10^{-5} - 10^{-1}	$>10^{-1}$

Expected ranges for dimensionless numbers in Tables 3.1 and 3.2 were calculated based on standard reactor dimensions and pressures and using gas property models found in the CRC Handbook of chemistry and physics³⁵ and in Poling³⁶. Transitional ranges are based on external reference²⁵, or in the cases of Knudsen number for heat transfer, and Peclet number for mass transfer, our own data presented in further sections. Our discussion of dimensionless numbers in heat transfer does not seek to be fully comprehensive over all potential iCVD conditions, but is a thorough accounting of normal conditions of operation.

3.3.2 Mixing simulations

Simulations were performed in Autodesk CFD to help visualize the mixing between two distinct gas flows entering the reactor through separate inlets. Two reactor geometries were used, the double chamber geometry matched the reactor used in experiments, and a single chamber equivalent was simulated to show the effects of design modification. While 2-chamber designs are common, the bottom chamber acts as a difficult to model dead-zone. Steady state mixing simulations were performed with constant flow of gas into the reactor through both inlets at 1 sccm. To obtain models of the full range of Peclet number, we varied the diffusion coefficient. Simulations were performed for Peclet number of 0.3, 1.3 and 5.3 in both the 1 chamber and 2-chamber designs, indicating a transitional Peclet number of roughly 0.1-10. Simulations were also performed for a Peclet number of 0.03, roughly corresponding to the mixing of argon and IPA studied in Section 3.3.3, however the system showed perfect mixing. Because poor mixing would be associated with non-uniform and difficult to repeat or characterize iCVD deposition, these simulations indicate that for good performance, iCVD reactors should be operated below a Peclet number of 0.3. Peclet number for IPA / Ar mixing system was calculated based on the binary diffusion coefficient of IPA and argon reported in Lee³⁷, scaled down to a pressure of 100 mtorr using the pressure dependence suggested by kinetic theory of gases³⁵. We have found that the Peclet number should be calculated from the individual gas flow-rates, not the overall gas flowrates such that each gas flow is associated with a different Peclet number. This treatment found better correspondence between mixing behavior with the step response than calculating the Peclet number from the total flowrate.

It is notable from simulations (figure 3.2) that the increased volume from the 2-chamber design has very little effect on the transitional Peclet number, as such, the appropriate length scale

to use in calculating the Peclet number is the distance from the inlet to the outlet. While we do not show simulations of reactors with aspect ratio differing significantly from one, if the Peclet number for such a reactor needs to be calculated, the longer length scale should be used. This is because during actual deposition, not only do inlet streams need to be well mixed, but degradation products of the pre-initiator must also be well-mixed (i.e. poor mixing and plug flow are both non-ideal). It should be noted that the Peclet number describes non-ideal mixing, however even at low Peclet number, non-uniform depositions can occur at high deposition rates²⁹.

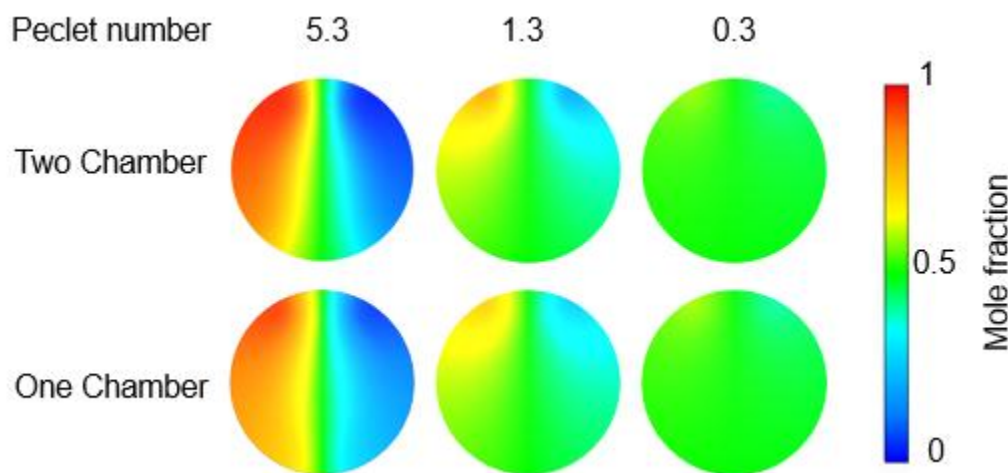


Figure 3.2. Binary gas mixing for a range of Peclet numbers over the transitional range of Peclet number in two-chamber and one-chamber reactor configurations.

Further transient simulations were performed to provide insight into the step response experiments performed in Section 3.3.3. While impulse response curves are generally preferred to step response in determining contacting patterns and mixing²¹, generating and detecting an impulse without significant pressure driven flows was infeasible in our setup. Simulations were run to exactly replicate the procedures used in Section 3.3.3. Figure 3.3 shows step response for all 3 Peclet numbers of 5.3 (figure 3.3(a)), 1.3 (figure 3.3(b)) and 0.3 (figure 3.3(c)). Simulations were also performed for a Peclet number of 0.03, roughly corresponding to the mixing of argon and IPA

studied in Section 3.3.3, and this curve could not be distinguished from the theoretical ideal. Videos of the simulations may be found in the supplementary information.

Step response curves in figure 3.3 enable two important conclusions. First, we see significant deviation from ideal mixing response for the same range of Peclet numbers we see poor mixing from figure 3.2, demonstrating that this experimental technique can be used to validate good mixing for a given set of conditions. Second, the data demonstrates clearly the sensitivity to sensor location. In practice, relying on the shape of the step response immediately following the step change is subject to significant errors, since generating the step change can induce moderate pressure driven flow in the system which is not easy to decouple from time delay at high Peclet number. Furthermore, precise step response data require accurate flow-rate calibration, which tends to be subject to experimental error. Even for an extreme poor mixing scenario (panel a) it may be difficult to positively confirm poor mixing if the Pirani gauge is placed on the outlet of the reactor, so this should be avoided. Gas sensing techniques like FTIR analyze the exhaust gas from the reactor²², and as such are much less valuable for real-time sensing of reactor dynamics. Furthermore, an experimenter working with a Peclet number of 1.3, consistent with the mixing in panel 'b', may have a very difficult time decoupling pressure shock from non-ideal behavior, even if sampling at the back-port or side-port is performed. The simulations in figure 3.3 (c) show that if precise mixing information is needed, sensors should be placed in multiple locations far from each other in the reactor, and their difference or ratio should be used to provide a sensitive measurement of deviation from ideal mixing. Step response curves in single chamber reactor configurations will differ from figure 3.3, however measurement at the outlet and side-port is expected to be nearly as sensitive to poor mixing.

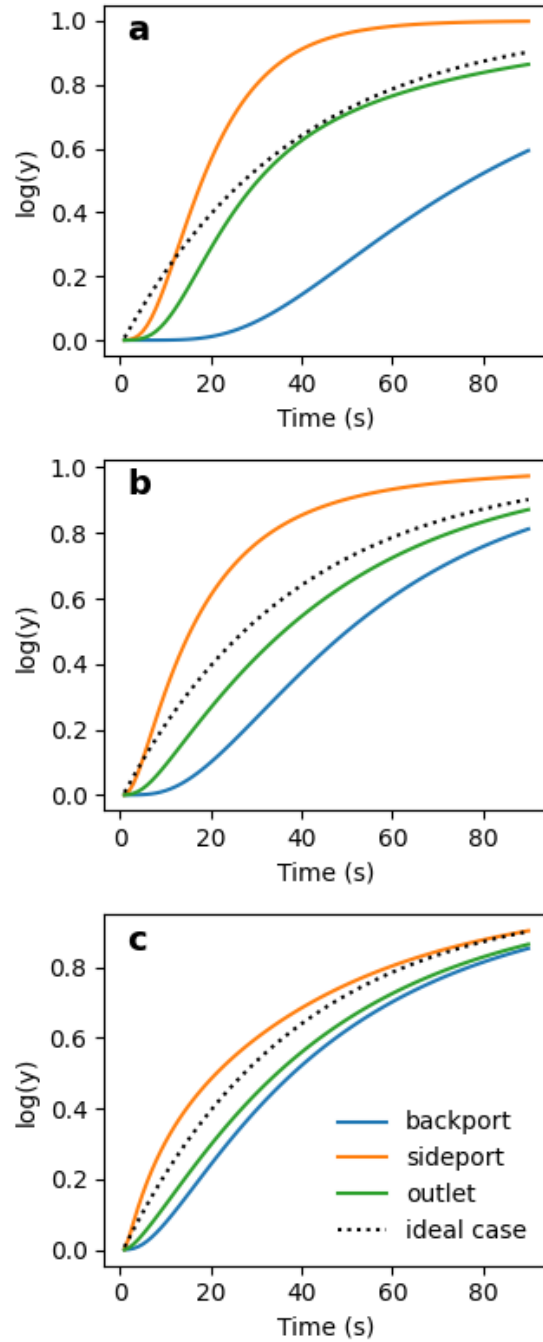


Figure 3.3. Step response curves for two chamber reactor configuration showing ideal mixing, measurement at the back-port of the reactor, measurement at the sideport and measurement at the reactor outlet. (a) Simulation at a Peclet number of 5.3 (b) Simulation at a Peclet number of 1.3 (c) Simulation at Peclet number of 0.3

3.3.3 Step response curves

In-situ composition sensing has been used on rare occasions in the iCVD literature to quantify batch mode decomposition of di-*tert*-butyl peroxide from in-situ mass spectrometry and measure qualitative response to pulsed input of monomer to the reactor using in situ FTIR, however a rigorous study of contacting patterns in iCVD and the phenomena responsible for them has been absent. In contrast to costly custom systems for composition sensing^{22,34}, we present a simple, off the shelf method of determining binary composition data using a convection assisted Pirani pressure gauge. The Pirani gauge uses the pressure dependence of thermal conductivity to determine pressure and provides a precise, but composition dependent, measurement of absolute pressure in the range of 1 mtorr to 1 torr, which corresponds to the pressure range for iCVD. Because heat transfer in low pressure gases depends on the composition of a gas, Pirani gauges are not appropriate for determining absolute pressure in an iCVD system, however when combined with a Baratron capacitance manometer, the Pirani gauge becomes an effective composition sensor. Thermal conductivity is routinely used for gas composition sensing³⁸, so use of the Pirani gauge is not a significant departure from existing methods, however it has never been used in the iCVD literature, so our methods represent an exciting new capability for monitoring and controlling the reactor environment. To measure the composition, a standard curve is first created (figure 3.2 (a)) in which IPA vapor and argon gas are flowed into the iCVD chamber to obtain a range of compositions, the pressure reported by the Pirani gauge is then recorded and divided by the pressure reported by the Baratron capacitance manometer to yield a proxy composition measurement we will refer to as the Pirani gauge ratio (c). The resulting standard curve is linear and can be used to convert the Pirani gauge measurement to composition.

Real time composition sensing from the Pirani gauge enables step-response experiments in

which argon is initially flowed at a steady-state, then it is quickly shut off and replaced with a flow of IPA vapor. This experiment maintains a constant total flow-rate and pressure. The resulting step response curve (figure 3.4 (b)) shows a characteristic CSTR or well mixed response, validating the expectation that a CSTR design equation describes the contacting pattern in the vapor phase of the iCVD reactor. The apparent presence of bypass in the residence time distribution is indicative of a slight pressure driven flow when the inlet composition is changed. While a pulsed response is generally more sensitive to non-ideal reactor conditions²¹, we use a step-response to maximize the sensing resolution, minimize the effect of pressure-driven flows, and importantly, to maintain the integrity of the signal – it is significantly easier to introduce a step change at constant flowrate and pressure than it is to introduce a pulse which approximates a delta function to an iCVD reactor system.

Based on the measured flowrate of 1.04 sccm into the reactor, the step-response curve indicates a reactor volume of 5.7 L, which is close to the expected reactor volume measured of 6.3 L. While these results are not surprising, given the expectation that the conditions measured represents a Peclet number of 0.03 (based on reactor dimensions and a calculated diffusion coefficient of $678.5 \text{ cm}^2/\text{s}$)³⁷, and that simulations indicated that the reactor would behave as a well-mixed system at these Peclet numbers, they are in marked contrast to the current literature²². Existing work has concluded that the iCVD reactor system is a plug flow system, however the experiments used pulse response curves and the data shown did not indicate when the pulse was introduced to the reactor, instead simply reporting a residence time distribution²², furthermore, there was no action taken to correct for the pitfalls discussed here which are intrinsic to the use of pulsed response experiments. Past simulations of iCVD reactors which may indicate poor mixing have generally utilized reactor geometries and conditions which differ significantly from standard

research conditions^{15,16}.

It is clear from dimensional analysis that the iCVD reactor mixing system is determined only by the Peclet and Knudsen numbers, with Reynolds and Mach numbers indicating that the flow is decisively laminar and incompressible. Step response curves illustrate that the reactor system is well mixed, in agreement with simulation results which provide us with a transitional Peclet number for the reactor system of 0.1 to 10 and illustrate, that the well-mixed state is the “ideal” iCVD reactor contacting pattern, and deviation from it (operation at high Peclet number) would result in poor coating thickness/composition uniformity.

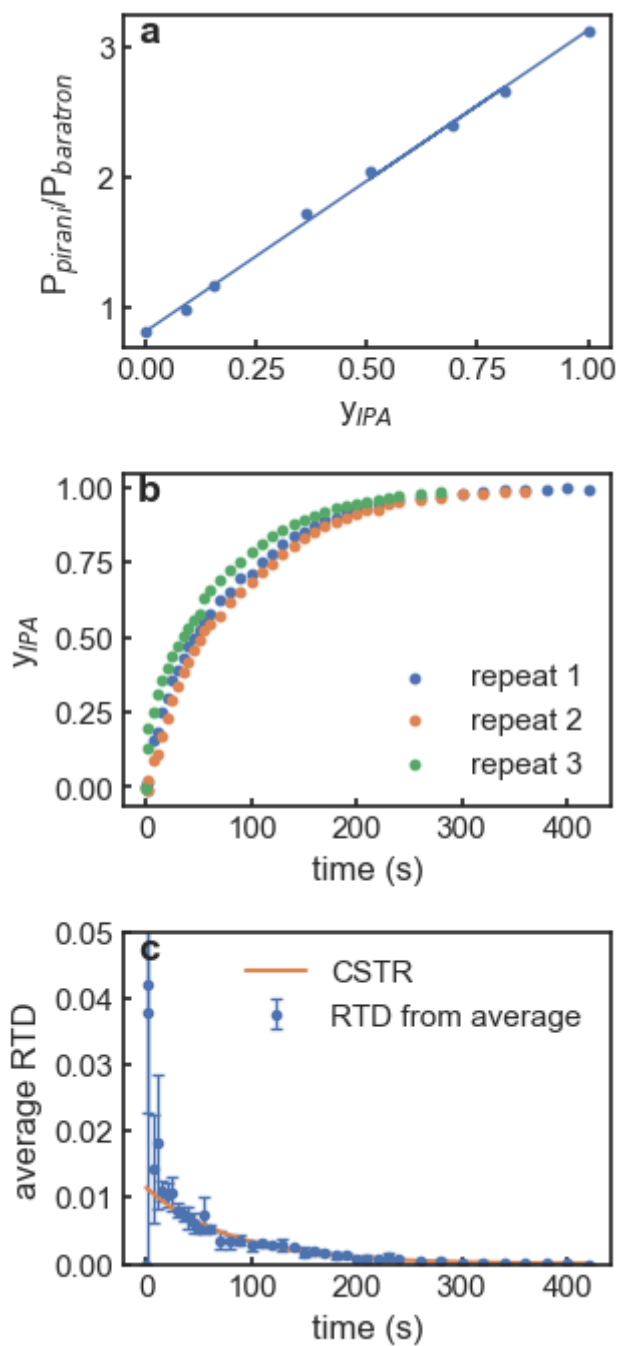


Figure 3.4. (a) Pirani gauge ratio standard curve for IPA and Ar (b) composition versus time calculated from the IPA/Ar standard curve for three repeats (c) derivative of composition versus time fit to ideal CSTR curve

3.3.4 Heat transfer

The work of Bakker *et al.* pioneers the measurement of heat transfer from the filament wires in iCVD, utilizing a useful radiation model to describe heat transfer at low pressure.¹⁹ Empirical models for the heat transfer from the filament were also introduced, however further fundamental explanation is useful¹⁹. We perform similar experiments to Bakker *et al.* in which we measure filament temperature while varying the pressure in the reactor and the power delivered to the array. Following from Bakker *et al.*, heat transfer by conduction from the filament array (q_{fil}) is given by equation (3.1) in accordance with radiation between grey bodies, when the visible area of one (the walls of the reactor) are much larger than the other (the filament)³⁹,

$$q_{fil} = \frac{RI^2}{L_{fil}} - 2\pi R_{fil}\sigma\epsilon_{fil}(T_{fil}^4 - T_{stage}^4) \quad (3.1)$$

where R is the resistance of the nichrome filament wire, I is the current through the filament, L_{fil} is the length of the filament wire, σ is the Stephan Boltzmann constant, R_{fil} is the radius of the filament wire and ϵ_{fil} is the emissivity of oxidized nichrome wire, which has a value of 0.95-0.98³⁹. At low pressure, we assume that q_{fil} is negligible, so we can solve for the emissivities from this limit¹⁹. This fit is shown in figure 3.7(a), where the fit gave an emissivity of 2.14, which is notably unphysical (emissivity must be less than 1). It is likely that this value is caused by an inaccurate current reading from the power supply, likely due to small leakage of current through the feedthroughs to the reactor. In all calculations, the current is adjusted by a constant factor such that the emissivity of the array is 0.95. From the known power and radiative heat loss, conductive heat loss is calculated. To model this heat transfer, we solve the equations of heat transfer in the domain shown in figure 3.5(a).

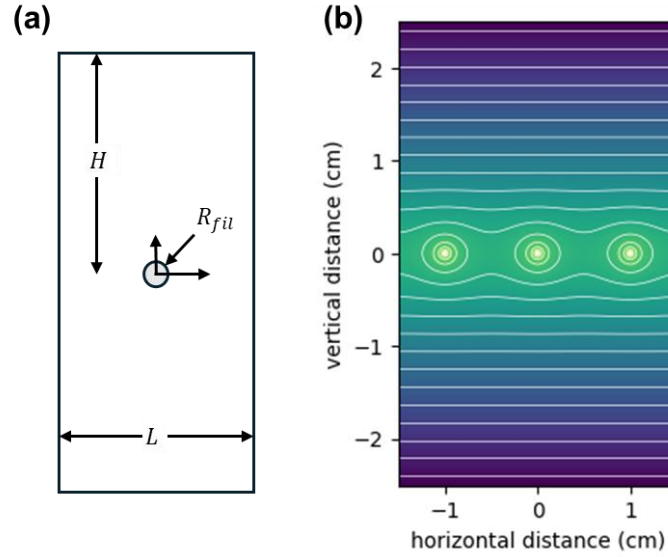


Figure 3.5 (a) Schematic showing the coordinate system used in modeling of temperature at the filament array (b) Plot showing equation (3.2) for three adjacent filament wires as a visual aid

Ignoring edge effects and noting that the filament wires are close to each other relative to the height of the chamber, we can model the temperature surrounding the filament using the method of images as an infinite line of 1D line sources on the domain shown in figure 3.5(a). Because of the correspondence between complex numbers and Laplace's equation established by the Cauchy-Reimann equations⁴⁰, a complex function with an infinite line of poles, and appropriate behavior at infinity, will correspond to the solution of Laplace's equation in our system. We can see by inspection that the real part of $\ln(\sin(z))$ has these characteristics, and so when complex numbers are converted to cartesian coordinates, and constants of integration are solved for, equation (3.2) is found to describe the solution to Laplace's equation for the filament array (See appendix A for derivation),

$$\phi = \phi_R + q_{fil} \frac{\ln \left(\frac{L}{R_{fil}\pi} \sqrt{\sin^2 \left(\frac{x\pi}{L} \right) \cosh^2 \left(\frac{y\pi}{L} \right) + \cos^2 \left(\frac{x\pi}{L} \right) \sinh^2 \left(\frac{y\pi}{L} \right)} \right)}{2\pi} \quad (3.2)$$

where H is the distance between the filament and the lid or stage (25 mm), and L is the distance between filaments (10 mm) as are represented in figure 3.5 (a). To help visualize the temperature distribution in the reactor, we have created a plot of equation (3.2) for three adjacent filament wires and our reactor dimensions. When we assume the filament is positioned midway between the stage and the lid, and that the stage and the lid are the same temperature (approximately true in our experiments), we find that the overall heat transfer resistance caused by the bulk gas can be described by equation (3.3),

$$\phi_{stage} - \phi_R = q_{fil} \frac{\ln \left(\frac{L \sinh \left(\frac{H\pi}{L} \right)}{R_{fil}\pi} \right)}{2\pi} \quad (3.3)$$

Having developed a theory for the heat transfer resistance due to conduction in the gas, we now turn our attention to heat transfer resistance due to the Knudsen jump at the filament array. From Su *et al.*²⁰, a description for the interfacial heat transfer is given by equation (3.4),

$$T_R = T_{fil} + \frac{\zeta_T \mu}{p} \sqrt{\frac{2k_B T_{fil}}{m}} \mathbf{n} \cdot \nabla T \quad (3.4)$$

where T_R corresponds to T_e in Su *et al.* and is the extrapolated temperature of the gas molecules at the surface of the filament array, T_{fil} is the temperature of the filament itself, and corresponds to T_w in Su *et al.* (see figure 3.6), p is the pressure, k_B is Boltzmann's constant, m is the mass of a gas

molecule, μ is the dynamic viscosity of the gas, n is the unit normal vector to the filament surface, and ζ_T is the temperature jump coefficient defined by equation (3.5).

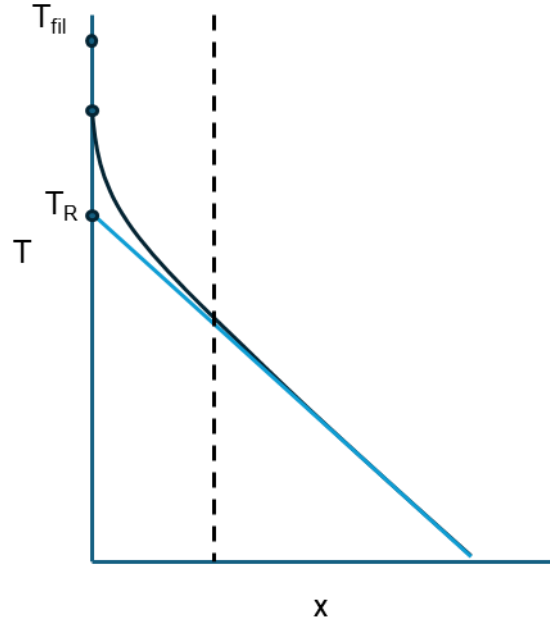


Figure 3.6. Temperature profile in the Knudsen layer, showing relevant quantities

$$\zeta_T = \frac{\gamma\sqrt{\pi}}{(\gamma+1)Pr} \frac{2-\alpha_0}{\alpha_0} \quad (3.5)$$

where Pr is the Prandtl number, defined as $Pr = \frac{c_p\mu}{m\kappa}$, where c_p is the heat capacity of a molecule, μ is the dynamic viscosity, κ is the thermal conductivity, and α_0 is the constant temperature jump coefficient, representing the fraction of molecules which are diffusely reflected at the wall.

We can linearize equation (3.4) in the heat transfer potential given by equation (3.6) to obtain equation (3.7).

$$\phi = -\frac{2}{3}\kappa'T^{\frac{3}{2}} \quad (3.6)$$

$$\phi_{fil} - \phi_R = \frac{8}{9} \frac{1}{(\gamma + 1)} \left(\frac{2 - \alpha_0}{\alpha_0} \right) \lambda \frac{q_{fil}}{2\pi R_{fil}} \quad (3.7)$$

Where the mean free path is given by equation (3.8)³⁵.

$$\lambda = \frac{k_B T}{\sqrt{2} \pi d^2 p} \quad (3.8)$$

We can now define equation (3.9) and (3.10), which relate the pressure-dependent heat transfer observed in the iCVD reactor to existing theory.

$$\phi_{fil} - \phi_R = \frac{h \lambda q_{fil}}{2\pi R_{fil}} \quad (3.9)$$

$$h = \frac{8}{9} \frac{2 - \alpha_0}{\alpha_0 (\gamma + 1)} \quad (3.10)$$

Overall, combining equations (3.9) and (3.3), gives equation (3.11),

$$\phi_{fil} = \phi_{stage} + \frac{q_{fil}}{2\pi} \left(\ln \left(\frac{L \sinh \left(\frac{H\pi}{L} \right)}{R_{fil} \pi} \right) + \frac{h \lambda}{R_{fil}} \right) \quad (3.11)$$

Equation (3.11) was fit to data, minimizing the mean squared error between the predicted and measured values of the filament temperature for the complete range of pressures and temperatures and for both IPA and argon gases. The resulting predicted versus measured values are plotted in figure 3.7 (c). Fitting parameters from equation (3.11) are included in Table 3.3. For better visualization of the data, for each pressure an effective heat transfer coefficient was obtained and plotted against the fitted model from figure 3.7 (c), where the mean free path was calculated

at a nominal temperature of 450 K. The data and model fit are presented in figure 3.7 (b). Note that because of heat losses, filament temperatures were lower at higher pressures, so the nominal 450 K was less appropriate causing an apparent over-prediction which is not present in the temperature-dependent data. To show the correspondence between the temperature independent and temperature dependent mean free path, we show the prediction and data for both in figure 3.7 (d) for IPA at 50mtorr over the full range of temperature, showing a close correspondence.

Table 3.3 Parameters used to fit equation (3.11)

	κ' (W m ⁻¹ K ^{-3/2})	α_0 (non-dim)
IPA	0.00146	0.28
Ar	0.00143	0.13

The thermal conductivity of IPA from Lange's Handbook⁴¹ is 0.025 W m⁻¹ K⁻¹ at 250°C giving a κ' of 0.0011 W m⁻¹ K^{-3/2}, while argon is reported to have a thermal conductivity of 0.0211 W m⁻¹ K⁻¹ at 100°C, giving a κ' of 0.00109 W m⁻¹ K^{-3/2}. The values of κ' reported in Table 3.3 vary from the true values by the same constant, indicating that the discrepancy is due to slight inaccuracy in the system parameters or edge effects which are not accounted for in equation (3.11), which would affect both the values by the same amount. The accommodation coefficient α_0 describes the impact of the Knudsen jump based on a calculation of the mean free path from equation (3.8), using diameters for IPA and argon of 0.58 nm and 0.358 nm³⁵ respectively. The diameter of IPA was calculated using $d = 0.809V_c^{1/3}$ (equation 9-4.8 in Poling³⁶) and a critical volume of 222 cm³/mol.³⁵ The values of α_0 show that, when correcting for the molecular diameter in the mean free path, the argon has a slightly less efficient accommodation at the surface of the filament. Notably, the points in figure 3.7 (c) which are not well predicted by the heat transfer

model correspond to argon gas at low pressure. Under these conditions, the assumption of equation (3.9), corresponding to a planar Knudsen jump is less appropriate, and geometry-dependent corrections⁴² can be introduced if necessary.

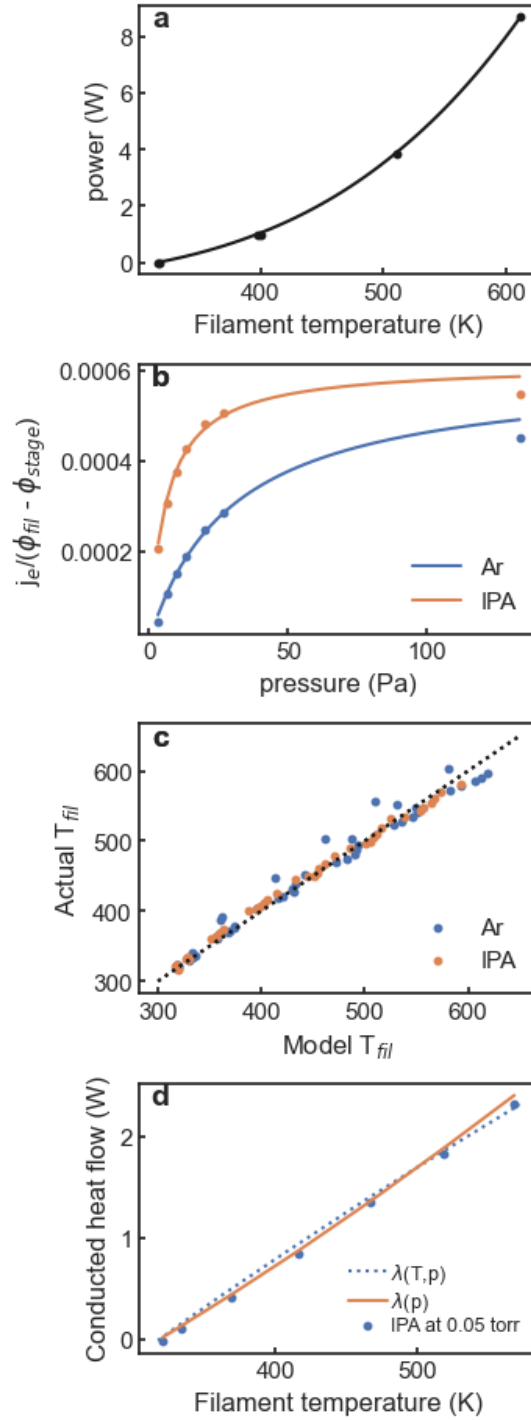


Figure 3.7 (a) power versus filament temperature in the low-pressure limit including a curve fit of equation (3.11) neglecting temperature dependence of mean free path (b) effective thermal conductivity versus pressure used to obtain a starting guess for fit of equation (3.11) to data

neglecting temperature dependence of mean free path (c) Actual filament temperature (K) versus predicted filament temperature (K) based on equation (3.11) for IPA and Ar (d) comparison between preliminary curve fit and final curve fit for conducted heat flow versus filament temperature, for equation (3.11) when we account for temperature dependence of the mean free path (dotted line), and ignoring the temperature dependence of the mean free path (solid line).

Our results for heat transfer from the filament in the iCVD system show that the pressure dependence of conduction can be well accounted for by simple models of heat transfer. Furthermore, they show that the same conduction model is valid for gases which absorb infrared radiation (IPA) and those which do not (argon). As such, the predictive ability of our model experimentally validates that infrared absorption can be neglected at iCVD conditions as was discussed in Section 3.3.1. We expect our results to be helpful in elucidating the thermal decomposition of di-*tert*-butyl peroxide at the filament array, which has largely gone un-explained.

3.3.5 Reactor Effective temperature

When operating the iCVD reactor in batch mode^{14,27,28,30}, the ideal gas law is a helpful model. In fact, essentially all studies of iCVD implicitly use the ideal gas law to perform flowrate calibration before running a deposition. Whenever the ideal gas law is used in these situations, an average temperature for the reactor must be specified, however, because the temperature in the reactor is non-uniform, the appropriate average is not immediately clear. The effect of temperature gradients becomes even more pronounced in any batch mode reaction or deposition, such as those that are used to perform condensed droplet polymerization^{27,28}. As batch processes become more common in the literature^{14,30}, defining an appropriate temperature becomes increasingly relevant. Herein, we will refer to the temperature consistent with treating the reactor with the ideal gas law

as the effective temperature.

To define the effective temperature, we begin by assuming (based on Section 3.3.1) that compositional and pressure gradients in the iCVD reactor are small, but we acknowledge that due to thermal gradients, vapor density ρ varies with location x . We assume this variation is described by the ideal gas law,

$$T(\mathbf{x}) = \frac{p}{R_G \rho(\mathbf{x})} \quad (3.12)$$

where R_G is the ideal gas constant, and p is the pressure. Integrating equation (3.12) over the volume of the reactor, we obtain equation (3.13),

$$\int \frac{1}{T} dV_r = \frac{nR_G}{p} \quad (3.13)$$

where n is the total number of moles of gas in the reactor. To appeal to the ideal gas law in the reactor system, we define the effective temperature of the reactor such that $pV_r = nR_G T_{eff}$, giving equation (3.14)

$$T_{eff} = \frac{V_r}{\int \frac{1}{T} dV_r} \quad (3.14)$$

Effective temperature changes in the reactor can be measured by measuring change in pressure, however the effects of leak need to be corrected for. This is enabled by equation (3.15),

$$T_{eff} = \frac{p T_{eff,leak}}{p_o + \frac{dp}{dt}_{leak} t} \quad (3.15)$$

where T_{eff} is the effective temperature, p is the pressure, $T_{eff,leak}$ is the effective temperature of the reactor during leak rate measurement, p_0 is the initial pressure in the reactor before the filament was turned on, $\frac{dp}{dt}_{leak}$ is the pressure change caused by leak into the reactor and t is the time since the filament was turned on.

To probe the impact of the filament temperature on the effective temperature, equation (3.15) must be fit to a model informed by our analysis of heat transfer (Section 3.3.4). For the purposes of predicting effective temperature, it is useful to break up the reactor into three volumes (equation 3.16),

$$T_{eff} = \frac{V}{\int \frac{1}{T} dV} = \frac{V_1 + V_2 + V_3}{\int \frac{1}{T} dV_1 + \int \frac{1}{T} dV_2 + \int \frac{1}{T} dV_3} \quad (3.16)$$

where V_1 is the volume of the bottom chamber of the reactor, which is effectively isolated from the filament, and remains at some low reference temperature throughout the experiment. V_2 is the volume of the reactor, which is well modeled by equation (3.2), and makes up most of the top chamber where the filament array is located. V_3 is the part of the reactor in the main chamber effected by the filament array through the impact of radiation, but not conduction. This is a correction for volume of the reactor near the edges of the filament array.

While the temperature in V_2 is accurately described by equation (3.2), for the purposes of calculating the effective temperature, it is expedient to treat the filament array as a planar heat source, such that the temperature profile is linear. Again, we assume the filament is located midway between the lid of the reactor and the reactor stage. Furthermore, even though heat transfer in a gaseous medium tends to go as $T^{3/2}$, instead of T over the temperature range of interest (20-200°C), approximating it as $\sim T$ yields a very close approximation.

For a linear temperature profile, we can derive equation (3.17) for the effective temperature,

$$\begin{aligned}
 T &= a + bx \\
 \frac{1}{T} &= \frac{1}{a + bx} \\
 A \int_0^X \frac{1}{T} dx &= \frac{AX \ln\left(\frac{T(X)}{T(0)}\right)}{T(X) - T(0)}
 \end{aligned}
 \tag{3.17}$$

where X is the length scale over which the linear profile occurs. To solve for V_3 , we plug in H for X and stage and lid temperatures for $T(X)$ and $T(0)$ to equation (3.17) noting that $2HA_3 = V_3$, where A_3 is area of the footprint taken up by V_3 .

$$\int_0^{2H} \frac{1}{T} dV_3 = \frac{V_3 \ln\left(\frac{T_{lid}}{T_{stage}}\right)}{T_{lid} - T_{stage}}$$

We can note that, when the lid and stage temperature are within 20 or 30°C, the result can be further simplified to equation (3.18):

$$\int_0^{H_{ref}} \frac{1}{T} dV_3 = \frac{V_3}{T_{av}}
 \tag{3.18}$$

Where we have defined an average temperature as $T_{av} = \frac{T_{stage} + T_{lid}}{2}$. Next, we can plug in H and $T(H)$ for X and $T(X)$, T_{lid} and T_{stage} for $T(0)$,

$$\int \frac{1}{T} dV_2 = \frac{A_2 H \ln\left(\frac{T(H)}{T_{stage}}\right)}{T(H) - T_{stage}} + \frac{A_2 H \ln\left(\frac{T(H)}{T_{lid}}\right)}{T(H) - T_{lid}}
 \tag{3.19}$$

where A_2 is the area of the footprint taken up by V_2 . In a similar way as for V_3 , V_2 can be approximated with good accuracy when the stage and lid temperatures are within 20-30°C noting that $2A_2H$ is equal to V_2 .

$$\int \frac{1}{T} dV_2 = \frac{V_2 \ln\left(\frac{T(H)}{T_{av}}\right)}{T(H) - T_{av}} \quad (3.20)$$

From our study of the heat transfer from the filament (Section 3.3.4), we know that heat transfer depends on pressure according to equation (3.11), which can be converted to temperature using the definition of the heat transfer potential (equation 3.6) to give (3.21):

$$j_e = \frac{4\pi\kappa' \left(T_{fil}^{\frac{3}{2}} - T_{stage}^{\frac{3}{2}} \right)}{3 \ln\left(\frac{L \sinh\left(\frac{H\pi}{L}\right)}{R_{fil}\pi} \right) + 3 \frac{h\lambda}{R_{fil}}} \quad (3.21)$$

We have also assumed that, for the purposes of calculating the effective temperature in equation (3.20), equation (3.2) can be approximated by a linear profile:

$$q_{fil} = 2L \frac{2\kappa'}{3H} \left(T(H)^{\frac{3}{2}} - T_{av}^{\frac{3}{2}} \right) \quad (3.22)$$

Equate equations (3.21) and (3.22), and solve for $T(H)$:

$$T(H)^{\frac{3}{2}} = T_{av}^{\frac{3}{2}} + \left(\frac{H\pi}{L}\right) \frac{\left(T_{fil}^{\frac{3}{2}} - T_{av}^{\frac{3}{2}}\right)}{\ln\left(\frac{L \sinh\left(\frac{H\pi}{L}\right)}{R_{fil}\pi}\right) + \frac{h\lambda}{R_{fil}}}$$

(3.23)

Only marginal error is incurred by linearizing as:

$$T(H) = T_{av} + \left(\frac{H\pi}{L}\right) \frac{(T_{fil} - T_{av})}{\ln\left(\frac{L \sinh\left(\frac{H\pi}{L}\right)}{R_{fil}\pi}\right) + \frac{h\lambda}{R_{fil}}}$$

Which we plug into equation (3.20).

$$\int \frac{1}{T} dV_2 = \frac{V_2 \ln\left(1 + h_{eff} \left(\frac{T_{fil}}{T_{av}} - 1\right)\right)}{h_{eff}(T_{fil} - T_{av})}$$

(3.24)

Where we define an effective heat transfer coefficient:

$$h_{eff} = \frac{\frac{H\pi}{L}}{\ln\left(\frac{L \sinh\left(\frac{H\pi}{L}\right)}{R_{fil}\pi}\right) + \frac{h\lambda}{R_{fil}}}$$

(3.25)

To obtain an expression for the overall effective temperature in the reactor, we substitute (3.24) into equation (3.16).

$$T_{eff} = \frac{V_1 + V_2 + V_3}{\frac{V_1}{T_{ref}} + \frac{V_2 \ln\left(1 + h_{eff} \left(\frac{T_{fil}}{T_{av}} - 1\right)\right)}{h_{eff}(T_{fil} - T_{av})} + \frac{V_3}{T_{av}}}$$

(3.26)

When the stage, lid and reference temperature are all within 30°C or so, we can write equation (3.27).

$$T_{eff} = \frac{1}{\frac{1-f}{T_{ref}} + \frac{f \ln \left(1 + h_{eff} \left(\frac{T_{fil}}{T_{av}} - 1 \right) \right)}{h_{eff}(T_{fil} - T_{av})}} \quad (3.27)$$

Curve fits in argon (figure 3.8 (a-d)) and IPA (figure 3.9 (a-d)) (gases which do not decompose at high temperature), show that equation (3.27) fits well, and importantly, that even over a large range in filament temperature, the effective temperature changes by only ~5-10%. Furthermore, the role of pressure change is minimal. There is some notable deviation from the shape of the model and the shape of the data at high temperature – due to radiative effects on the stage temperature. We suggest that this effect be quantified by the Boltzmann number. Accounting for these phenomena is outside the scope of work, and such high temperatures are rarely used. To fit equation (3.27), distance between filament wires L was 1 cm, distance from filament to stage H was 2.5 cm, and radius of the filament wire R_{fil} was 0.2 mm which correspond to Section 3.3.4. The fraction of the reactor f described by equation (3.2) was assumed to be 0.3, and the reference temperature T_{ref} was assumed to be 40°C. Details on the fitting procedure can be found in Section 3.2.6. Fitting coefficient h is reported in Table 3.4. The accommodation coefficient α_0 from the effective temperature treatment corresponds excellently with the same value found from the heat transfer analysis for both argon and IPA. This remarkable match lends credibility to the heat transfer model presented.

**Table 3.4 Dimensionless interfacial heat transfer resistance for Ar and IPA fit from
effective temperature**

	α_0 (non-dim)
IPA	0.28
Ar	0.13

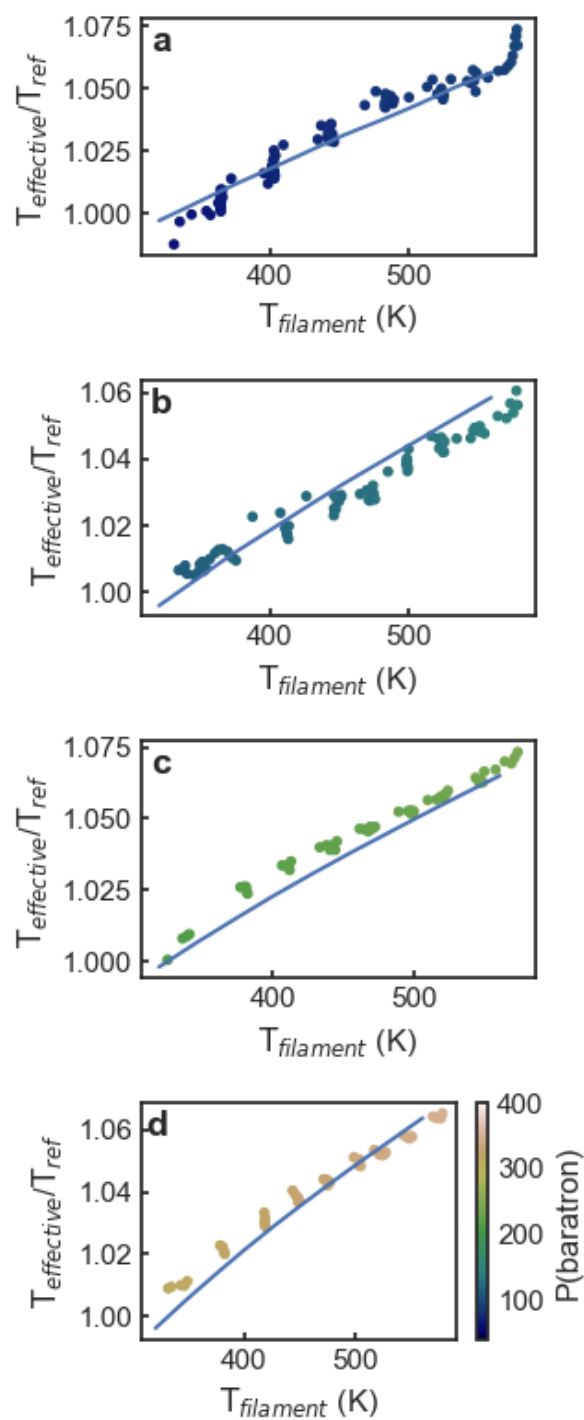


Figure 3.8 Ratio of effective temperature to reference temperature versus filament temperature for argon at nominal pressures of (a) 50 mtorr (b) 100 mtorr (c) 200 mtorr and (d) 300 mtorr

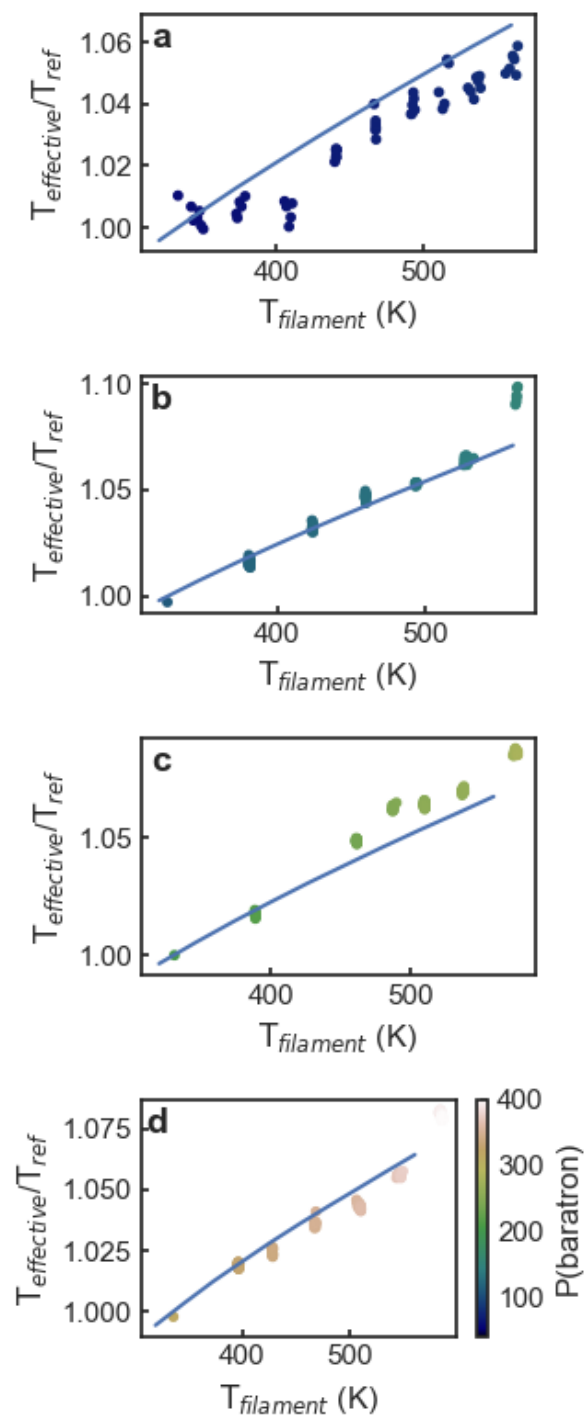


Figure 3.9 Ratio of effective temperature to reference temperature versus filament temperature for IPA at nominal pressures of (a) 50 mtorr (b) 100 mtorr (c) 200 mtorr and (d) 300 mtorr

3.4. Conclusions

The published iCVD literature contains very little discussion of the fundamental processes underlying mixing and heat transfer in the reactor. This study attempts to clarify the fundamental processes which underly contacting patterns in the vapor phase, heat transfer from the filament and heat transfer from the reactor body. We present an overview of the dimensionless numbers important to these three cases. Detailed analysis of the Peclet number in mass transfer and the Knudsen number in heat transfer from the array have been considered, but we have left experimental investigation of non-ideality in mass flow at high Knudsen number (termed Knudsen flow), the effects of high thermal Peclet number and high Boltzmann number on heat transfer to future work.

For mass transfer, we find that the iCVD reactor is ideally well mixed, and that deviations from perfect mixing can be fully quantified by the Peclet number. We validate the perspective that under ideal conditions the reactor is well mixed by performing step-response experiments with IPA and argon using a Pirani gauge for real-time composition sensing, finding that the resulting step response curve matches a well-mixed design equation²¹. To help visualize the role of mixing in the reactor, we perform simulations in Autodesk CFD at a range of Peclet numbers to replicate the step response experiments, finding that measuring gas composition at the outlet of the reactor has poor sensitivity in determining the degree of mixing, but that by placing two sensors at disparate locations in the reactor, comparison of their readout will give strong sensitivity to mixing non-ideality.

We also performed simulations of binary gas mixing at the same range of Peclet number with both a 1-chamber and 2-chamber reactor design. These simulations demonstrated that the mixing was essentially the same in the 1-chamber and 2-chamber cases. Furthermore, these mixing

simulations represent the potential effect of poor mixing on an iCVD deposition and provide a clear visualization of what the effect will be at the full range of transitional Peclet numbers. Our analysis shows that simple simulations accounting for only incompressible viscous flow and diffusion provide an excellent tool for understanding mixing. We find that below a Peclet number of 0.3 the reactor can be considered as well mixed based on our simulations, but that film non-uniformities may arise above this point. Finally, our work illustrates the potential of the Pirani gauge in characterization of composition, as a simple, low cost, precise measurement technique.

For heat transfer, analysis of dimensionless numbers shows that ideal heat transfer to the process gas occurs via Fourier conduction with no interfacial heat transfer resistances. Interfacial heat transfer resistances developing at low pressures can be quantified by the Knudsen number, and advective effects can be quantified by the thermal Peclet number (not investigated). To validate this perspective of ideal heat transfer, we build on past work from Bakker *et al.* to measure heat transfer conducted from the filament wires and find strong agreement with a first-order approximation of the Knudsen jump based on the planar geometry²⁰, though the system geometry likely causes some deviation for low pressure argon. Temperature-independent thermal conductivities of 0.00146 and 0.00143 Wm⁻¹K^{-3/2} were obtained for IPA and Ar respectively, compared with values of 0.0011 and 0.00109 Wm⁻¹K^{-3/2} from the literature⁴¹. Both measured values differ from the literature by a constant factor, indicating that there may be some impact from edge effects, or inaccurate dimensions used in our model.

While it appears from our data that heat transfer is strongly effected by pressure at normal iCVD operating pressures, it should be noted that due to the larger collision diameters of iCVD process gases, iCVD deposition largely avoids the Knudsen effect under normal conditions¹⁹. The heat transfer model presented provides a fundamental understanding of the heat transfer process

from the filament arrays, enabling future insight into methods of improving the initiation efficiency during iCVD. The predictive ability of this model further supports our neglect of absorption of thermal radiation by the process vapor, and thermal convection.

To make the application of the ideal gas law rigorous in the iCVD reactor system, we introduce the concept of an effective reactor temperature and provide a model for it. By turning on the filament array while it is closed off from the pump, we measured the change in the effective temperature caused by the filament array, finding that it is predicted well by our model, and that its overall effect is limited to a 5-10% change in the reactor pressure. Notably, dimensionless heat transfer coefficients obtained by fitting the effective temperature of 2.63 and 4.89 for IPA and argon respectively matched those obtained from heat transfer modelling which gave values of 2.61 and 4.89 for IPA and argon respectively. This correspondence supports the assumption that effects of the thermal Peclet number can be neglected, since heat transfer coefficients were measured with gases flowing, while effective temperature modelling was performed in the batch mode. Furthermore, the correspondence for both infrared absorbing gas (IPA) and non-absorbing gas (argon) provides evidence that the gas can be treated as transparent to radiation from the filament.

Overall, our results show that the ideal iCVD reactor should be considered as well mixed, with conduction dominating thermal advection, with temperature surrounding the filament defined by equation (3.2), and overall heat transfer from the filament described by equation (3.11). At large Knudsen numbers, non-idealities related to the filament temperature are expected to appear and at large Peclet numbers, non-idealities related to poor mixing and advection of heat are expected to occur. We have used simple simulations to show how mixing can readily be understood in different reactor configurations, and how this might be used to make reactor design choices. This understanding will enable more precise measurements from the iCVD reactor, a concrete

understanding of the conditions under which film uniformity or kinetics may be effected by poor mixing, and the temperature profile and heat transfer from the filament array in order to guide a better understanding of the initiation process, which have not yet been explained.¹³

Noting the surprising diversity of innovation from efforts to push the feasible boundaries of deposition at high monomer saturation^{26–29}, we hope that our work inspires and enables the exploration of other boundaries such as high Peclet number or Knudsen number. We also note that further investigations are needed to quantify the transitional region of the thermal Peclet number as well as develop a fundamental understanding of the apparent increase in effective temperature observed at high Boltzmann number, which is likely to play a large role in deposition kinetics, due to the strong dependence on stage temperature. Additionally, innovative reactor design is needed to obtain uniform films from iCVD at high Peclet number. Having introduced the Pirani gauge as a thermal conductivity-based composition sensing device, it is also interesting to consider drawing correlations between the Pirani gauge reading and the dimensionless interfacial heat transfer coefficient ‘ h ’. Finally, it is important to underscore the importance of utilizing the fundamental understanding gained from these experiments in further developing an understanding of the chemical reactions which occur in iCVD by leveraging the temperature profile and CSTR design equation for computational modeling of reactions in the vapor phase.

3.5. References

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CHAPTER 4:

CHARACTERIZATION OF THE VAPOR PHASE DURING ICVD POLYMERIZATION OF CYCLOHEXYL METHACRYLATE

Abstract

While initiated chemical vapor deposition (iCVD) has been used in a wide range of fundamental and applied studies in the past two decades, little is known about the reactive vapor phase, leading to inconsistent treatment of the initiation process in the existing literature. Di-*tert*-butyl peroxide (TBPO) is commonly used as an initiator in iCVD, by leveraging radical intermediates produced during thermal decomposition. Despite its widespread use, no models exist for TBPO decomposition in iCVD. To bridge this knowledge gap, we integrated the Pirani pressure gauge into the iCVD setup to measure the real-time vapor-phase composition in the reactor. After establishing the relationship between the Pirani gauge pressure and composition from batch data, we use it to quantify the fractional conversion of pure TBPO in the reactor in the continuous mode. We find that under iCVD conditions, TBPO fully decomposes to acetone and ethane following a first order Arrhenius kinetics, consistent with prior literature. This means that (i) the partial pressures of radicals are small (below the detection limit of the Pirani gauge) and (ii) there is a stoichiometric increase in the number of gas molecules in the reactor which has not been accounted for in existing models. This causes overestimation of monomer pressure, which is a key parameter in the current understanding of iCVD. We analyze the decomposition of TBPO in the presence of cyclohexyl methacrylate (CHMA) monomer to show that the monomer does not react appreciably in the vapor phase. To show how this model of the vapor phase is related to the deposition kinetics, we compare our data to the classical linear polymerization theory, which describes free radical polymerization of CHMA in solution and find an acceptable fit. Notably, in Arrhenius plots of the deposition rate, there is a well-documented, but unexplained decrease in the apparent activation energy at high filament temperature ($>300^{\circ}\text{C}$) which is predicted by our models.

4.1. Introduction

A century ago, free radical polymerization (initiated by peroxides) was used to produce the first synthetic polymers.¹ Since then, polymers have become increasingly important in every major industry. One major application is polymer thin films, which are foundational in technologies such as surface chemistry modification, thin-film composite membranes, organic electronic devices, and passivating coatings. The most popular polymer thin film synthesis methods are solution phase methods like spin coating, spin casting, and dip coating. Unfortunately, these methods only produce high quality films on flat and chemically compatible substrates. In response to these limitations, a diverse set of polymer chemical vapor deposition (CVD) technologies have been developed. Polymer CVD eliminates the use of solvents, produces films which conform to nanostructures and performs well with most substrate surface chemistries. While these properties lend distinct advantages, many types of polymer CVD (like parylene CVD) are restricted to specific polymer chemistries or (like plasma-enhanced CVD) require harsh processing conditions that are incompatible with delicate substrates.² Unlike other polymer CVD methods, Initiated Chemical Vapor Deposition (iCVD) allows independent control over the initiator and monomer chemistries. This expands the library of possible polymer chemistries while enabling mild processing conditions.^{3,4}

Since the invention of iCVD, studies investigating its kinetics have proliferated.^{5–20} These studies aim to model the iCVD process, with special focus on the roles of the initiator and monomer. The current understanding of initiator dependence originates primarily from studies of iCVD in nanoconfinement, where deposition in pores and trenches is described using models of the initiating radicals undergoing Knudsen diffusion.^{10,12,20} These studies support an Eley-Rideal initiation model in which the rate of initiation is independent of the substrate temperature. This

finding suggests that changes in deposition rate with sample temperature are due only to the monomer.

Using this idea, the role of monomer in deposition has been examined by measuring deposition rate over a range of stage temperatures.^{11,15,17,19,20} The results of these studies show that deposition rate follows an Arrhenius dependence on stage temperature which matches the temperature dependence of monomer adsorption. The dimensionless ratio of monomer partial pressure to its saturation pressure (p_m/p_{sat}) is used to account for the role of monomer in deposition consistent with the Brunauer-Emmett-Teller (BET) theory of adsorption.^{11,15,17,19,20} No consensus has been reached on the exact form this dependence takes. However, the relationship between deposition rate and p_m/p_{sat} is fit as linear with either a positive^{15,20} or negative^{11,19} y-intercept, and some discussion of a quadratic dependence at low p_m/p_{sat} .^{11,19} This existing work provides strong evidence that the surface concentration of monomer is determined by adsorption, and that initiating radicals likely react upon collision with the surface (Eley-Rideal mechanism). Critically however, this literature does not establish the order of reaction with respect to either initiator or monomer, and does not comment on the potential role of vapor phase phenomena (e.g. TBPO decomposition and mixing).

So far, direct measurement of vapor phase composition during deposition is rare,²¹ and existing efforts to quantify it^{6,21} do not validate any specific model with their data. While computational simulations of iCVD have been performed,^{22,23} their scope is limited, and they do not simulate typical iCVD conditions. With no experimentally validated model of the vapor phase to draw from, kinetic studies rely on largely un-examined assumptions when attempting to predict how vapor phase composition depends on flowrate or filament temperature.⁵⁻¹⁵ This issue impacts every study which has compared complete quantitative mechanistic models to their experimental

results.^{5,8,9} By complete, we mean models which describe how all process variables impact deposition, not those which isolate only a single phenomenon. The volume of work which relies on unverified assumptions of the iCVD vapor phase illustrates a critical vulnerability and underscores the need for an experimentally validated model.

While a model of the vapor phase is necessary to further understand the mechanisms underlying the iCVD process, it is also necessary to guide the choice of initiator, understand and optimize additives like vapor solvents,²⁴ measure and engineer monomer yield,²⁵ and improve reactor control.²¹ In the field of inorganic CVD, extensive reactor simulations have been instrumental in guiding the optimization of reactor design parameters for next-generation reactors and enabling the direct study of surface reactions.²⁶

The key vapor-phase reaction in iCVD is the thermal decomposition of the initiator. Among the various initiators used in iCVD,^{5,14} di-*tert*-butyl peroxide (TBPO) is by far the most widely studied and utilized.³ While TBPO decomposition has been characterized experimentally,^{6,21} a complete model of its decomposition has not yet been developed for the iCVD system. Outside of the iCVD literature however, fundamental studies of reaction kinetics have extensively examined TBPO as a model gas-phase decomposition reaction.^{27–29} This work shows that TBPO decomposes via three steps (Scheme 1): (i) the primary decomposition, in which the peroxide bond is cleaved to produce *tert*-butoxy radicals (TBO*); (ii) rapid β -scission, in which TBO* radicals decompose into methyl radicals and acetone; (iii) recombination of methyl radicals to form ethane. The primary decomposition is sufficiently slower than the β -scission and recombination reactions that the overall reaction is observed to be first order in the concentration of TBPO.²⁷

$$\text{Primary decomposition} \quad TBPO \xrightarrow{r_1} 2TBO^* \quad r_1 = \frac{k_1 p_{TBPO}}{R_G T} \quad (4.1)$$

$$\beta\text{-scission} \quad TBO^* \xrightarrow{r_\beta} \text{Acetone} + Me^* \quad r_\beta = \frac{k_\beta p_{TBO^*}}{R_G T} \quad (4.2)$$

$$\text{Recombination} \quad 2Me^* \xrightarrow{r_r} \text{Ethane} \quad r_r = k_r \left(\frac{p_{Me^*}}{R_G T} \right)^2 \quad (4.3)$$

Scheme 1. Thermal decomposition of TBPO

In scheme 1, r_1 is the rate of primary decomposition in $\text{mol m}^{-3} \text{s}^{-1}$, k_1 is the rate constant for primary decomposition in s^{-1} , p_{TBPO} is the partial pressure of TBPO in s^{-1} , T is the temperature in Kelvin, R_G is the ideal gas constant ($8.314 \text{ J mol}^{-1} \text{K}^{-1}$), r_β is the rate of β -scission in $\text{mol m}^{-3} \text{s}^{-1}$, k_β is the rate constant for β -scission in s^{-1} , p_{TBO^*} is the partial pressure of TBO^* radicals, r_r is the rate of recombination in $\text{mol m}^{-3} \text{s}^{-1}$, k_r is the rate constant for recombination in $\text{m}^3 \text{mol}^{-1} \text{s}^{-1}$ and p_{Me^*} is the partial pressure of methyl radicals in pascals.

Compiled results from previous studies show the Arrhenius kinetics for the primary decomposition reaction has an activation energy, $E_{A,1}$, of $158.2 \pm 0.2 \text{ kJ mol}^{-1}$ and a pre-exponential factor, A_1 , of $10^{15.8 \pm 0.2} \text{ s}^{-1}$.²⁷ Though TBPO decomposition is an effective method of generating initiating radicals,²⁹ rapid β -scission and methyl radical recombination lead to partial pressures of the radical intermediates which are difficult to directly measure. The partial pressures of the initiating radical species are determined by the relative rates of β -scission and recombination, so an understanding of these reactions is necessary to estimate radical availability. Because the recombination of methyl radicals is a fundamental gas phase radical reaction, it has been well characterized outside the context of TBPO thermal decomposition.^{30–32} A notable qualitative insight from this work is that recombination proceeds more rapidly at low temperatures because of its negative reaction entropy.²⁷ Applying these data quantitatively to TBPO decomposition is

uncertain because TBPO decomposition products may influence recombination.³⁰ Unlike methyl radical recombination, β -scission is difficult to study outside the context of TBPO decomposition, so it is less well studied.³³

Characterization of the vapor phase is instrumental to developing an understanding of the decomposition of TBPO and other vapor phase processes in iCVD. We have introduced a method in Chapter 3 by which vapor phase composition can be deduced from the Pirani pressure gauge measurements, which we summarize here. A Pirani gauge reports pressure by measuring heat transfer from a fine wire. At low pressures (less than 0.5 torr for most gases), the rate of heat transfer is limited by the rate at which gas molecules collide with the wire, so it is proportional to pressure. Larger gas molecules transfer more energy per collision however, so the Pirani gauge pressure will vary with composition (larger for larger gases and smaller for smaller gases). Leveraging this dependence, this work develops a model of how the Pirani gauge pressure depends on vapor-phase composition to enable real-time composition measurement based on the gauge reading.

To move closer to a fundamental understanding of iCVD kinetics and address the current knowledge gap in understanding the vapor phase, we characterize the thermal decomposition of TBPO in the iCVD reactor system using Pirani gauge-based composition sensing and compare it to the current understanding of the reaction. In agreement with the literature, we find that TBPO decomposes fully into acetone and ethane following Arrhenius kinetics. We further find that the iCVD reactor should be treated as a well-mixed system (frequently referred to as a continuous stirred tank model) for the purposes of TBPO decomposition in agreement with Chapter 3. CHMA is a commonly studied monomer in iCVD, and is well known to undergo ideal linear polymerization in the solution phase,^{35–37} making a good model system for investigating deposition

kinetics. We model and characterize the thermal decomposition of TBPO in the presence of CHMA vapor with a high-stage temperature while deposition is not occurring. We find that CHMA does not react considerably with TBPO in the vapor phase. This flow-through experiment is repeated with a low stage temperature, where the effect of deposition on the composition of the vapor can be estimated.

To further explore the role of initiation in the deposition process and root our investigation in existing studies of iCVD kinetics, we fit deposition rate data to the classical linear polymerization model,³⁸ adapted for iCVD. To adapt this model from the solution phase to iCVD, we assume that surface concentrations of monomer and initiating radicals are proportional to their partial pressures. Similar assumptions are made in existing models of iCVD¹⁹ kinetics and are expected to be rigorous at low p_m/p_{sat} , such that non-linearity from monomer adsorption is negligible. As previous work has done, the model assumes polymerization is initiated by TBO*. We hope our work will enable a deeper understanding of iCVD kinetics by elucidating the role of vapor phase reactions. We expect that revisiting iCVD kinetics with a deeper understanding of the vapor phase will yield fruitful insight into fundamental processes.

4.2. Materials and Methods

4.2.1 Materials

Cyclohexyl methacrylate (CHMA) (99.98% purity, AmBeed) was used as a monomer and di-*tert*-butyl peroxide (TBPO) (98%, from Sigma Aldrich) was used as a pre-initiator in all depositions and mixed-gas flowthrough experiments. Both were used without further purification. Depositions were performed on polished P/Boron doped silicon wafers (from University Wafer).

4.2.2 Filament temperature calibration

To measure the temperature of the filament array, a k-type thermocouple was attached to the filament such that it measured temperature independently of the direction current was flowing through the array, ensuring the thermocouple wasn't being shorted by the array. Because the measured temperature was found to change slightly from day to day, we have standardized all temperature measurements (unless otherwise noted), by measuring the Arrhenius curve for TBPO flowing at 1 sccm and 100 mtorr. A standard Arrhenius curve was collected (see section 4.4.3 for details), which serves as a calibration dataset to which all other temperature curves were calibrated. Equation (4.6) adjusts measured temperatures onto the consistent temperature scale from the calibration data. Equation (4.6) is purely empirical, and its validity as a temperature correction requires a good fit of both the calibration data and the calibrated data to the Arrhenius curve at both adjusted and un-adjusted filament temperature. All calibrated and un-calibrated data showed strong fit to the Arrhenius model. $E_{A,1,CAL}^{eff}$ and $A_{1,CAL}^{eff}$ were $91.531 \text{ kJ mol}^{-1}$ and $4.232 \times 10^4 \text{ m}^3 \text{ s}^{-1}$ respectively. An example of the calibration is shown in figure (4.1).

4.2.3 Vapor phase composition measurements using a Pirani gauge

A Granville-Phillips Company convection gauge (catalog no. 275) was used for composition sensing and will be referred to throughout as the Pirani gauge. The pressure reported by it will be referred to as the Pirani pressure, while actual mechanical pressure is sensed using an MKS Baratron Capacitance Manometer model 626C, and this pressure will be referred to as the Baratron pressure.

To measure the composition using the Pirani gauge, a standard curve is created by measuring the Pirani and Baratron pressures under controlled conditions where total flowrate and reactor pressure is held constant, but makeup of the inlet gas is changed. The ratio of the Pirani

pressure to the Baratron pressure is then plotted against the reactor composition (mole fraction in the case of binary gas mixing, or fractional conversion in the case of a first order reaction) and fit to a model (generally linear). In scenarios where the reactor composition is not known, but the Pirani gauge pressure and Baratron pressures can be read, the standard curve can then be used to determine the reactor composition.

4.2.4 Batch mode TBPO decomposition

The reactor was charged with TBPO at a fixed pressure of 100 mtorr at which point it was held under these conditions for 1 minute to allow for measurement of the leak rate. Keeping the reactor isolated from vacuum, the filament array was turned on, and the Baratron and Pirani gauge pressures were measured as a function of time. The reactor was then pumped down to base pressure again and these procedures were repeated for decomposition at filament temperatures from 180-280°C. The pressure curves are fitted to a first order kinetic model to obtain primary decomposition rate constants for the reactor. The ratio of Pirani pressure to Baratron pressure is plotted against fractional conversion generating a standard curve to measure fractional conversion given measurement of Pirani and Baratron pressures.

4.2.5 Continuous flow-through TBPO decomposition

Using the form of the standard curve generated from batch mode TBPO decomposition, Baratron and Pirani gauge pressures for continuous flow-through TBPO decomposition can be used to better understand the iCVD process (which occurs in a flow-through mode). To test if the rate of decomposition depends on the flowrate of TBPO, the composition in the reactor was measured using the Pirani gauge at a controlled pressure of 100 mtorr, and filament temperature

of 240°C while varying the flow of TBPO into the reactor over its full range from 0.11 to 1.479 sccm.

At a constant flow-rate of 1 sccm and total pressure of 100 mtorr, the power to the array was then turned on so filament temperature reached ~160°C (a low value below which reaction is too slow to measure). Power was then increased incrementally until a filament temperature of 360°C was reached (the highest temperature our power supply could reliably reach) while recording pressures and temperatures. To ensure that sufficient time was given for thermal equilibration while stepping up the temperature, the power to the array was stepped back down while pressures and temperatures were recorded. No significant hysteresis was observed in fractional conversion, however there was some hysteresis in measured stage temperatures. This procedure was performed at pressures of 50, 100, 200 and 300 mtorr.

A similar flow through experiment was then performed for TBPO and CHMA mixtures at inlet TBPO mole fractions of 0, 0.2, 0.4, 0.6, 0.8, and 1, at a reactor pressure of 100 mtorr and a reactor stage temperature of 50°C such that deposition was unlikely to impact the Pirani gauge measurement. Pirani gauge was measured for filament temperatures from room temperature to 430°C for each flow condition (the full range of filament temperatures accessible to our system). Time for equilibration was allowed between flow conditions.

4.2.6 iCVD depositions

All iCVD depositions were performed in a custom-built two-chamber iCVD reactor. All samples were cut from silicon wafers to be the same size 50 cm², setting aside a reference sample from each wafer for baselining FTIR absorbance spectra. Samples were placed in the center of the cooled stage and a K-type thermocouple was affixed to the stage adjacent to the sample using

Kapton tape and Cool-Grease™ thermal heat transfer compound from AI technologies. Once the reactor reached a base pressure of 3-6 mtorr, the leak rate was recorded by measuring the Baratron pressure every 10 seconds for 2 minutes and performing a linear regression to obtain the slope. This change in pressure over time is then converted to a molar leak-rate using the ideal gas law. For the leak rate and all flow-rate measurements the effective temperature was assumed to be 42.5°C as an average of the stage temperature of 30°C and the wall/lid temperature of 50°C. Depositions with leak-rate over 0.03 sccm were discarded from the data analysis.

Flow rates of CHMA and TBPO were determined by closing the throttling valve connecting the chamber and the vacuum pump and measuring pressure every 2 seconds for 20 seconds. The rate of change of pressure was then used to obtain a molar flow rate using the ideal gas law.

To calibrate the Pirani gauge measurement before the deposition, Pirani pressure, filament temperature, stage temperature, voltage and current to the array (as read by the BK precision DC power supply) were all recorded at the following set of conditions.

- Filament off, with a 1 sccm CHMA flowrate at a Baratron pressure of 100 mtorr
- Filament off, with no CHMA flowing, with a 1 sccm TBPO flowrate at 100 mtorr
- Filament power set to a nominal setpoint for the deposition, with 1 sccm TBPO at 100 mtorr
- Filament power set to maximum with 1 sccm TBPO at 100 mtorr

The deposition was then performed for 1 hour at a pressure of 100mtorr and a total flow rate of 1 sccm. Depositions were performed at a range of initiator/monomer inlet compositions and filament temperatures, with all other variables held constant. Deposition conditions are provided in Table 4.1.

Table 4.1 Deposition Conditions

F_{LEAK} (sccm)	F_{TBPO} (sccm)	F_{CHMA} (sccm)	T_{STAGE} (°C)	T_{fil} (°C)	y_{TBPO}^{in}	$DR \times 10^{11}$ (mol cm ⁻² s ⁻¹)	y_{TBPO}	y_{CHMA}	T_{fil} (K)	p_m/p_{sat}
0.0237±0.0012	0.204±0.0035	0.749±0.0094	33.7	497.15	0.214	2.68	0.194	0.763	509	0.0817
0.0278±0.00078	0.2±0.0040	0.827±0.017	32.9	474.85	0.195	3.09	0.159	0.763	532	0.0909
0.0168±0.00078	0.202±0.0028	0.8±0.0053	32.8	492.15	0.202	2.6	0.172	0.765	523	0.0923
0.0265±0.0017	0.807±0.0071	0.196±0.0036	34.9	494.05	0.805	0.351	0.782	0.192	467	0.0175
0.0248±0.00035	0.791±0.0054	0.197±0.0043	36.6	541.15	0.801	0.429	0.648	0.176	517	0.0128
0.0217±0.00023	0.805±0.005	0.194±0.00037	45.7	595.15	0.806	0.591	0.382	0.131	569	0.00300
0.0127±0.00062	0.598±0.0030	0.394±0.0024	44.5	493.95	0.603	0.519	0.581	0.389	476	0.0104
0.0116±0.0012	0.605±0.0049	0.398±0.0036	38.7	515.05	0.603	1.2	0.517	0.366	512	0.0203
0.0202±0.0017	0.609±0.0060	0.398±0.0064	41.2	550.55	0.605	1.02	0.323	0.295	564	0.0119
0.0297±0.00057	0.396±0.0036	0.593±0.0033	38.3	495.95	0.400	2.02	0.360	0.573	506	0.0335
0.0179±0.0013	0.21±0.0062	0.804±0.0085	35.3	506.15	0.207	3.11	0.182	0.765	517	0.0662
0.0247±0.0012	0.195±0.0050	0.803±0.0089	35.2	575.95	0.195	6.09	0.0441	0.630	624	0.0552
0.027±0.00027	0.195±0.0040	0.802±0.0081	36.2	490.15	0.196	1.89	0.187	0.795	490	0.0611
0.0275±0.00047	0.195±0.01	0.817±0.010	34.2	503.15	0.193	4	0.163	0.773	525	0.0773
0.0277±0.00082	0.193±0.0032	0.807±0.0051	35.2	519.75	0.193	5.57	0.142	0.747	545	0.0655
0.0217±0.00067	0.305±0.0022	0.709±0.016	35.6	503.25	0.301	2.66	0.285	0.686	492	0.0570
0.0186±0.0011	0.298±0.0058	0.703±0.014	32.9	527.95	0.298	3.43	0.258	0.667	518	0.0794
0.0219±0.00055	0.295±0.0056	0.7±0.0085	35.1	541.85	0.296	5.39	0.247	0.660	524	0.0586
0.0233±0.00069	0.3±0.0097	0.709±0.0024	33.7	545.35	0.297	5.71	0.214	0.629	544	0.0673
0.0268±0.00020	0.3±0.01	0.706±0.0081	36	558.85	0.298	4.33	0.167	0.586	567	0.0463
0.0265±0.00093	0.105±0.0038	0.889±0.012	36.9	500.65	0.106	1.81	0.0995	0.885	501	0.0621
0.0188±0.00017	0.099±0.0034	0.894±0.00079	37.5	531.15	0.0997	3.2	0.0830	0.875	531	0.0567
0.0272±0.00096	0.093±0.0011	0.89±0.0030	39.9	544.95	0.0946	4.35	0.0584	0.850	564	0.0404
0.0175±0.00035	0.1±0.0030	0.904±0.0078	46.6	559.85	0.0996	4.91	0.0737	0.861	547	0.0177
0.0195±0.00088	0.198±0.00059	0.797±0.0043	33.9	552.85	0.199	5.31	0.122	0.713	562	0.0743

4.2.7 Thin film characterization

A JA Woollam alpha-SE spectroscopic ellipsometer collected spectra at 65°, 70° and 75° angle of incidence on poly(CHMA) thin films. Data was fitted using Complete EASE 6 software, version 6.51 with a 3-layer model of a Cauchy film on a silicon thermal oxide on a silicon metal substrate. For many samples, the Cauchy model was not appropriate due to large mean squared error but for consistency, the same model is used throughout.

A Thermo Scientific Nicolet iS50 FT-IR collected transmission spectra of all polymer film samples and their respective reference wafers using a DGTS detector. A single grazing incidence spectrum of CHMA monomer was taken using a variable angle grazing incidence attenuated total reflectance module with a germanium crystal for reference also using the DGTS detector. All spectra were collected from 700-4000 cm^{-1} with a resolution of 4 cm^{-1} . Omnic software was used for atmospheric suppression, which eliminated spurious peaks due to CO_2 . To determine the sample absorbance from the FTIR spectral data, quartic baseline subtraction was performed by taking the derivative of the spectrum, then fitting a 3rd order polynomial to the baseline, and subtracting it. After removing outlier spectra, spectra were then averaged, and the average spectrum was multiplied by a coefficient and subtracted from each of the spectra individually such that the residual was minimized. The coefficient for subtraction was then reported as the “optimal scale”, which is directly proportional to the total absorbance of CHMA. No evidence of contamination was observed in sample spectra, and all residual spectra resulting from subtraction were checked to ensure minimal error. Total absorbance of CHMA was related to mass per unit area using the density of bulk poly(CHMA) and the average absorbance per nanometer of film for samples thicker than 250 nm, having MSE less than 150.

4.3. Theory

4.3.1 Temperature Calibration

To ensure that all temperature measurements are consistent with each other, for most of our experiments (excluding batch decomposition), we have re-calibrated the temperature. The re-calibration procedure we use assumes that the rate constant and activation energy for the decomposition of TBPO is the same on different days when the process conditions are the same. Any differences are attributed to temperature measurement inaccuracy.

The procedure for re-calibration is as follows. First, we collect rate constant data at a range of filament temperatures to make an Arrhenius plot (figure 4.1 (a)), reflecting the behavior in equation (4.4),

$$\ln(k_1^{eff}) = \ln A_{1,CAL}^{eff} - \frac{E_{A,1}\eta_{1,CAL}}{R_G T_{fil}} \quad (4.4)$$

where k_1^{eff} is the effective rate constant for decomposition, A_1^{eff} is the effective pre-exponential factor and $E_{A,1}$ is the activation energy for primary decomposition, η_1 accounts for the averaging over the volume of the reactor, and is an empirical quantity which relates the apparent activation energy to the actual activation energy for the reaction. Subscript ‘CAL’ refers to the calibration data. The nature of the Arrhenius fit to data is discussed in Section 4.3.2 in more detail.

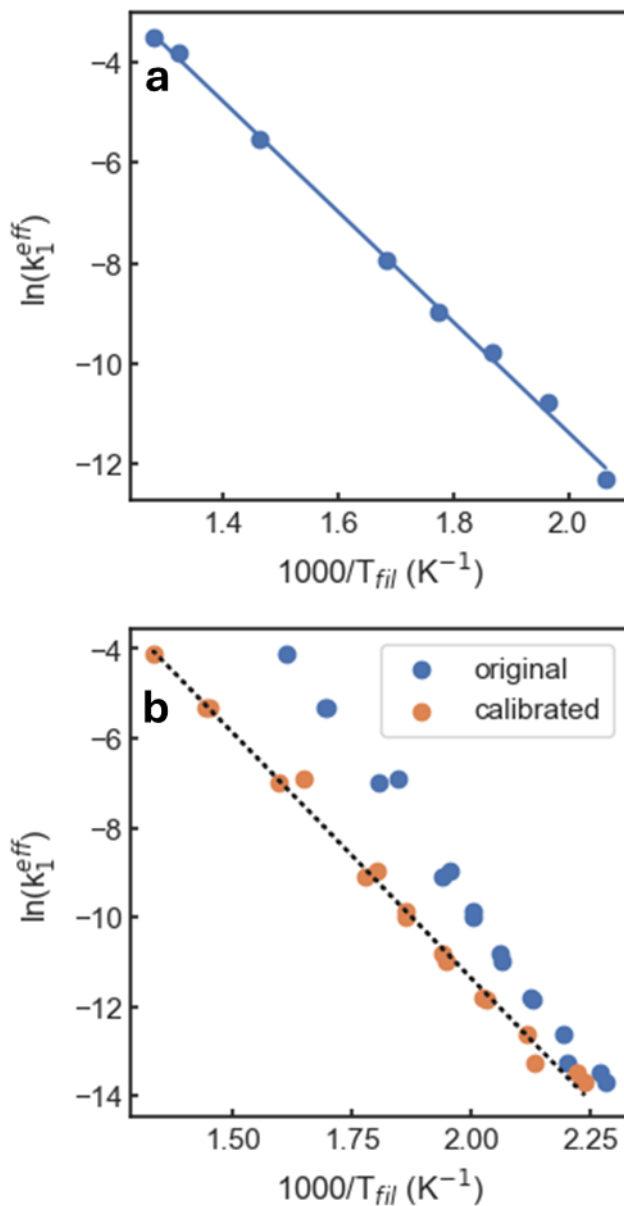


Figure 4.1. (a) Arrhenius curve used for temperature calibration **(b)** Calibration of temperature example showing flowthrough TBPO at 100mtorr

Next, on a different day, and for a different experiment, the same set of data is collected at the same pressure (100 mtorr) and flow-rate (1 sccm), resulting in the blue data in figure 4.1 (b) labeled “original”. While both datasets show a good Arrhenius fit, because of variability in the thermocouple connection to the filament array, they do not show the same Arrhenius fit. To correct

for this, we assume that the temperature measurement on the data is incorrect, but that the rate constants are the same, giving Equation (4.5),

$$\frac{-E_{A,1}\eta_1}{T_{fil,meas}R_G} + \ln A_1^{eff} = \frac{-E_{A,1}\eta_{1,CAL}}{T_{fil}R_G} + \ln A_{1,CAL}^{eff} \quad (4.5)$$

where $T_{fil,meas}$ is the measured filament temperature in Kelvin and T_{fil} is the “true” filament temperature which would have been recorded on the day that the data in figure 4.1 (a) was collected, which we want to solve for.

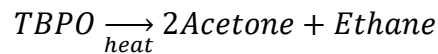
To calibrate the data, we solve equation (4.5) for T_{fil} to get equation (4.6).

$$T_{fil} = \frac{E_{A,1}\eta_{1,CAL}}{\frac{E_{A,1}\eta_1}{T_{fil,meas}} - R_G(\ln A_1^{eff} - \ln A_{1,CAL}^{eff})} \quad (4.6)$$

Equation (4.6) is used to re-calibrate temperature so that the dataset gives an activation energy and pre-exponential factor which matches the calibration data (figure 4.1 (a)). This is shown with the orange data from figure 4.1 (b) with a line of best fit having activation energy and logA corresponding to figure 4.1 (a).

4.3.2 Kinetic model of batch decomposition

This section derives the governing equations for a model of TBPO decomposition in the batch mode which is tested in Section 4.4.1 against data. For batch mode operation of the iCVD reactor, we consider the overall equation for decomposition of TBPO into acetone and ethane.



Mole balances for TBPO, air, ethane and acetone are given by equation (4.7),

$$\frac{dn_{TBPO}}{dt} = -r_1^{tot} \quad (4.7a)$$

$$\frac{dn_{air}}{dt} = \dot{n}_{leak} \quad (4.7b)$$

$$\frac{dn_{ethane}}{dt} = r_1^{tot} \quad (4.7c)$$

$$\frac{dn_{acetone}}{dt} = 2r_1^{tot} \quad (4.7d)$$

where n_{TBPO} is the number of moles of TBPO in the reactor, r_1^{tot} is the rate of decomposition of TBPO in mol s^{-1} , n_{air} is the number of moles of air in the reactor, $\dot{n}_{leak}$ is the leak rate in mol s^{-1} , n_{ethane} is the number of moles of ethane in the reactor, and $n_{acetone}$ is the number of moles of acetone in the reactor.

Equations (4.7) (a)-(d) can be combined to give the total mole balance on the reactor given by equation (4.8),

$$\frac{dn}{dt} = 2r_1^{tot} + \dot{n}_{leak} \quad (4.8)$$

where n is the total number of moles of gas in the reactor. equation (4.8) can be integrated to give equation (4.9),

$$n = n^i + 2 \int r_1^{tot} dt + \dot{n}_{leak} t \quad (4.9)$$

where n^i is the initial number of moles in the reactor. Now, consider that the ideal gas law can be

defined for the reactor using the effective temperature according to equation (4.10).

$$n = \frac{pV_R}{R_G T_{eff}} \quad (4.10)$$

Giving equation (4.11).

$$\frac{pV_R}{R_G T_{eff}} = 2 \int r_1^{tot} dt + \dot{n}_{leak} t + n^i \quad (4.11)$$

The total number of TBPO molecules which have decomposed, $\int r_1^{tot} dt$, is related to the fractional conversion by equation (4.12).

$$\chi = \frac{\int r_1^{tot} dt}{n_{TBPO}^i} \quad (4.12)$$

Substitute equation (4.12) into equation (4.11) and solve for the fractional conversion.

$$\chi = \frac{1}{2} \left(\frac{pV_R}{R_G T_{eff} n_{TBPO}^i} - \frac{\dot{n}_{leak} t}{n_{TBPO}^i} - \frac{n^i}{n_{TBPO}^i} \right) \quad (4.13)$$

Substitute the initial mole fraction of TBPO, and use equation (4.10) to further write in terms of pressures and temperatures to give equation (4.14),

$$\begin{aligned} \chi &= \frac{1}{2} \left(\frac{pV_R}{R_G T_{eff} n_{TBPO}^i} - \frac{\dot{n}_{leak} t}{n_{TBPO}^i} - \frac{1}{y_{TBPO}^i} \right) \\ \chi &= \frac{1}{2y_{TBPO}^i} \left(\frac{pV_R}{R_G T_{eff} n^i} - \frac{\dot{n}_{leak} t}{n^i} - 1 \right) \\ \chi &= \frac{1}{2y_{TBPO}^i} \left(\frac{p T_{eff}^i}{p^i T_{eff}} - \frac{1}{p^i} \left(\frac{dp}{dt} \right)_{leak} t - 1 \right) \end{aligned} \quad (4.14)$$

where χ is the fractional conversion of TBPO, y_{TBPO}^i is the initial mole fraction of TBPO in the reactor (generally over 0.99), p^i is the initial pressure in the reactor, $\left(\frac{dp}{dt}\right)_{leak}$ is the measured pressure change caused by leak at base pressure, with the filament off and T_{eff}^i is the initial effective temperature of the reactor (obtained from equation (4.36) and the experimental process conditions).

To model the decomposition kinetics measured by equation (4.14), the rate equation at uniform temperature (equation (4.1)) must be adapted for the iCVD geometry. When adapting equation (4.1) for the iCVD geometry, we hypothesize that the total rate of primary decomposition (r_1^{tot}) is proportional to the product of the (measurable) mole fraction of TBPO far from the array and the concentration of ideal gas at the filament temperature. We introduce an effective rate constant for primary decomposition (k_1^{eff}) as a constant of proportionality which accounts for the poorly defined region close to the array. This expression is given by equation (4.15),

$$r_1^{tot} = k_1^{eff} \frac{p}{R_G T_{fil}} y_{TBPO} \quad (4.15)$$

where $r_1^{tot} = \int_{V_r} r_1 dV$ in mol s^{-1} . In equation (4.15), the effective rate constant, k_1^{eff} , is determined empirically, but represents the average of k_1 , from equation (4.1) over the reactor. It's likely that coupled heat and mass transfer induce compositional gradients close to the filament, meaning TBPO decomposition occurs over a range of concentrations and temperatures. Estimating k_1^{eff} from first principles would require modelling efforts outside the scope of this paper. Instead, we simply assume that since k_1 is known to obey Arrhenius kinetics,²⁸ k_1^{eff} will as well. Though, the pre-exponentiation factor and apparent activation energy will differ, and are determined empirically.

Because primary decomposition occurs close to the array, we use the filament temperature as the decomposition temperature so k_1^{eff} is given by equation (4.16),

$$k_1^{eff} = A_1^{eff} \exp\left(-\frac{E_{A,1}\eta_1}{R_G T_{fil}}\right) \quad (4.16)$$

where A_1^{eff} is an empirically determined pre-exponentiation factor and $E_{A,1}$ is the “true” activation energy for the decomposition of TBPO at uniform temperature (measured in the literature). To account for thermal gradients at the array, η_1 is a number between zero and one which relates the true activation energy $E_{A,1}$ to the apparent activation energy from Arrhenius plots.

In order to predict the composition of TBPO in the reactor, we consider the mole balance on decomposition of TBPO alone (equation 4.7a), and substitute equation (4.15) to get equation (4.17).

$$\frac{dn_{TBPO}}{dt} = -\frac{k_1^{eff} p_{TBPO} V_R}{R_G T_{fil}} \quad (4.17)$$

Substitute equation (4.10) into equation (4.17) to get equation (4.18).

$$\frac{dn_{TBPO}}{dt} = -\frac{k_1^{eff} n_{TBPO} T_{eff}}{T_{fil}} \quad (4.18)$$

At constant temperature, equation (4.18) is integrated to give equation (4.19),

$$n_{TBPO} = n_{TBPO}^i \exp\left(-\frac{k_1^{eff} T_{eff}}{T_{fil}} t\right) \quad (4.19)$$

Solve for fractional conversion $\chi = \frac{n_{TBPO}^i - n_{TBPO}}{n_{TBPO}^i}$, to give equation (4.20).

$$\chi = 1 - \exp\left(-\frac{k_1^{eff} T_{eff}}{T_{fil}} t\right) \quad (4.20)$$

Now, we turn our attention to how we might determine composition (mole fraction). We begin by writing out the definition of the mole fraction.

$$y_{TBPO} = \frac{n_{TBPO}}{n}$$

Substitute total number of moles (equation (4.9)).

$$y_{TBPO} = \frac{n_{TBPO}}{n^i + 2 \int r_1^{tot} dt + \dot{n}_{leak} t}$$

Substitute in the fractional conversion $n_{TBPO} = n_{TBPO}^i (1 - \chi)$ and equation (4.12).

$$y_{TBPO} = \frac{n_{TBPO}^i (1 - \chi)}{n^i + 2 n_{TBPO}^i \chi + \dot{n}_{leak} t}$$

Simplify.

$$y_{TBPO} = \frac{1 - \chi}{\frac{1}{y_{TBPO}^i} + 2\chi + \frac{\dot{n}_{leak} t}{n_{TBPO}^i}}$$

Finally, we can re-express the leak-rate in terms of the pressure change to obtain equation (4.21).

$$y_{TBPO} = y_{TBPO}^i \frac{1 - \chi}{1 + y_{TBPO}^i 2\chi + \frac{1}{p^i} \left(\frac{dp}{dt}\right)_{leak} t} \quad (4.21)$$

4.3.3 Kinetic model for continuous decomposition

In this section we will present a model of the reactor which enables the derivation of equations which are relevant to continuous operation of the reactor. We will assume in our derivations that monomer is present, and deposition is occurring. To relate the reactor composition

to the kinetic model, the total rate of reaction needs to be determined from a mole balance on the reactor. Because this mole balance requires the composition of the reactor outlet, degree of mixing in the reactor must be known.⁴² For the iCVD vapor phase, Chapter 3 has shown that the degree of mixing is determined by the Peclet number, and that the reactor can generally be approximated as well-mixed. Throughout this work, we will make the well-mixed assumption (for all species except radical species which are present in low concentrations), such that composition is constant throughout the volume of the reactor. To measure decomposition in the continuous mode, we can perform mass balances on TBPO, TBO*, acetone, Me*, ethane and monomer presented in equation (4.22).

$$F_{TBPO}^{out} - F_{TBPO}^{in} = -r_1^{tot} \quad (4.22a)$$

$$F_{TBO^*}^{out} = 2r_1^{tot} - r_\beta^{tot} \quad (4.22b)$$

$$F_{acetone}^{out} = r_\beta^{tot} \quad (4.22c)$$

$$F_{Me^*}^{out} = r_\beta^{tot} - 2r_r^{tot} \quad (4.22d)$$

$$F_{ethane}^{out} = r_r^{tot} \quad (4.22e)$$

$$F_{CHMA}^{out} = F_{CHMA}^{in} - r_p^{tot} \quad (4.22f)$$

Note that β -scission and recombination are fast relative to primary decomposition (see Section 4.4.1), so we can simplify the mass balances by setting the flow of methyl TBO* and Me* out of

the reactor to zero. This converts equation (4.22b) and equation (4.22d) to equation (4.23) and equation (4.24),

$$2r_1^{tot} = r_\beta^{tot} \quad (4.23)$$

$$r_\beta^{tot} = 2r_r^{tot} \quad (4.24)$$

Using equations (4.23), (4.24) and equation (4.15), we can convert equation (4.22) into equation (4.25),

$$F_{TBPO}^{out} - F_{TBPO}^{in} = -\frac{k_1^{eff} p_{TBPO}}{R_G T_{fil}} \quad (4.25a)$$

$$F_{acetone}^{out} = 2 \frac{k_1^{eff} p_{TBPO}}{R_G T_{fil}} \quad (4.25b)$$

$$F_{ethane}^{out} = \frac{k_1^{eff} p_{TBPO}}{R_G T_{fil}} \quad (4.25c)$$

$$F_{CHMA}^{out} = F_{CHMA}^{in} - r_p^{tot} \quad (4.25d)$$

Now, from the well-mixed assumption, we define the mole fraction of TBPO in the reactor from equation (4.26).

$$y_{TBPO} = \frac{F_{TBPO}^{out}}{F_{TBPO}^{out} + F_{acetone}^{out} + F_{ethane}^{out} + F_{CHMA}^{out}} \quad (4.26)$$

We can now substitute in equation (4.25) into equation (4.26) to get equation (4.27).

$$y_{TBPO} = \frac{F_{TBPO}^{in} - \frac{k_1^{eff} p y_{TBPO}}{R_G T_{fil}}}{F_{TBPO}^{in} + 2 \frac{k_1^{eff} p y_{TBPO}}{R_G T_{fil}} + F_{CHMA}^{in} - r_p^{tot}} \quad (4.27)$$

Now, we can set F_{CHMA}^{in} and r_p^{tot} to zero, and solve equation (4.27) for k_1^{eff} to get equation (4.28).

$$k_1^{eff} = \frac{F_{TBPO}^{in} R_G T_{fil} (1 - y_{TBPO})}{p y_{TBPO} (2 y_{TBPO} + 1)} \quad (4.28)$$

Next, we will solve equation (4.27) for y_{TBPO} .

$$2 \frac{k_1^{eff} p}{R_G T_{fil}} y_{TBPO}^2 + y_{TBPO} \left(F_{TBPO}^{in} + F_{CHMA}^{in} - r_p^{tot} + \frac{k_1^{eff} p}{R_G T_{fil}} \right) - F_{TBPO}^{in} = 0$$

Substituting F_{tot}^{in} for $F_{TBPO}^{in} + F_{CHMA}^{in}$.

$$2 \frac{k_1^{eff} p}{R_G T_{fil}} y_{TBPO}^2 + y_{TBPO} \left(F_{tot}^{in} - r_p^{tot} + \frac{k_1^{eff} p}{R_G T_{fil}} \right) - F_{TBPO}^{in} = 0$$

Solve the quadratic to get equation (4.29).

$$y_{TBPO} = \frac{-\left(F_{tot}^{in} - r_p^{tot} + \frac{k_1^{eff} p}{R_G T_{fil}}\right) + \sqrt{\left(F_{tot}^{in} - r_p^{tot} + \frac{k_1^{eff} p}{R_G T_{fil}}\right)^2 + F_{tot}^{in} y_{DTBP}^{in} 8 \frac{k_1^{eff} p}{R_G T_{fil}}}}{4 \frac{k_1^{eff} p}{R_G T_{fil}}} \quad (4.29)$$

To derive an expression for the mole fraction of monomer, we will define the mole fraction of monomer in the reactor as equation (4.30).

$$y_{CHMA} = \frac{F_{CHMA}^{out}}{F_{TBPO}^{out} + F_{acetone}^{out} + F_{ethane}^{out} + F_{CHMA}^{out}} \quad (4.30)$$

Substituting equation (4.25), we get equation (4.31).

$$y_{CHMA} = \frac{F_{CHMA}^{in} - r_p^{tot}}{F_{TBPO}^{in} + 2 \frac{k_1^{eff} p y_{TBPO}}{R_G T_{fil}} + F_{CHMA}^{in} - r_p^{tot}} \quad (4.31)$$

Simplifying and re-writing flowrates, we get equation (4.32).

$$y_{CHMA} = \frac{F_{tot}^{in} (1 - y_{TBPO}^{in}) - r_p^{tot}}{F_{tot}^{in} - r_p^{tot} + 2 \frac{k_1^{eff} p}{R_G T_{fil}} y_{TBPO}} \quad (4.32)$$

Finally, we will turn our attention to the derivation of partial pressure of TBO* radicals. This requires a kinetic definition for the rate of β -scission, which is given by equation (4.33).

$$r_{\beta}^{tot} = \frac{k_{\beta}^{eff} p_{TBO^*}}{R_G T_{fil}} \quad (4.33)$$

We can now plug equation (4.33) and equation (4.15) into equation (4.23) to give equation (4.34).

$$2 \frac{k_1^{eff} p_{TBPO}}{R_G T_{fil}} = \frac{k_\beta^{eff} p_{TBO^*}}{R_G T_{fil}} \quad (4.34)$$

Simplifying, and solving for p_{TBO^*} , we get equation (4.35)

$$p_{TBO^*} = 2 \frac{k_1^{eff}}{k_\beta^{eff}} p_{TBPO} \quad (4.35)$$

4.4. Results and Discussion

4.4.1 Batch mode decomposition of pure TBPO

The key vapor-phase reaction in iCVD is TBPO decomposition. In a closed chamber with a fixed initial quantity of TBPO, pressure increases as TBPO decomposes because each molecule of TBPO decomposes into two molecules of acetone and one molecule of ethane. This pressure change is frequently used to characterize TBPO decomposition kinetics.²⁸ However, the conditions it is typically studied at (atmospheric pressure and uniform reactor temperature) differ significantly from iCVD.^{27,28} In iCVD, (i) the filament array introduces significant temperature non-uniformity, and (ii) the leak rate contributes measurably to pressure changes. Because these factors affect the pressure, reaction kinetics must be validated under iCVD-specific conditions. While batch experiments are illuminating, *continuous flow-through* experiments more accurately represent iCVD. Unfortunately, they do not allow kinetic measurements based on pressure change. Therefore, this section also establishes the relationship between the Pirani gauge reading and

vapor-phase composition during TBPO decomposition so the Pirani gauge can be used as a composition sensor.

The following steps correct for the effects of leak and non-uniform temperature, such that pressure change during *batch mode* decomposition experiments can be used to deduce composition in the reactor. We start by accounting for the effect of changing gas temperature on the total pressure in the reactor (i.e., effect (i) mentioned above). In Chapter 3, we established a method to correct for temperature changes in batch mode using an effective temperature, defined such that the ideal gas law (which is used routinely in iCVD research such as precursor flow-rate calibration) holds for the reactor according to equation (4.10). Chapter 3 has measured the effective temperature using inert argon and IPA, respectively, showing that changes in gas temperature are responsible for a 5-10% change in pressure after the filament is turned on. This work also validates a model of the effective temperature obtained by averaging T^{-1} (in Kelvin) over the volume of a 2-chamber reactor (matching our experimental setup). In this average, the bottom chamber is isolated from the filament array, the filament array is described by a planar heat source, and temperature profiles are linearized. The result is given by equation (4.36),

$$T_{eff} = \frac{1}{\frac{1-f}{T_{ref}} + \frac{f \ln \left(1 + h_{eff} \left(\frac{T_{fil}}{T_{av}} - 1 \right) \right)}{h_{eff}(T_{fil} - T_{av})}} \quad (4.36)$$

where T_{av} is the average temperature in Kelvin of the lid and the stage when the filament is off (assumed to be equal to the stage temperature), f is the fraction of the total reactor volume which is directly affected by the filament array (estimated from the length of the filament, the height of the chamber and the separation between filament wires divided by the total volume of the reactor which rounds to approximately 0.3), and T_{ref} is the temperature of the bottom chamber of the

reactor (40°C). At especially low pressures (or Knudsen numbers), the Knudsen temperature jump^{39,40} may introduce pressure dependence into the overall heat transfer from the array. However, the data of Bakker⁴¹ and Chapter 3 suggest that this effect is negligible at our conditions, and that heat transfer between the filament and reactor body is described by an effective heat transfer coefficient (h_{eff}), given by equation (4.37) (details of the derivation of equations (4.36) and (4.37) may be found in Chapter 3),

$$h_{eff} = \frac{\frac{H\pi}{L}}{\ln\left(\frac{L \sinh\left(\frac{H\pi}{L}\right)}{R_{fil}\pi}\right)} \quad (4.37)$$

where H is the distance from the filament to the stage (25 mm), L is the distance between the array wires (10 mm), R_{fil} is the radius of the filament (0.2 mm).

Batch decompositions were performed by pressurizing the reactor with TBPO, then closing the outlet to the vacuum pump and turning on the filament heating. As discussed previously, there is a stoichiometric pressure increase associated with decomposition of TBPO. Side reactions such as hydrogen abstraction to form *tert*-butanol are not known to occur over the pressure range considered,²⁹ so the pressure change is due only to conversion of TBPO to acetone and ethane. Correcting for (i) temperature effects and (ii) leak, the resultant pressure change can be used to experimentally determine fractional conversion (χ) using equation (4.14).

Because thermal decomposition kinetics are limited by the (unimolecular) primary decomposition of TBPO into TBO* radicals, the fractional conversion is expected to exponentially decrease according to equation (4.21). Plotting fractional conversion (calculated using equation (4.14)) against time on linearized axes (figure 4.2 (a)) shows that when k_1^{eff} is treated as a fitting parameter, equation (4.21) delivers an excellent fit to the data over a range of filament temperatures

($r^2 > 0.97$ for all cases except $T_{fil} = 186.4^\circ\text{C}$ where $r^2 = 0.95$). The success of this fit supports the assumption that TBPO decomposes directly into acetone and ethane, and therefore, that the TBO* and Me* intermediates instrumental to iCVD are only present in trace quantities. This conclusion has significant implications for kinetic modeling and reporting of deposition conditions since we are aware of no past work which corrects for TBPO decomposition in modeling.

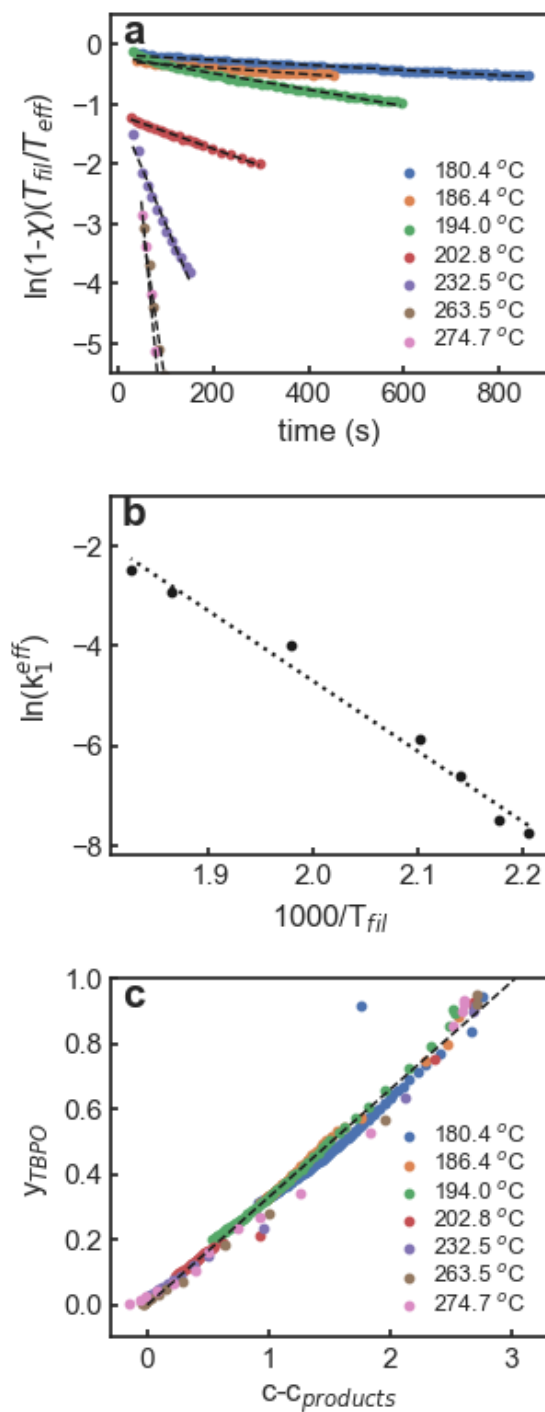


Figure 4.2. (a) Fractional conversion calculated from equation (4.14) plotted on linearized axes against time for batch-mode TBPO decomposition, for a range of filament temperatures showing a good fit to data (r^2 at least 0.95 for all cases). The fit indicates that equation (4.15) is appropriate

to describe TBPO thermal decomposition in iCVD. **(b)** Arrhenius plot of k_1^{eff} from the slopes of fits in (a) in terms of filament temperature showing an apparent activation energy of 117 kJ mol^{-1} ($\eta = 0.74$) and a pre-exponential factor $\ln(A_1^{eff})$ of $23.4 \text{ (m}^3 \text{ s}^{-1}\text{)}$ with $r^2 = 0.98$. **(c)** Mole fraction of TBPO during the thermal decomposition of pure TBPO in batch mode calculated from Equation (4.21) plotted against the adjusted Pirani gauge ratio $c - c_{products}$ for a range of filament temperatures including a dotted line representing equation (4.39) to show that equation (4.39) is suitable for predicting composition ($r^2 = 0.96$).

Plotting $\ln(k_1^{eff})$ versus $1000/T_{fil}$ shows that Arrhenius kinetics (equation (4.16)) are an accurate approximation of the overall decomposition rate (figure 4.2 (b)). This fit shows an apparent activation energy of 117 kJ mol^{-1} ($\eta_1 = 0.74$) and a pre-exponential factor $\ln(A_1^{eff})$ of $23.4 \text{ (m}^3 \text{ s}^{-1}\text{)}$ with $r^2 = 0.98$. Though our data analysis corrects for the potential for thermally driven pressure changes and leak these factors reduce accuracy of batch measurements relative to continuous flow through measurements. Therefore, we leave a more thorough analysis of Arrhenius parameters for our discussion of decomposition in the continuous mode (Section 4.4.2).

We now turn our attention to the relationship between gas composition and the Pirani gauge pressure during the thermal decomposition of TBPO. In Chapter 3, we showed that the Pirani gauge is effective for measuring the composition of simple two-vapor (binary) mixtures. TBPO decomposition parallels this scenario. TBPO decomposition produces acetone and ethane in a 2:1 ratio, so the resulting compositions in the reactor are the same as if TBPO were being mixed with a 2:1 ratio of acetone and ethane. Since the Pirani gauge varies linearly with mole fraction in mixtures, we expect it to vary linearly with the mole fraction of TBPO during decomposition, which follows from the fractional conversion by equation (4.21).

To correlate the composition calculated by equation (4.21) from the pressure change to that derived from the Pirani gauge reading, we first calculated a leak-corrected Pirani gauge ratio (the ratio of the Pirani pressure to the Baratron pressure). Leak correction is implemented in equation (4.38) by subtracting the rate of pressure change caused by leak for the Pirani and Baratron gauges from their respective measured pressures,

$$c = \frac{p_{Pirani} - \left(\frac{dp_{Pirani}}{dt} \right)_{leak} \frac{T_{eff}}{T_{eff}^i} t}{p - \left(\frac{dp}{dt} \right)_{leak} \frac{T_{eff}}{T_{eff}^i} t} \quad (4.38)$$

where $\left(\frac{dp_{Pirani}}{dt} \right)_{leak}$ is the pressure change during leak-rate measurement for the Pirani gauge, c is the leak-corrected Pirani gauge ratio and p_{pirani} is the pressure measured by the Pirani gauge. Following the analogy to binary mixing, we propose equation (4.39) relates the Pirani gauge ratio, c to the mole fraction of TBPO in the reactor when pure TBPO is decomposing,

$$y_{TBPO} = \frac{c - c_{products}}{c_{TBPO} - c_{products}} \quad (4.39)$$

where *products* subscript refers to the degradation products of TBPO decomposition (2:1 acetone to ethane).

Figure 4.2 (b) shows that the composition predicted by equations (4.14) and (4.21), based on pressure change, exhibits a linear relationship with the Pirani gauge ratio ($r^2 = 0.96$), consistent with equation (4.39). On different days, the Pirani gauge reports slightly different pressures for the same vapor at identical conditions. This drift is likely caused by small quantities of polymeric material which have an outsized impact on the gauge reading. Drift is corrected in figure 4.2 (b)

by subtracting the Pirani gauge ratio of the fully decomposed TBPO ($c_{products}$) so that the data falls on a line. In figure 4.2 (b), c_{TBPO} ranged from 5.27 to 5.67, while $c_{products}$ ranged from 2.24 to 2.63. Figure 4.2 (b) suggests two primary conclusions. First, the Pirani gauge can be used to quantify the fractional conversion of TBPO during thermal decomposition. Second, the relationship between the Pirani gauge pressure and composition is given by equation (4.39). Notable curvature in some datasets is explained by contamination by leak and latency in the filament temperature readings. To ensure accurate results, the Pirani gauge is recalibrated before performing experiments, so c_{TBPO} and $c_{products}$ are determined from the Pirani gauge pressure of pure TBPO before performing experiments. The ability to measure TBPO fractional conversion with the Pirani gauge suggested by figure 4.2 (b) enables us to study the thermal decomposition of TBPO in steady-state continuous operation of the reactor, which is covered in Section 4.4.3.

4.4.2 Flow-through decomposition of pure TBPO

The strong correlation provided by equation (4.39) in figure 4.2 (c) validates the use of a linear model relating the mole fraction of unreacted TBPO to the Pirani gauge ratio. This section capitalizes on this relationship to extend the study of TBPO decomposition to the continuous mode. The continuous mode corresponds more closely to normal iCVD operation and provides explicit insight into the role flow rate and pressure play in TBPO decomposition during iCVD.

Though the Pirani gauge provides a reliable composition measurement during continuous operation of the reactor, we're ultimately interested in evaluating the kinetic model given by equation (4.15). The well-mixed assumption enables prediction of k_1^{eff} according to equation (4.28) to determine the mole fraction of TBPO in the reactor (y_{TBPO}). This is compared to the value calculated from c (the Pirani gauge ratio) using equation (4.39). To determine y_{TBPO} from c using equation (4.39) requires c_{TBPO} and $c_{products}$. The value for c_{TBPO} can be measured directly from the

unreacted TBPO, however determining $c_{products}$ accurately is less straight forward. When the filament is at its maximum temperature ($\sim 400^{\circ}\text{C}$), almost all TBPO in the reactor has been converted to acetone and ethane, so the gauge ratio at these conditions is a close approximation of $c_{products}$. However, because $c_{products}$ cannot be measured exactly, it is treated as a free parameter in future data fitting. The fitted values of $c_{products}$ were all found to differ by only about 0.05 from the measured value of c at the maximum filament temperature.

With a way to calculate k_1^{eff} from c , we must now determine how it depends on temperature, pressure and flowrate. In section 4.4.1, equation (4.16) was shown to accurately represent TBPO temperature dependence, however the dependence of A_1^{eff} and η_1 on pressure and flowrate was not investigated. Experimentally, we leveraged the Pirani gauge ratio to determine the dependence of thermal decomposition on filament temperature and chamber pressure. The measurements were performed with a TBPO flow rate of 1 sccm, spanning filament temperatures from 40-360°C, and chamber pressures from 50-300 mtorr. Using equations (4.28) and (4.39), we plot $\ln(k_1^{eff})$ against $1000/T_{fil}$, showing clear Arrhenius kinetics and no pressure dependence (figure 4.3 (a)). Pressure independence is notable as it indicates that for pressures above 50 mtorr, pressure-dependent heat transfer, which is often observed in rarefied gases, can be neglected.^{40,41} This supports a simplified view of heat transfer and TBPO thermal decomposition for iCVD under most deposition conditions and lends credence to the use of equation (4.36) for the effective temperature in Section 4.4.1. Pressure is held constant in all other flow-through experiments, so pressure dependence is not considered outside of figure 4.3 (a).

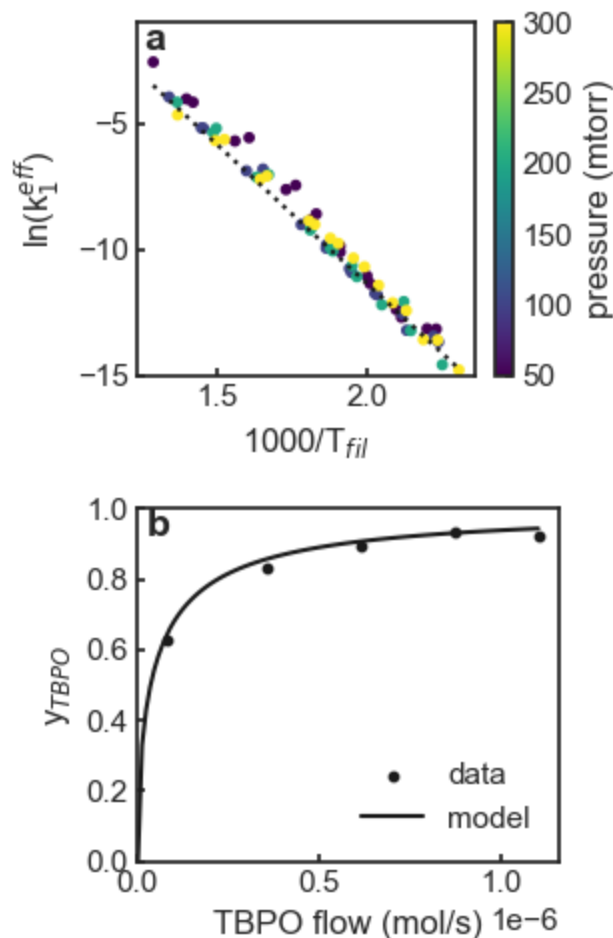


Figure 4.3. (a) Arrhenius plot for thermal decomposition of TBPO in flow-through model showing pressure independence. **(b)** Mole fraction of TBPO from equation (4.39) versus flow of TBPO into the reactor including a fit from equations (4.28) and (4.16) (solid line) where A' and η were determined from the fit performed in (a) showing the CSTR model describes flow-rate dependence of TBPO decomposition kinetics.

When we turn our attention to the temperature dependence of TBPO decomposition, an issue arises. As is standard in iCVD literature, we have measured the temperature of the filament array using a thermocouple. However, because the energy balance on the filament is extremely sensitive, and a thermocouple disrupts that energy balance in slightly different ways during

different experiments, temperatures reported across experiments are different (though they are consistent within experiments). The mechanisms by which errors are introduced primarily lower the measured temperature, therefore simply averaging many observations will not improve accuracy. Instead, the same TBPO decomposition is performed before every flow-through experiment (the 100mtorr condition in figure 4.3 (a)), assuming any differences in the Arrhenius parameters are caused by temperature bias, the temperature for all-experiments is recalibrated to a consistent temperature scale using equation (4.7) (methods described in Section 4.2.2 and 4.3.1). The conditions with the highest filament temperature (assumed to be least biased) were used for calibration (Section 4.3.1), having an apparent activation energy of 91.5 kJ mol^{-1} ($\eta_1 = 0.578$) and a pre-exponential factor $\ln(A_I^{eff})$ of $10.7 \text{ (m}^3 \text{ s}^{-1})$. The activation energy reported in literature for the thermal decomposition of TBPO at uniform temperature is $158.2 \text{ kJ mol}^{-1}$.²⁷ The depressed activation energy is primarily due to averaging over a range of temperatures - which is implicit in equations (4.15) and (4.16). We did not re-calibrate batch experiments since the calibration experiment was performed in the continuous mode, therefore the lower activation energy and larger pre-exponential factor in batch mode is likely caused by bias in temperature measurement. Having addressed the issue of measurement error, we can now move onto discussion of pressure and flow-rate dependence.

Pressure dependence of Arrhenius kinetics was investigated by comparing $\ln(k_I^{eff})$ versus $1000/T_{fil}$ for pure TBPO flowing at 1 sccm and 50, 100, 200 and 300 mtorr in figure 4.3 (a), showing they correspond closely. Comparing the Arrhenius parameters for the 100 mtorr case ($\eta_1 = 0.578$, $\ln(A_I^{eff}) = 10.7 \text{ (m}^3 \text{ s}^{-1})$) the aggregate data at all four pressures gives an r^2 value of 0.98, showing a strong fit.

While the fit to Arrhenius kinetics indicates that the well-mixed assumption is valid for the reactor (because of its use of equation (4.28)), the data was all collected at a single flow rate of 1 sccm, so verification over a range of flow rates is also warranted. To do this, the Pirani gauge pressure was measured for TBPO flow rates from 0.1-1.5 sccm (the maximum range our system is capable of) and used to determine the mole fraction of TBPO. The rate constant for the primary decomposition of TBPO was then calculated using equation (4.28). Figure 4.3 (b) shows mole fraction of TBPO plotted against flow-rate with model prediction from equation (4.28) and (4.16), where A_1^{eff} and η_1 were taken from the fit in figure 4.3 (a). This fit validates the use of the well-mixed assumption over the flow rates of interest. To quantify the effect of poor mixing and determine the onset of non-ideality, similar experiments should be performed at higher flow rates (see Chapter 3 for details on the effects of flow rates on mixing).

4.4.3 Relationship between Pirani gauge and composition

Batch TBPO decomposition data (Section 4.4.1) shows that a simple linear model relates the Pirani gauge ratio to the mole fraction of TBPO during decomposition. Additionally, we examined the dependence of TBPO decomposition on filament temperature, pressure, and flow rate under continuous flow-through conditions, mimicking deposition conditions. To fully understand the vapor phase during iCVD however, it is necessary to understand thermal decomposition of TBPO in the presence of monomer. Because CHMA is well studied in iCVD^{14,17} and is known to follow classical linear polymerization kinetics in solution,³⁵⁻³⁷ we have used it as a model monomer. This section extends equation (4.39) to incorporate the presence of monomer during TBPO decomposition, so that the Pirani gauge can be used to determine vapor-phase composition when CHMA is present in addition to TBPO. In the model, we hypothesize that CHMA does not react significantly with TBPO or any of its degradation products in the vapor

phase. Section 4.4.4 evaluates this hypothesis, and the model developed in this section against experimental data.

Mixing of CHMA and TBPO is directly measurable using methods discussed in Chapter 3, but warrants additional background on the operating principle of the Pirani gauge. The Pirani gauge measures heat loss from a fine wire, and is useful as a pressure sensor at low pressures where heat transfer is proportional to pressure.⁴³ At these conditions, the Pirani gauge coefficient for pure vapors does not vary with pressure, and is expected to vary linearly with mole fraction in binary mixtures.⁴⁴ For relatively small, weakly interacting molecules (IPA and argon) we have directly observed this linear relationship (Chapter 3). However, when we plot the Pirani gauge ratio against the inlet mole fraction determined by flow-rate calibration for CHMA and TBPO at 80, 100 and 120 mtorr (figure 4.4 (a)) the relationship is distinctly non-linear for the 100 and 120 mtorr conditions. These results were repeatable on separate days. The clear pressure dependence of the Pirani gauge ratio for pure CHMA (figure 4.4 (a)) suggests that the nonlinear relationship between composition and the Pirani gauge ratio is because the Pirani gauge is operating outside its ideal range.

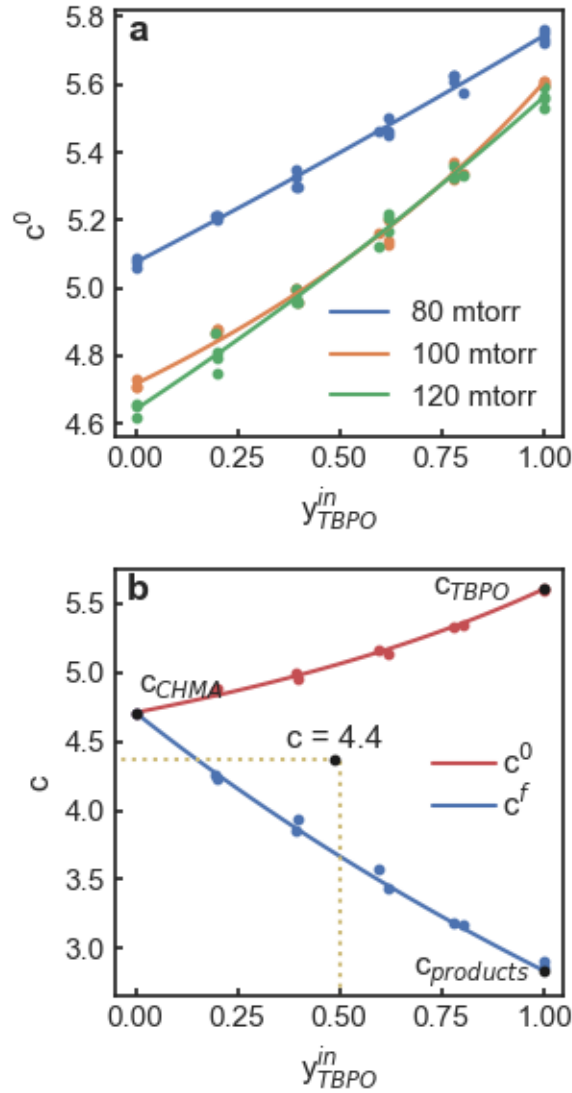


Figure 4.4. (a) Pirani gauge ratio versus mole fraction of TBPO for a range of total reactor pressures from 80 mtorr to 120 mtorr showing that the Pirani gauge ratio for pure CHMA varies significantly with pressure and that there is a non-linear relationship between Pirani gauge ratio and composition. Solid lines are fits from equation (4.40). (b) Pirani gauge ratio versus inlet TBPO mole fraction for unreacted TBPO with a fit of equation (4.40) in red and fully reacted TBPO with a fit of equation (4.41) in blue. Pirani gauge ratio for pure CHMA (c_{CHMA}), pure TBPO (c_{TBPO}), degradation products of TBPO ($c_{products}$) and a test point to show how equation (4.43) is used are also included in black and labeled.

We have found empirically that equation (4.40) describes the non-linearity observed in figure 4.4 (a) and has been included as a curve fit to data for all three pressures,

$$\frac{c^0 - c_{CHMA}}{c_{TBPO} - c_{CHMA}} = \frac{y_{TBPO}^{in}}{y_{TBPO}^{in} + D^0(1 - y_{TBPO}^{in})} \quad (4.40)$$

where D is a fitting coefficient and superscript ‘0’ indicates the quantity refers to mixtures of unreacted vapors, c_{CHMA} is the Pirani gauge ratio of pure CHMA, c_{TBPO} is the Pirani gauge ratio of pure TBPO, and y_{TBPO}^{in} is the mole fraction of TBPO at the inlet of the reactor, calculated from flow rate calibration. A fit of equation (4.40) to data at 100 mtorr is shown in figure 4.4 (a) in orange and figure 4.4 (b) in red with $D^0 = 1.53$. Figure 4.4 (b) also labels the points corresponding to c_{CHMA} , c_{TBPO} and $c_{products}$ for clarity.

Analogous to equation (4.40), a similar mixing curve applies to the reaction products of TBPO (2:1 acetone to ethane mixture) with CHMA. Correcting for the stoichiometry of TBPO decomposition, we obtain equation (4.41),

$$\frac{c^f - c_{CHMA}}{c_{products} - c_{CHMA}} = \frac{y_{TBPO}^{in,eff}}{y_{TBPO}^{in,eff} + \frac{D^f}{3}(1 - y_{TBPO}^{in,eff})} \quad (4.41)$$

where superscript ‘ f ’ indicates the quantity applies to the mixing of CHMA with the reaction products of TBPO, and $y_{TBPO}^{in,eff}$ is the effective inlet mole fraction of TBPO. When there is no deposition, $y_{TBPO}^{in} = y_{TBPO}^{in,eff}$, however $y_{TBPO}^{in,eff}$ needs to be adjusted when deposition is occurring. In the reactor mass balance, loss of monomer to deposition is equivalent to reducing the inlet monomer flow rate, so the same mixing model can be used (equation 4.41), provided that $y_{TBPO}^{in,eff}$ is adjusted for the loss of monomer to deposition using equation (4.42),

$$y_{TBPO}^{in,eff} = \frac{y_{TBPO}^{in} F_{tot}^{in}}{F_{tot}^{in} - r_p^{tot}} \quad (4.42)$$

where r_p^{tot} is the overall rate of polymerization in the reactor in mol s^{-1} . Use of equation (4.42) fully accounts for deposition in the mixing model. A fit of equation (4.41) to data at 100 mtorr and a stage temperature of 50°C (r_p^{tot} is zero) is shown in figure 4.4 (b) in blue with $D^f = 2.40$. Also note that the point corresponding to $c_{products}$ is labeled in figure 4.4 (b) for clarity.

Equations (4.40), (4.41), and (4.42) relate the Pirani gauge coefficient to composition for unreacted and fully reacted gases, but do not allow interpolation to intermediately decomposed states. Recall from Section 4.4.2 that the linear model of neat TBPO decomposition (equation (4.39)) was justified by considering that the evolution of the vapor-phase composition during the decomposition of TBPO is analogous to a binary mixing of TBPO and a 2:1 mixture of acetone to ethane. With the addition of monomer to the reactor, the change in c as TBPO decomposes is now influenced by the mole fraction of monomer in the reactor. A detailed model of these relationships requires further work, however we assume the relationship between the Pirani gauge ratio and the mole fraction of TBPO is approximately linear, such that it is described by equation (4.43).

$$y_{TBPO} = \frac{c - c^f}{c^0 - c^f} y_{TBPO}^{in,eff} \quad (4.43)$$

When pure TBPO is decomposing, $y_{TBPO}^{in,eff}$ is one, and equation (4.43) is identical to equation (4.39).

When taken together, equations (4.40)-(4.43) provide a model which relates the value of the Pirani gauge coefficient to the composition in the reactor for any mixture of CHMA and TBPO at any (accessible) filament temperature, provided c_{TBPO} , $c_{products}$, c_{CHMA} , D^0 and D^f , are known at

the pressure data is being taken at (100 mtorr in our case from figure 4.4 (a) and A(b)). The value of r_p^{tot} at each condition must be determined independently. We have not combined equations (4.40)-(4.43), since the result is cumbersome and does not provide additional insights. However, figure 4.4 (b) is included to show how these equations relate ‘ c ’ and y_{TBPO}^{in} to y_{TBPO} . The red line in figure 4.4 (b) represents c^0 and the blue line represents c^f . Consider the situation in which y_{TBPO}^{in} is known to be 0.5, c is measured to be 4.4 and there is no deposition occurring (shown by the labeled point in figure 4.4 (b)). First, since $r_p^{tot} = 0$, $y_{TBPO}^{in} = y_{TBPO}^{in,eff}$. To find y_{TBPO} , we see that the point of interest is mid-way between the red and blue lines, so we can use equation (4.43) to find $\frac{y_{TBPO}^{in,eff}}{y_{TBPO}} = 0.5$. Plugging in $y_{TBPO}^{in,eff}$, we find y_{TBPO} is 0.25 for this case.

4.4.4 Flow-through decomposition of CHMA / TBPO mixtures

The investigation of TBPO decomposition in the continuous flow-through mode in Section 4.4.2 validates the use of Arrhenius kinetics and the well-mixed assumption for TBPO and its decomposition products. However, these experiments don’t provide insight into the potential interactions between TBPO and CHMA occurring in the vapor phase. Understanding how TBPO and CHMA interact is paramount, as TBPO is always accompanied by monomer vapor during iCVD. Overall, we aim to test the null hypothesis that CHMA is present as an inert gas (neither reacts with TBPO, nor influences its Arrhenius kinetics) by comparing predicted values of the Pirani gauge ratio to measured values. The model developed in Section 4.4.3 extends the relationship between the Pirani gauge ratio and mole fraction of TBPO valid for decomposition of pure TBPO (equation 4.39) to TBPO decomposition in the presence of CHMA (equations (4.40)-(4.43)). This section extends the *kinetic* relationship for neat TBPO decomposition (equation 4.28) to TBPO decomposition in the presence of CHMA. Doing so means that the Pirani gauge ratio during TBPO decomposition in the presence of CHMA can be predicted by combining the

relationship between the Pirani gauge and composition (equations (4.40)-(4.43)), and the kinetic model discussed in Section 4.3.3. This prediction allows us to test if CHMA can be described as an inert gas and provides a useful model for predicting vapor composition.

Extending the kinetic model for neat TBPO to mixed TBPO and CHMA is relatively simple, only requiring additional terms for CHMA flow into and out of the reactor and consumption of monomer by deposition. Combining the rate equations given in scheme 1 and the vapor-phase mole balance on the reactor yields equation (4.29) for the mole fraction of TBPO. We tested the validity of equations (4.29), (4.40)-(4.43) (and by extension the hypothesis that CHMA is inert) by considering the decomposition of TBPO / CHMA mixtures at a stage temperature of 50°C such that negligible deposition occurs (i.e. there is no visible build-up of polymer on any surfaces in the reactor). Equation (4.29) can then be used to predict y_{TBPO} based only on the inlet mole fraction of TBPO and the rate constant for primary decomposition determined from Arrhenius kinetics for pure TBPO (Section 4.4.2). Equations (4.40)-(4.43) from Section 4.4.3 predict the Pirani gauge ratio from the TBPO mole fraction, enabling comparison with data. To validate equation (4.29), we measured Pirani gauge ratios for the full range of inlet TBPO mole fractions (0 to 1), and the full range of filament temperatures accessible to our system (25-450°C). Together with the flow rate and pressure dependence established in Section 4.4.2, this covers the common vapor phase conditions in iCVD.

Figure 4.5 (a) plots the expected Pirani gauge ratio against the measured gauge ratio, showing an excellent correspondence. While mass spectrometry or FTIR is needed for further confirmation, there is no evidence in our data suggesting appreciable interaction between CHMA and TBPO. Note that the prediction in figure 4.5 (a) relies on parameters derived from three earlier fits, one in Section 4.4.2 to determine Arrhenius kinetics of pure TBPO decomposition, and the

other two from fits of equations (4.40) and (4.41) discussed in Section 4.4.3 to obtain D^0 and D^f . The fit for D^0 was performed by using a gradient descent method to minimize the mean squared error between the data and prediction in figure 4.4 (a), while the fit for D^f was performed by assuming that TBPO was fully decomposed at the highest achievable temperature and performing an analogous fit of equation (4.41) to the Pirani gauge data for the fully reacted state. The values of c_{CHMA} , c_{TBPO} , $c_{products}$, D^0 , and D^f used in figure 4.5 (a) were 4.71, 5.61, 2.86, 1.53, 2.40, respectively. The relationships between these quantities can be seen clearly in figure 4.4 (b).

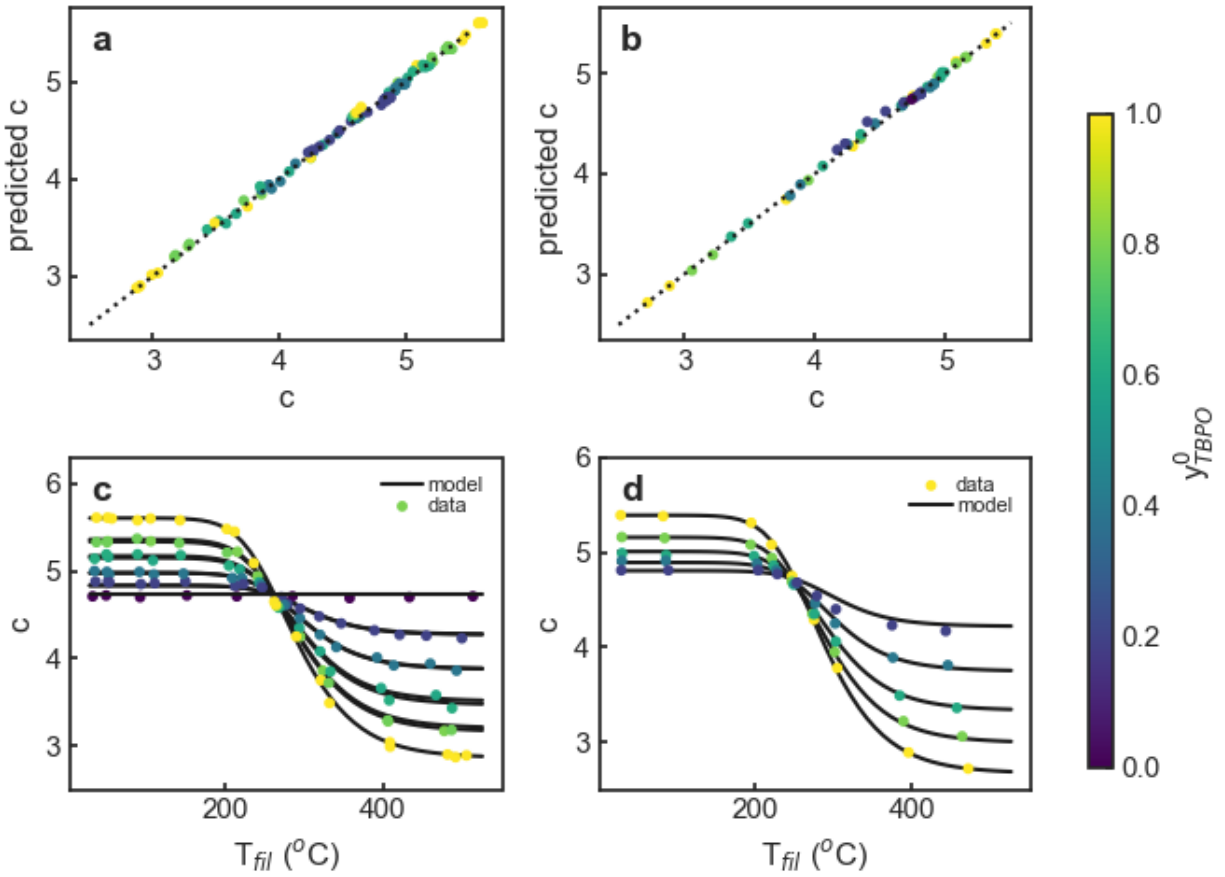


Figure 4.5. (a) Predicted Pirani gauge ratio (using theory from section 4.3.3) versus actual Pirani gauge ratio for thermal decomposition of TBPO in the presence of CHMA at stage temperature of 50°C assuming no deposition or vapor-phase reactions other than TBPO decomposition (b)

Predicted Pirani gauge ratio (using theory from Section 4.3.3) versus actual Pirani gauge ratio for thermal decomposition of TBPO in the presence of CHMA at stage temperature of 30°C assuming no gas phase reactions other than TBPO decomposition or deposition, showing a slight deviation for low inlet TBPO mole fraction **(c)** Measured Pirani gauge ratio (color dots) versus filament temperature for data taken at a stage temperature of 50°C; model prediction (solid lines) was obtained using , which assumes no gas phase reactions other than TBPO decomposition and that deposition does not affect vapor **(d)** Measured Pirani gauge ratio (color dots) versus filament temperature for data taken at a stage temperature of 50°C; model prediction (solid lines) was obtained using theory from Section 4.3.3, which assumes no gas phase reactions other than TBPO decomposition and that deposition does not affect vapor.

While prior literature has theorized that the rate of deposition is much slower than the flow of monomer into the reactor,⁴⁵ the Pirani gauge enables a direct test of this by observing deviation from equation (4.29) when the Pirani gauge ratio is measured under conditions which favor deposition. When we reduce the stage temperature to 30°C (Figure 4.5(b)) we see slight disagreement between the predicted gauge ratio and the measured gauge ratio at an inlet monomer fraction of 0.8 The values of C_{CHMA} , C_{TBPO} , $C_{products}$, D^0 , and D^f used in figure 4.5 (b) were 4.74, 5.39, 2.67, 2.19, and 2.19 respectively.

To help visualize the data in figure 4.5 (a) and (b), we have also plotted the Pirani gauge ratio versus filament temperature for both cases in figure 4.5 (c) and (d). Figure 4.5 (c) shows excellent agreement over all conditions between the model and data. As expected, figure 4.5 (d) shows slight underprediction of the Pirani gauge ratio under conditions of high monomer partial pressure, and high filament temperature, which are the conditions which favor deposition.

Disagreement between data and prediction in figure 4.5 (b) and (d) illustrates that deposition likely impacts the vapor phase composition, but it is not immediately clear whether the effect on the vapor phase is significant. Furthermore, figure 4.5 does not show how the vapor phase model discussed in Section 4.3.3 compares with existing methods. Because the magnitude of the impacts of deposition and TBPO decomposition on vapor phase compositions directly motivates the use of vapor phase models, further analysis is warranted. For clarity, and to minimize the impact of measurement error, we will focus on conditions which most favored deposition, 30°C stage temperature (figure 4.5 (b) and (d)) and inlet TBPO mole fraction of 0.2. Because of the conclusive importance of the monomer partial pressure in iCVD literature³, we will compare monomer mole fraction predicted using 3 methods to demonstrate the relative impact of vapor phase processes. The current mainstay method of predicting vapor phase composition (monomer composition) in iCVD assumes the composition is equal to that of the reactor feed regardless of filament temperature³. This method neglects the stoichiometry of TBPO decomposition and the impact of deposition. The mole fraction of CHMA predicted using the standard method is represented by the dotted line in figure 4.6.

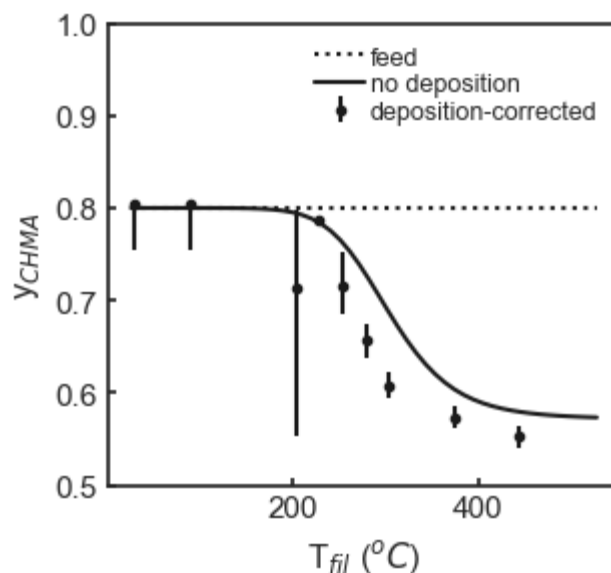


Figure 4.6. Plot showing the mole fraction of CHMA predicted by the feed composition alone, accounting for stoichiometry of TBPO decomposition, but not deposition, and accounting for deposition quantified by the experimentally obtained deviations from the prediction in figure 4.5 (b) and (d).

Correcting for TBPO decomposition using equations (4.29) and (4.32), with r_p^{tot} equal zero yields the solid line in figure 4.6, showing significantly lower monomer mole fraction (by up to 0.2) compared with the standard method.

To estimate the impact of deposition on the vapor phase composition, we first predict the deposition rate necessary to cause the deviation between the measured Pirani gauge ratio and the model prediction seen in figure 4.5 (b) and (d). To accomplish this, a gradient descent algorithm varied r_p^{tot} to minimize the difference between y_{TBPO} calculated from the Pirani gauge reading (using equations N, M, P and T) and y_{TBPO} calculated from equation (4.29) (derived using TBPO decomposition kinetics and species balance) until a difference of less than 10^{-5} was achieved. The values of r_p^{tot} obtained represented at most 25% of the inlet monomer flow-rate, or 0.2 sccm of monomer lost to deposition (for the 0.8 sccm CHMA flow). To illustrate the degree to which

deposition removes monomer from the vapor phase, we calculate the mole fraction of monomer in the vapor from equation (4.32) for an inlet TBPO mole fraction of 0.2. The deposition-corrected CHMA mole fraction from the Pirani gauge data obtained at low-stage-temperature (30°C) is shown by data points in figure 4.6, highlighting that the potential effects of deposition on the mole fraction of monomer in the reactor may be significant, further reducing the mole fraction by 0.05 in this case.

Error bars are based on measurement uncertainty for the Pirani gauge determined from spread of datapoints in figure 4.4, giving error in c of ± 0.02 . Note that the large error for data taken at a filament temperature of 200°C (figure 4.6) occurs because the Pirani gauge reading is relatively composition-independent and will have the same value for a range of compositions (figure 4.5 (c) and (d) shows this clearly). This illustrates the limitation of the Pirani gauge for handling uncertainty in inlet composition at this filament temperature. If the feed composition is known with good confidence however, the Pirani gauge remains an effective metric for quantifying vapor phase composition using equations (4.40)-(4.43).

Figure 4.6 enables two notable insights. First, the standard method of approximating the partial pressure of monomer from the feed composition leads to significant errors at conditions relevant to iCVD because it neglects the stoichiometric expansion of decomposing TBPO. Second, the effect of deposition (estimated using the method discussed above) is much smaller, but still non-negligible.

4.4.5 Deposition Kinetics

One of the primary motivations in developing the model presented in Sections 4.4.2-4.4.4, is to help improve our understanding of the polymerization process occurring in iCVD. Therefore,

this section further expands the model of the vapor phase to include radical species and develops a simple surface polymerization model which can be compared with measured deposition rates.

Because the characterization of deposited polymer requires use of FTIR and ellipsometry, some discussion of these methods is needed. Figure 4.7 shows the FTIR of CHMA monomer and polymer, showing the disappearance of the vinyl bond peak at around 1700 cm^{-1} indicating polymerization has occurred. The FTIR spectrum for monomeric CHMA was obtained using attenuated total reflectance (ATR), while the poly(CHMA) spectrum was collected in transmission mode. ATR causes distortion in relative peak areas⁴⁶, which likely causes the apparent increase in intensity of C-H peaks at 3000 cm^{-1} between monomer and polymer spectra. A more detailed FTIR peak assignment and composition analysis for poly(CHMA) can be found in Xu and Gleason (2011)¹⁴.

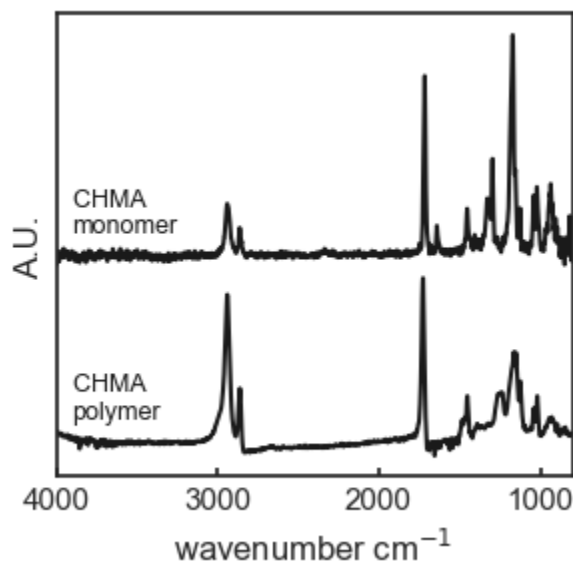


Figure 4.7. FTIR absorbance spectra for CHMA monomer and CHMA polymer showing the disappearance of the vinyl peak at 1700 cm^{-1} consistent with complete polymerization.

Deposition rate data was determined for CHMA at a range of inlet mole fractions (y_{TBPO}^{in}) from 0.1 to 0.8 and filament temperatures from 200°C to 350°C . Deposition conditions including

p_m/p_{sat} for each condition may be found in Table 4.1. Above an inlet TBPO mole fraction of 0.6, deposition was slow, and many samples showed a foggy appearance, which may be caused by low molecular-weight poly(CHMA). The inlet composition used in calculations was determined from flow calibration, however the Pirani gauge ratio could also be used to determine composition (see figure 4.4 (a)). Figure 4.8 (a) plots the composition derived from the Pirani gauge ratio against the TBPO inlet composition for all depositions. All points were taken at 100 mtorr and at a stage temperature of 30°C, so they have been compared to the model prediction of Equation (4.40), using a value of D^0 (2.19) corresponding to 30°C stage temperature experiments (figure 4.5 (b) and (d)). The values of c_{CHMA} and c_{TBPO} were measured on the same day and are reported along with other deposition conditions in Table 4.1. While the fit in figure 4.8 (a) is reasonably good, deviation between the model and the data may be due to insufficient equilibration time.

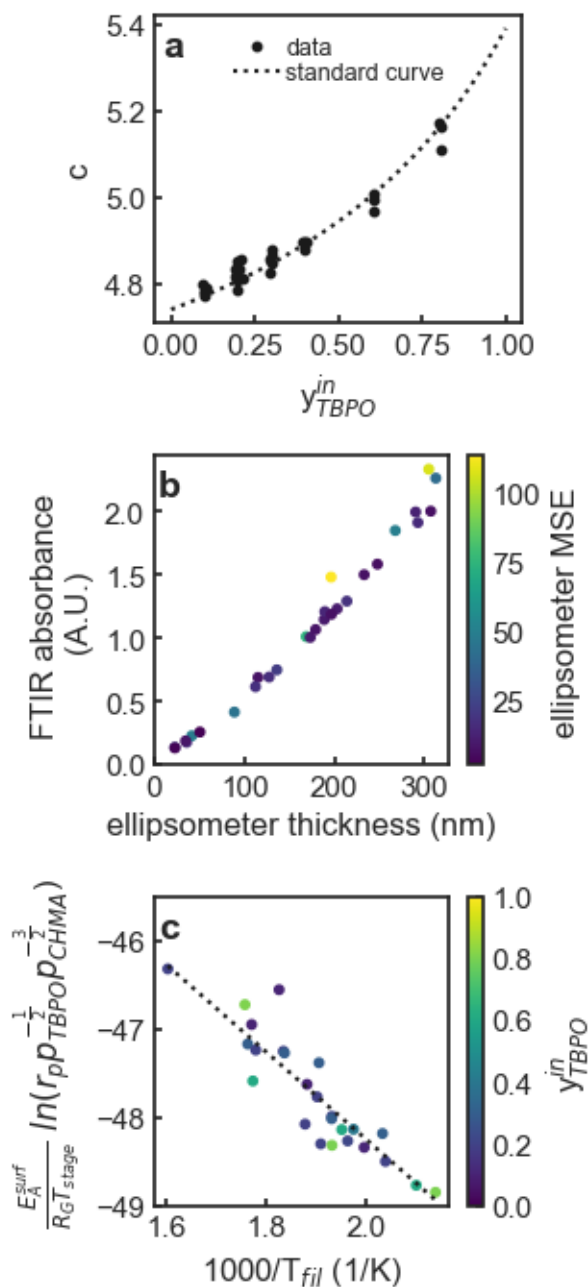


Figure 4.8. (a) Pirani gauge ratio versus inlet mole fraction of TBPO aligns well with predicted value from figure 4.4 (a) (100 mtorr case). Deviation from the model may be caused by insufficient equilibration time before measurement. (b) Dimensionless FTIR absorbance versus thickness measured by ellipsometer showing a clear non-linear relationship, indicative of density variation

with thickness, high MSE correlates with deviation. **(c)** Arrhenius plot for deposition rate constant based on equations (4.46) and (4.47).

Since deposition rate is obtained by dividing the total quantity of accumulated polymer (per unit area) by the deposition time, an accurate measurement of film mass is needed. If film density is constant and equal to the bulk polymer density, then film thickness (from ellipsometry) can be used to estimate deposition rate, however this method is subject to error since polymer thin film density often varies with film thickness,⁴⁷ and film non-ideality can make ellipsometric measurements unreliable. To avoid these issues, FTIR absorbance, which is directly proportional to the area density of polymer sub-units (polymer mass) by the Beer-Lambert law, was used to determine deposition rate instead. Section 4.2.7 details the procedures for obtaining FTIR absorbance (in arbitrary units). Figure 4.7 (b) shows the plot of thickness measured by ellipsometry versus FTIR absorbance, showing apparent curvature for films less than 200 nm. This indicates that density increases with film thickness, a well-known phenomenon caused by surface excesses,⁴⁸ though variability in molecular weight may play a small role as well. Deposition rate in $\text{mol s}^{-1} \text{m}^{-2}$ was calculated from FTIR absorption using the mean absorption per nm of thickness for films thicker than 250 nm with ellipsometer MSE less than 150, molecular weight of CHMA monomer (168.2 g mol^{-1}) and the density of bulk poly(CHMA), taken as 1.1 g ml^{-1} based on supplier specifications for commercially available poly(CHMA) to give a conversion factor of $9.38 \times 10^{-8} \text{ mol absorbance}^{-1} \text{ cm}^{-2}$.

We start our effort to model deposition with classical linear polymerization theory. Classical linear free radical polymerization kinetics (i.e. no branching or crosslinking) is well established in the solution phase,³⁸ where it has been demonstrated to apply to free radical polymerization of CHMA.^{36,37} While significant differences likely exist between iCVD

polymerization and solution phase polymerization, existing studies of iCVD kinetics which propose complete quantitative models appeal to a model of the vapor phase which is incompatible with the results in Section 4.4.4, and used monomers which do not follow classical linear polymerization theory in solution^{5,8,9} (e.g., the monomer used is prone to branching^{49,50}). As such, these models are not suitable for our system. The kinetic equation for the rate of polymerization derived from classical linear free radical polymerization in solution can be adapted for iCVD by substituting surface concentrations instead of volumetric concentrations (similar assumptions are frequently made in kinetic models of iCVD⁵) to give equation (4.44),

$$r_p = k_p \sqrt{\frac{k_i}{k_t}} [I^*]^{\frac{1}{2}} [M]^{\frac{3}{2}} \quad (4.44)$$

where r_p is the rate of polymerization (deposition) in $\text{mol s}^{-1} \text{ cm}^{-2}$, k_p is the propagation rate constant in $\text{cm}^2 \text{ mol}^{-1} \text{ s}^{-1}$, k_i is the initiation rate constant in $\text{cm}^2 \text{ mol}^{-1} \text{ s}^{-1}$, k_t is the termination rate constant in $\text{cm}^2 \text{ mol}^{-1} \text{ s}^{-1}$, $[I^*]$ is the surface concentration of initiating radicals in mol cm^{-2} , and $[M]$ is the surface concentration of monomer in mol cm^{-2} . Details of the derivation may be found in Odian³⁸.

While TBO* and Me* radicals are both present in the reactor, we assume that $[I^*]$ refers to TBO* radicals in equation (4.44). TBO* are frequently assumed to play a greater role in iCVD than Me* radicals due to their lower volatility.⁶ This idea is supported by data showing that an initiator with lower volatility than TBPO (*tert*-butyl peroxy-benzoate) has higher initiating efficiency than TBPO.¹⁴ Furthermore, study of iCVD in nanoconfinement has found the initiator volatility ($p_i/p_{i,sat}$), necessary to describe kinetics.¹² In systems which are similar to iCVD, such as photo-initiated polymerization of butadiene vapor by TBPO, a strong preference for TBO* over

Me* radicals has also been observed.^{29,51} While future work is needed to directly evaluate the factors which might drive preference for different initiators in iCVD, the existing work discussed provides reasonable support for our assumption.

To implement equation (4.44) in iCVD, relationships between partial pressure and surface concentration are needed for CHMA and TBO*. While the BET isotherm is widely believed to describe monomer surface concentration in iCVD^{3,17}, and the Eley-Rideal model for initiator suggests a kinetic relationship like the Hertz-Knudsen equation,⁵² both these equations can be approximated by a direct proportionality according to equation (4.45),

$$[X] = k_X p_X \quad (4.45)$$

where $[X]$ is the surface concentration of species X in mol cm⁻² (CHMA or TBO*), p_X is the partial pressure in Pa and k_X (either k_{CHMA} or k_{TBPO}) is the proportionality-constant relating partial pressure to surface concentration in mol cm⁻² Pa⁻¹.

When we substitute equation (4.15) for r_1^{tot} , and equation (4.33) for r_β^{tot} , we obtain equation (4.35). Polymerization rate can now be obtained by substituting equations (4.45) and (4.35) into equation (4.44) to obtain equation (4.46),

$$r_p = k_{surf} p_{TBPO}^{\frac{1}{2}} p_{CHMA}^{\frac{3}{2}} \quad (4.46)$$

where $k_{surf} = k_p \sqrt{\frac{2k_t k_{TBO*} k_{CHMA}^3 k_1^{eff}}{k_t k_\beta^{eff}}}$ is a lumped rate constant for deposition. Note that k_p , k_i , k_t ,

k_{TBO*} and k_{CHMA} all depend on stage temperature, while k_1^{eff} and k_β^{eff} depend on filament temperature, allowing us to consider these dependences independently. The work of Chan and Gleason¹⁷ investigates the stage temperature dependence of CHMA deposition with TBPO and

their data suggests an approximate Arrhenius dependence over the temperature range of interest, such that the overall temperature dependence of k_{surf} can be approximated by equation (4.47),

$$k_{surf} = A_{surf} \exp \left(-\frac{E_A^{surf}}{R_G T_{stage}} - \frac{E_{A,1}\eta_1 - E_{A,\beta}\eta_\beta}{2R_G T_{fil}} \right) \quad (4.47)$$

where A_{surf} is an empirical pre-exponential factor with units of $\text{mol cm}^{-2} \text{s}^{-1} \text{Pa}^{-2}$ and E_A^{surf} is the apparent activation energy describing the stage temperature dependence of deposition. Fitting the data of Chan and Gleason suggests an apparent activation energy (E_A^{surf}) of 51 kJ mol^{-1} . An Arrhenius fit to the data of Chan and Gleason is shown in figure 4.9, showing an acceptable r^2 value of 0.97. Details on the mechanism underlying stage temperature dependence, and a more precise model of stage temperature dependence are contained in Chan and Gleason.¹⁷

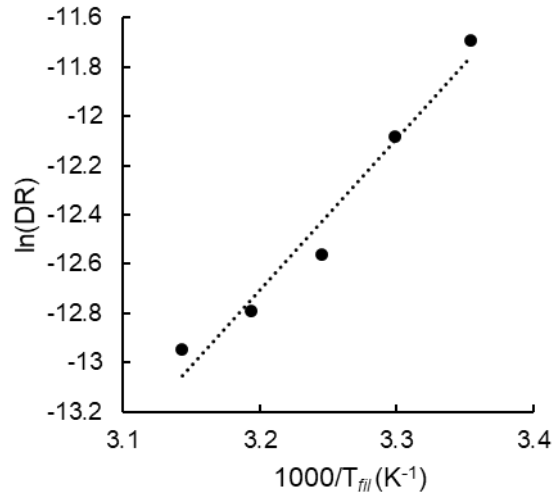


Figure 4.9. Stage temperature dependence fit from data of Chan and Gleason.

By re-arranging equation (4.47) and (4.46), the quantity $r_p p_{TBP}^{-\frac{1}{2}} p_{CHMA}^{-\frac{3}{2}} \exp \left(\frac{E_A^{surf}}{R_G T_{stage}} \right)$ is expected to follow an Arrhenius relationship in the filament temperature. Figure 4.7 (c) shows a

plot of $\frac{E_A^{surf}}{R_G T_{stage}} \ln \left(r_p p_{TBPO}^{-\frac{1}{2}} p_{CHMA}^{-\frac{3}{2}} \right)$ versus $1000/T_{fil}$, showing a roughly linear trend ($r^2 = 0.77$).

Deposition rate (r_p) was determined from FTIR absorption using the methods discussed above, p_{TBPO} was estimated using equation (4.29), and p_{CHMA} was estimated using equation (4.32). The slope of the linear fit shown in figure 4.7 (c) corresponds to an apparent activation energy of 41.0 kJ mol⁻¹, and a pre-exponentiation factor $\ln(A_{surf})$ of -38.37 (mol Pa⁻² s⁻¹ cm⁻²). Comparison with the apparent activation energy for primary decomposition observed in Section 4.4.2 indicates that the apparent activation energy of β -scission is roughly 13.3 kJ mol⁻¹ ($\eta_\beta = 0.26$), significantly smaller than the literature value of 51 kJ mol⁻¹.⁵ To explain this, we consider that at uniform temperature, β -scission is much less temperature sensitive than primary decomposition. This means that in the reactor, it will not be as locally constrained to the filament array as primary decomposition, and the temperature of the filament will be a much poorer descriptor of the temperature at which β -scission is most frequently occurring (though still the closest we can measure). Further analysis of this effect would require theoretical or computational efforts to solve the simultaneous heat transfer, reaction and diffusion problem at the array. The notable scatter in the data shown in figure 4.8 (c) is due to experimental error and suggests that this work should be taken as a preliminary indication of how the vapor phase modeling can be implemented in deposition models. Further work is needed, which will likely find it necessary to modify the simple deposition model proposed here.

While the fit shown in figure 4.8 (c) is effective for relating the vapor phase model we propose to deposition at a range of compositions and temperatures, it obscures how the proposed model relates to past work. Without the model of the vapor phase we develop, past efforts to investigate the dependence of iCVD on filament temperature have plotted deposition rate directly on Arrhenius axes. It was observed in many studies that at high filament temperatures (>350°C)

the Arrhenius fit begins to fail, but no explanation was forthcoming.^{6,14} This type of plot is presented in figure 4.10, where figure 4.10 (a) reproduces our data for CHMA deposition with TBPO from figure 4.8 (c) at y_{TBPO}^{in} of 0.2, including the fit from equation (4.46). Note that the y-axis is identical to figure 4.8 (c), except that the dependence on partial pressures of monomer and initiator are removed. As a comparison with existing work, we have also reproduced the data of Xu and Gleason¹⁴ for deposition of CHMA with TBPO at a range of filament temperatures, showing the characteristic kink at high temperature.

Because the reactor system used by Xu and Gleason is not known to us and their process conditions differed somewhat from ours (both factors which impact TBPO decomposition), when fitting equation (4.46) to their data, η_1 , A_1^{eff} , η_β and A^{surf} were all varied using a gradient descent algorithm to minimize the mean squared error between the model and prediction. To obtain the solid line shown, fit values of 0.92, $8.1 \times 10^9 \text{ m}^3 \text{ s}^{-1} \text{ mol}^{-1}$, 0.20 and $1.68 \times 10^{-8} \text{ mol Pa}^{-2} \text{ s}^{-1} \text{ cm}^{-2}$ were obtained for η_1 , A_1^{eff} , η_β and A^{surf} respectively. The larger value of η_1 is likely due in part to difference in temperature measurement, while the value for η_β matches well with our findings.

Qualitatively, the kink occurs at high temperature because of the decomposition and mixing processes modeled in Sections 4.4.1 – 4.4.4. When the rate of TBPO decomposition is relatively small, it is primarily determined by the filament temperature and since the rate of TBPO decomposition is proportional to the partial pressure of radicals (equation BF), deposition follows an Arrhenius dependence under these conditions. However, when the rate of TBPO decomposition reaches the same order of magnitude as the total TBPO flow into the reactor, the rate of decomposition becomes limited by the amount of TBPO flowing into the reactor and it begins to plateau. When the rate of TBPO decomposition plateaus, so does the deposition rate, causing the kink shown in figure 4.10. It is important to note that this phenomenon cannot be explained without

the vapor phase model we present in this paper, demonstrating that this added understanding is a marked advance in our understanding of the iCVD process.

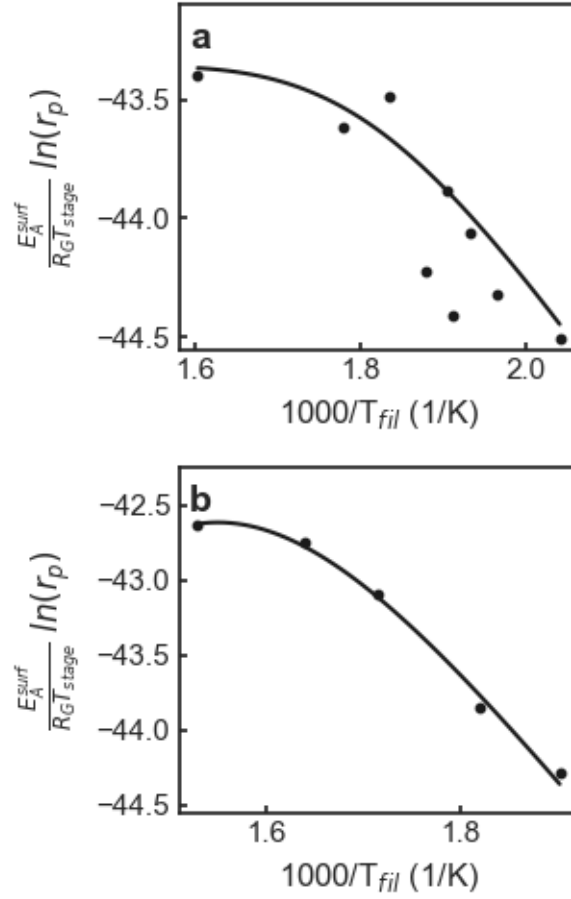


Figure 4.10. (a) Arrhenius plot of deposition rate data collected at y_{TBPO}^{in} of 0.2 corrected for effects of changing stage temperature by data of Chan¹⁷, plotted against the model fit of equation (4.46), which is also given as the dotted line in figure 4.8 (c). **(b)** Arrhenius plot of deposition rate data from Xu and Gleason¹⁴, similarly corrected for the effects of changing stage temperature and plotted against the model fit of equation (4.46).

4.5. Conclusions

This study aims to provide a thorough characterization of the vapor phase during iCVD and relate this to the deposition rate. By elucidating the fundamental processes occurring in the vapor phase, we hope our work can lay the groundwork for stronger models of the surface processes in iCVD.

Quantifying composition in the vapor phase begins with a thorough characterization of TBPO decomposition. During batch decomposition of TBPO, the change in reactor pressure can be used to determine the fractional conversion of TBPO. We use this to show that TBPO undergoes first-order decomposition with an Arrhenius temperature dependence in agreement with the literature.^{27–29} This observation suggests that the rate of decomposition is limited by primary decomposition, and that the radical species capable of initiating polymerization (TBO* and Me*) are only present in trace amounts. When we plot the mole fraction of TBPO against the Pirani gauge ratio, we observe a consistent linear relationship. When pure TBPO is decomposing, this relationship allows us to determine real-time composition in the reactor from the Pirani gauge pressure.

The ability of the Pirani gauge to sense composition allows us to study the decomposition of TBPO as it flows through the reactor at steady-state, mirroring normal iCVD operation. Comparing composition determined by the Pirani gauge to the well-mixed assumption closely predicts decomposition kinetics over a range of inlet flowrates. Varying temperature and pressure showed that decomposition follows Arrhenius kinetics, with very little pressure dependence above pressures of 50 mtorr. The pressure independence of kinetics is consistent with heat transfer experiments performed by Bakker⁴¹ in which it was found that heat transfer from the array in TBPO vapor varied little above 50 mtorr. While we have observed significant variability in

temperature measurement, we estimate TBPO decomposition in our reactor to have an apparent activation energy of 91.5 kJ mol^{-1} ($\eta_1 = 0.578$) and $\ln(A_1^{eff})$ of $10.7 \text{ (m}^3 \text{ s}^{-1}\text{)}$. The difference between these kinetics and those at uniform temperature (discussed in the literature) arise because our kinetics (equation A) are averaged over the non-uniform conditions of our reactor. Future efforts devoted to solving the heat and mass transfer problem close to the array are needed to accurately predict deposition from first principles.

Steady-state flowthrough decomposition studies were also performed with mixed TBPO and CHMA at a range of mole fractions. At a stage temperature of 50°C , when there is no deposition, a model of the Pirani gauge ratio based only on the pure TBPO mixing, and unreacted CHMA/TBPO mixing is an excellent predictor of the Pirani gauge ratio. The excellent fit of the well-mixed model following first order Arrhenius kinetics to the measured Pirani gauge ratio provides no evidence to suggest monomer and initiator interact in the vapor phase, however this possibility is not ruled out by our data. Since the monomer partial pressure is well established as a key parameter in iCVD kinetics, we compared the performance of the model we've presented, with the existing methods to show that significant error is incurred when the processes we've studied are ignored (up to 20% for the case plotted). When combined TBPO and CHMA decomposition experiments were performed at a stage temperature of 30°C , slight overprediction of the Pirani gauge ratio was observed at conditions which favor deposition. This overprediction was used to estimate the overall rate of deposition in the reactor, and the effect of this on the monomer composition showed up to a 5% bias on the mole fraction of monomer in the reactor. Such an effect could be important if a precise model is needed.

To relate the proposed model of the vapor phase to deposition itself, depositions were performed at a range of filament temperatures and inlet mole fractions. Deposition rates were fitted

to a simple model of linear (i.e. no branching) free radical polymerization in which TBO* was assumed to be the primary initiating species and stage temperature dependence was corrected using the data of Chan and Gleason.¹⁷ This model fit data reasonably well, however there was significant experimental error. Based on this fitting, we were able to estimate the effective activation energy for β -scission in the reactor, which was 13.3 kJ mol^{-1} ($\eta_\beta = 0.26$), suggesting that β -scission is much less localized to the array than primary decomposition. Importantly, our simple deposition model predicts a well-known decrease in the activation energy of iCVD deposition at high filament temperature (300°C),^{6,14} illustrating its necessity in predicting deposition rate. To show this, we compare our deposition model to data from Xu and Gleason studying the same system,¹⁴ showing an excellent fit.

Though the use of thermocouples for measuring filament temperature is standard in iCVD, the issues with repeatability in this study suggest a broader issue within the iCVD literature. In recognition of this, there is a need to develop non-contact temperature measurement based on overall resistance of the filament or an infrared thermometer. Neither overall resistance nor radiation-based temperature measurements are established in iCVD and will require a dedicated effort to develop and verify. A good implementation of a contact-free filament temperature measurement would allow for consistency in filament temperature measurements across research labs, and reactors, significantly improving progress by ensuring published data can be compared.

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CHAPTER 5:

HYDROLYSIS OF POLY(FLUOROACRYLATE) THIN FILMS SYNTHESIZED FROM THE VAPOR PHASE

Abstract

Post-synthesis surface reaction of vapor deposited polymer thin films is a promising technique in engineering heterogeneous surface chemistry. Because existing research has neglected marginally reactive precursor films in preference of their highly reactive counterparts, our knowledge of kinetics and loss of film integrity during reaction are limited. To address these limitations, we characterize hydrolysis of two fluoroacrylates, poly(1H,1H,2H,2H-perfluorooctyl acrylate) (pPFOA) and poly(2,2,3,4,4,4-hexafluorobutyl acrylate) (pHFBA), with sodium hydroxide using x-ray photoelectron spectroscopy. Without crosslinking with di(ethylene glycol) divinyl ether (DEGDVE) and grafting with trichlorovinyl silane, the films degrade rapidly during hydrolysis. An S_N2 mechanism describes hydrolysis well, with rate constants of $0.0029 \pm 0.0004 \text{ L mol}^{-1} \text{ s}^{-1}$ and $0.011 \pm 0.001 \text{ L mol}^{-1} \text{ s}^{-1}$ at 30°C for p(PFOA-co-DEGDVE) and p(HFBA-co-DEGDVE) respectively. Our detailed study of hydrolysis kinetics of marginally reactive fluoroacrylates demonstrates the full capability and limitations of post-synthesis reaction. Importantly, copolymers are characterized using a density correction new to polymer chemical vapor deposition.

5.1. Introduction

Polymer coatings and thin films are frequently used to modify surface chemistry^{1,2}. Fluoro-acrylic polymers often find applications for their low dielectric constant, low friction coefficient, low refractive index, low surface energy, low flammability, inertness to chemicals, and interesting repellency of both polar and non-polar liquids³. They are often used as coatings to prevent weathering⁴ and other forms of aging. Two representative fluoroacrylates are poly(1H,1H,2H,2H-perfluorooctyl acrylate) (pPFOA) and poly(2,2,3,4,4,4-hexafluorobutyl acrylate) (pHFBA), which have both found applications as self-cleaning hydrophobic coatings^{5,6}. The base-catalyzed hydrolysis of these fluoro-acrylates is of interest for three reasons. First, polymers with long perfluorinated side chains (like pPFOA) are unique for their low surface energy but can introduce bio-persistent, carcinogenic perfluorinated acids (PFAs) into the environment (notably 8 carbon PFAs)⁷. Therefore, understanding the hydrolysis of fluoro-acrylates is important in estimating environmental release of PFAs from polymeric sources. Second, the partial hydrolysis of fluoro-acrylates may lead to novel amphiphilic surfaces, which have been useful in antifouling applications⁸. Finally, hydrolysis of fluoro-acrylic polymers has interesting applications in the fabrication of spatially non-uniform coatings. These applications are the primary motivation for our study and warrant a more in-depth discussion.

Techniques which enable in-plane surface chemistry non-uniformity (distinct from film non-uniformity normal to the surface) are a significant development in our ability to engineer surfaces. Due to their hydrophobicity, fluoropolymer thin films are difficult to synthesize using traditional methods, so this discussion of film surface chemistry non-uniformity will be limited to polymer chemical vapor deposition (CVD) techniques (in-depth discussion of polymer CVD will follow). Existing polymer CVD research focuses on three main applications of in-plane surface

chemistry non-uniformity, Janus membranes^{9–12}, templated (or patterned) surfaces^{13–15} and chemical gradients^{16–18}. Janus membranes are valuable in applications such as membrane distillation, water collection in arid environments, breathable fabric for activewear, and wound dressings with improved blood barrier and wound protection properties^{19,20}. Templating is used to produce patterns of two contrasting chemistries, which may be used for drug release, biosensors and artificial skins^{13–15}. Films with chemical gradients have been used to mimic in vivo cellular growth conditions^{16–18}.

Spatially selective reaction of thin films can be applied to any of these applications of spatially non-uniform coatings as a post-synthesis modification, but research has been limited. In the solution phase, post synthesis modification is common, and frequently performed to obtain chemistries which do not polymerize directly²¹. Broadly, this paradigm extends into the heterogeneous (surface) reaction of solid polymers in which unique surface chemistries are generated and bulk properties are decoupled from surface properties^{1,22}. Because these applications are often unconcerned with the surface chemistry of the precursor polymer, polymer thin film research remains limited to highly reactive precursors^{23,24}. A common example of this is the decoration of a surface with large biomolecules such as enzymes^{25–28} (often by nucleophilic substitution of reactive esters).

There are two main gaps in our understanding of post deposition thin film reaction. First, established techniques, while powerful, treat the precursor reactive polymer chemistry as sacrificial, meaning that the primary design constraint for choice of precursor is to maximize reactivity. When patterning a surface, or creating a Janus material, the precursor chemistry is left behind following modification, so unless extra steps are added to change the precursor chemistry, the requirement that one use a highly reactive precursor limits the range of possible surfaces we

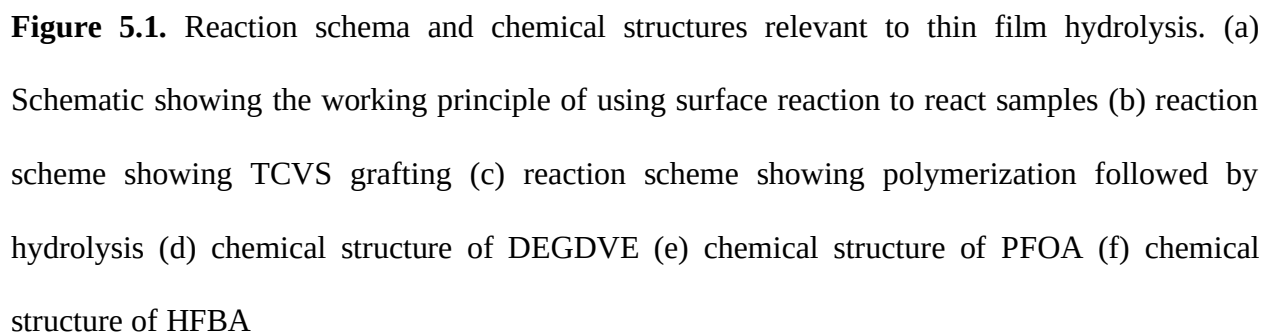
can create. By exploring hydrolysis of marginally reactive fluoroacrylates we can expand the domain of precursor chemistries, and by extension, expand the domain of potential non-uniform surface chemistries obtained by post-synthesis reaction. Second, within the field of polymer CVD, there exist numerous examples of post-deposition reaction, however because the existing literature is made up primarily of applications-oriented studies of highly reactive films, kinetics of these reactions remain poorly understood^{12,14,16,17,25-44}. Overall, the precursor surface chemistry has been rarely considered when using post-deposition reaction of polymer thin films to obtain non-uniform surface chemistry¹².

Investigation of pHFBA and pPFOA thin film hydrolysis is an excellent place to start to understand limitations of post-synthesis reaction. Because the primary utility of fluoropolymers is their passivating properties, they present a difficulty for post-deposition reaction. Because of the current lack of research in post synthesis reaction of marginally reactive polymers, further research is required to use post-synthesis reaction on passivating or hydrophobic precursors.

Solution phase processing methods like spin coating and dip coating are effective in many applications, unfortunately, coating defects in fluoropolymer films are common due to dewetting caused by low surface energy⁴⁵, so it is challenging to produce uniform fluoropolymer thin films^{46,47}. Polymer chemical vapor deposition (CVD) is a powerful method of synthesizing hydrophobic polymer films, and has demonstrated excellent conformality on micro- and nano-structures^{46,47}. Polymer CVD eliminates the use of solvents, so post polymerization extraction or solvent removal are no longer required, reducing the environmental impact of solvents, and improving the cost-effectiveness of the synthesis. Because polymer CVD occurs in the vapor phase, there are no solubility restrictions on the chemistry of the precursor polymers, yielding uniform coatings with even the most hydrophobic monomers^{46,47}. While polymer CVD refers to a

range of techniques including plasma enhanced CVD⁵, parylene CVD¹⁶ and oxidative CVD⁴⁸, this investigation focuses on initiated CVD (iCVD)⁴⁹ due to its compatibility with acrylate chemistry⁵⁰ including PFOA and HFBA.

To better understand the limits of surface reactions of thin films we hydrolyze crosslinked and un-crosslinked pHFBA and pPFOA with sodium hydroxide (NaOH) to determine reaction kinetics and the effect of crosslinking on film stability during hydrolysis. Di(ethylene glycol) divinyl ether (DEGDVE) (Figure 5.1 d) was chosen as a crosslinker because it is reasonably inert to NaOH⁵¹ and hydrophilic, enhancing the difference in hydrophilicity following hydrolysis. Grafting with trichlorovinyl silane (TCVS) (Figure 5.1 b) is frequently used to prevent film delamination and is used to stabilize films during hydrolysis. We choose base catalyzed hydrolysis for three reasons. The -OH ion is small and is therefore an effective nucleophile for penetrating into the fluoropolymer. While hydrolysis will always yield carboxylic acids, it is made significantly more versatile by the fact that the carboxylic acid moiety can readily undergo secondary functionalization reactions⁵²⁻⁵⁴. Base catalyzed hydrolysis is favored over acid catalyzed hydrolysis because siloxane bonds and ethers in the grafting agent and crosslinker are more easily hydrolyzed in acidic environments.



170

5.2 Materials and Methods

5.2.1 Materials

Di(ethylene glycol) divinyl ether (DEGDVE) (99%, from Aldrich) (figure 5.1 d), 1H,1H,2H,2H-perfluorooctyl acrylate (PFOA) (97%, from Oakwood Chemical) (figure 5.1 e) and 2,2,3,4,4,4 – hexafluorobutyl acrylate (HFBA) (95%, from Aldrich) (figure 5.1 f) were used as monomers during deposition. Di-tert-butyl peroxide (TBPO) (98%, from Aldrich) was used as an initiator. All chemicals were used without purification. Deposition was performed on polished p/boron doped silicon wafers as a reference to monitor thickness with an in-situ interferometer, for ellipsometry and for fourier transform infrared spectroscopy (FTIR) characterization. Deposition of samples destined to be hydrolyzed were performed on 2.2 cm square glass cover slips (Fisherbrand). Glass cover slips were used for hydrolysis experiments because the reaction of silicon wafer with NaOH interfered with analysis, while the reaction of NaOH with glass was too slow to impact our experiment.

Compressed oxygen for plasma cleaning and ultrahigh purity compressed argon as iCVD patch gas were obtained from Airgas.

Trichlorovinyl silane (97%, from Aldrich chemical) was used for grafting of polymer film to the substrate.

NaOH solution was prepared in a volumetric flask with Milli-Q water from a Q-POD Biopak® polisher (cat. no. CDUFB1001) and NaOH pellets (99%) used without purification from Sigma-Aldrich. Milli-Q water was also used for sample cleaning following hydrolysis.

5.2.2 iCVD deposition

Vapor deposition of polymer thin films occurred in a custom-built iCVD reactor. Pressure in the reactor was monitored with an MKS Baratron capacitance manometer model 626C and was controlled with an MKS throttle valve (model no 253B) at the outlet of the reactor. TBPO was used as an initiator and flowrate of vapor was controlled using an MKS mass flow controller type 1159B. Flow of each monomer was controlled using a needle valve. All monomers and TBPO vapors were introduced to the reactor from liquid contained in glass vials attached to the reactor. To ensure sufficient volatility, DEGDVE and PFOA were heated at 80°C and 60°C respectively. HFBA and TBPO were not heated. Temperature of the reactor filament array was monitored with a k-type thermocouple during the deposition (220°C during all depositions). The leak rate of the reactor was measured as less than 0.05 sccm before each deposition. The reactor stage was cooled using an Accel 500 LT recirculating chiller (fisher scientific). The filament was resistively heated using a BK Precision DC power supply.

Gas flowrates and leak rate were measured by closing the throttle valve to the reactor and measuring the linear increase in pressure as gas flowed into the chamber. The increase in pressure was related to the flow using the volume of the reactor and the ideal gas law. The average and standard deviation of three measurements is reported.

Copolymer p(HFBA-co-DEGDVE) was deposited on glass slides and silicon wafer. Both grafted and un-grafted samples were prepared in the same deposition for consistency of results. Homopolymer p(HFBA) film grafted with TCVS was deposited on glass slides and silicon wafers. Ungrafted homopolymer p(HFBA) film was deposited on silicon wafer to be used solely as a reference sample in FTIR composition analysis. Grafted homopolymer p(PFOA) was deposited on silicon wafer and glass slides for hydrolysis and as a reference sample in FTIR analysis. Grafted

and ungrafted p(PFOA-*co*-DEGDVE) was deposited on glass slides and silicon wafer. See Table 5.1 for deposition conditions corresponding to each run.

5.2.3. *Grafting*

Samples were cleaned in oxygen plasma using a PDC-001-HP plasma cleaner with PDC – FMG plasmaflo from Harrick plasma on the high setting for 2 minutes. Samples were then transferred directly to a chamber with 0.5 ml TCVS and air was evacuated using house vacuum. Samples were removed after 2 minutes and moved directly to the iCVD reactor.

5.2.4. *Hydrolysis*

Grafted pHFBA, pPFOA, and grafted and ungrafted p(HFBA-*co*-DEGDVE) and p(PFOA-*co*-DEGDVE) samples deposited on glass slides were floated on roughly 20 mL of NaOH in a beaker, set stirring at 200rpm on a hotplate. We assumed constant NaOH concentration over the course of the reaction based on the quantity of NaOH used, as there are $\sim 10^{-2}$ mol of -OH ions in the 20 mL of 1M NaOH solution used for reaction, while we estimate $\sim 10^{-7}$ mol of functional groups in the polymer film. Even so, fresh NaOH was used between samples. Temperature of the hotplate was controlled at 30°C for all reactions and temperature equilibrated for 30 minutes prior to reaction. 30°C was chosen since it was the lowest temperature on the hot plate, and therefore the lowest temperature we could reliably control since room temperature might fluctuate from day to day. Evaporation was prevented by sealing the beaker with parafilm. At the end of the reaction, samples were removed, quickly dried with nitrogen gas to prevent further reaction, rinsed thoroughly with Milli-Q water, and dried again with nitrogen.

5.2.5 Characterization

Infrared spectra were collected using a Nicolet iS50 FTIR spectrophotometer in transmission mode at a resolution of 4 cm^{-1} with a DGTS detector from 600 to 4000 cm^{-1} . A variable angle grazing angle attenuated total reflectance attachment with a Germanium crystal was used to collect monomer spectra by placing a few drops of monomer on the crystal. Omnic software was used for analysis and peak integration. Omnic was also used for baseline correction, a comparison between uncorrected and corrected spectra can be found for pHFBA in figure 5.2.

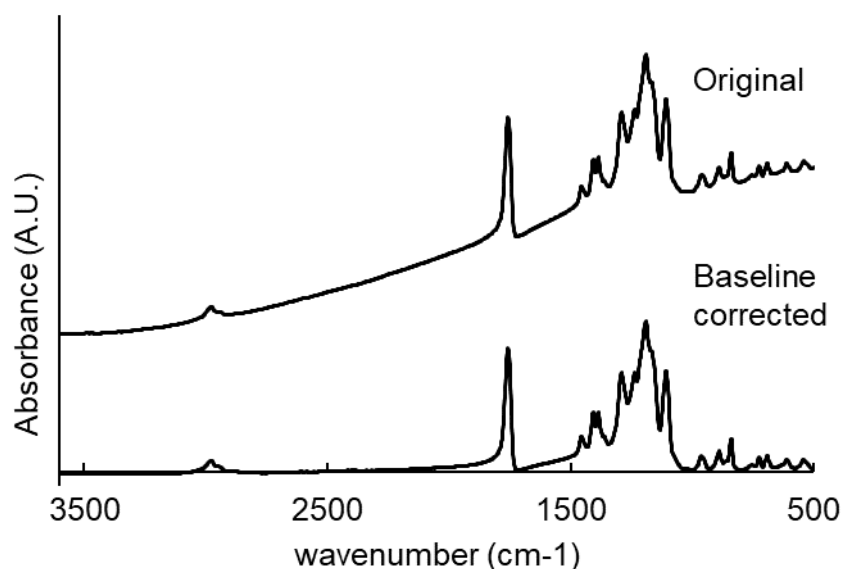


Figure 5.2. FTIR spectrum of pHFBA with and without baseline correction

Film thickness was characterized using a JA Woolam alpha-SE spectroscopic ellipsometer with data taken at 65° , 70° and 75° . Data was fitted using Complete EASE 6 software version 6.51. Polymer films were modeled as Cauchy films.

Dynamic contact angle was characterized using the sessile drop method with a Ramé-Hart goniometer with 150 W fiber optic illuminator. Water was pumped out until a sufficiently accurate advancing contact angle was observed, then water was pumped back in to measure the receding

contact angle. Measurements over ~20 datapoints were averaged and standard deviation was reported as measurement error.

Optical microscope images were captured with a VHX-970F digital microscope from Keyence in reflectance mode. Depending on the scale desired either 500x magnification or 50x magnification was used.

X-ray photoelectron spectroscopy (XPS) survey scans were collected from samples using a ScientaOmicron ESCA 2SR XPS. 150 eV pass energy, 10kV filament voltage with a step size of 1 eV. A floodgun was used for sample neutralization. Angle resolved XPS (AR-XPS) was performed by tilting the sample relative to the detector to vary the emission angle. The sample was tilted at 0° (standard orientation), 20°, 40°, and 50°. Data analysis and peak integration were performed using Casa XPS software. Inelastic mean free path (IMFP), used to estimate sampling depth, was calculated using QUASES-IMFP-TPP2M Ver. 3.0 software⁵⁵.

Table 5.1. Deposition conditions

	Fluoro- acrylate fraction	P_{tot} (torr)	T_{stage} (°C)	F_{TBPO} (sccm)	F_{DEGDVE} (sccm)	F_{HFBA} (sccm)	F_{PFOA} (sccm)	F_{ar} (sccm)	Thickness (nm)	Deposition rate (nm/min)
p(HFBA- <i>co</i> - DEGDVE)	0.46±0.03	0.1	10	0.10	0.05	0.48	-	-	189.6	7.0
p(PFOA- <i>co</i> - DEGDVE)	0.87±0.02	0.1	15	0.14	0.08	-	0.1	-	153.4	1.3
pHFBA	1	0.5	30	1.06	-	1.08	-	-	~150	-
pPFOA	1	0.05	10	0.06	-	-	0.30	-	360.3	45.0
pHFBA	1	1.1	30	0.99	-	1.99	-	0.99	459.1	-

5.3 Theory

5.3.1 Derivation of Fractional conversion from XPS data

We want to calculate the fractional conversion of a reacting sample on a fluorine basis to reduce error introduced by adventitious carbon. The only source of fluorine in the film is from HFBA or PFOA, so we will calculate fractional conversion on a fluorine basis. Begin by defining the mole fraction of fluorine x_F :

$$x_F = \frac{n_F}{n_F + n_C + n_O} \quad (5.1)$$

Where n_F , n_O , and n_C are the number density of fluorine oxygen and carbon atoms at the surface of the film.

Now, re-write in terms of ' N_r ', the number of times the reaction has happened per unit volume:

$$x_F = \frac{n_F^0 - 6N_r}{n_F^0 + n_C^0 + n_O^0 - \phi_m r} \quad (5.2)$$

Where superscript '0' indicates the number density of a given atom at the beginning of the reaction, and ϕ_m is the number of total atoms lost when the reaction occurs once. ϕ_m is 21 for PFOA and 10 for HFBA. This assumes that during the reaction, any side chain re-orientation doesn't significantly change the mole fraction of fluorine atoms in the XPS signal.

We can write the number density of fluorine atoms in the top surface in terms of the number density of fluoro-acrylate moieties in the top surface by multiplying by the number of fluorine atoms in the HFBA or PFOA functional group, which we will call π_m :

$$n_F = \pi_m n_m$$

Where n_m is the number density of monomer (HFBA or PFOA) at the surface. For HFBA $\pi_m =$

6, for PFOA $\pi_m = 13$.

The fractional conversion, χ_m , can be written as the number of times the reaction has happened, r , divided by the original number density of monomer n_m^0 :

$$\chi_m = \frac{N_r}{n_m^0} = \frac{\pi_m r}{n_F^0} \quad (5.3)$$

Now we can solve equation (5.2) for ' N_r ':

$$\frac{x_F(n_F^0 + n_C^0 + n_O^0) - n_F^0}{\phi_m x_F - \pi_m} = N_r$$

Now we can plug this expression into equation (5.3) to get:

$$\chi_m = \frac{x_F - x_F^0}{\left(\frac{\phi_m}{\pi_m} x_F - 1\right) x_F^0} \quad (5.4)$$

Equation (5.4) assumes that during the reaction, any side chain re-orientation doesn't significantly change the mole fraction of fluorine atoms in the XPS signal. We accept this assumption, because detailed characterization of the side chain reorientation is outside the scope of this study.

5.3.2 Equations for determination of reaction order

We want to find the solution for the fractional conversion in terms of time for a first-order reaction. The definition of a first order reaction is that the rate of consumption of the species is directly proportional to the number of species present. Assume that the rate is first order in both concentration of fluoro-acrylate functionality, n_m (either PFOA or HFBA) and bulk concentration of sodium hydroxide c_{NaOH} .

$$-\frac{dn_m}{dt} = kn_m^\alpha c_{NaOH}^\beta \quad (5.5)$$

Where k is the rate constant. We can rewrite this in terms of the fractional conversion $X_m = \frac{n_m^0 - n_m}{n_m^0}$ assuming the concentration of NaOH stays constant over the course of the reaction

$$\frac{d\chi_m}{dt} = kn_m^0{}^{\alpha-1} c_{NaOH}^\beta (1 - \chi_m)^\alpha$$

Integrating at constant sodium hydroxide concentration we get:

$$-\ln(1 - \chi_m) = \begin{cases} kc_{NaOH}^\beta t & \alpha = 1 \\ \frac{1}{1 - \alpha} \ln((\alpha - 1)kn_m^0{}^{\alpha-1} c_{NaOH}^\beta t + 1) & \alpha \neq 1 \end{cases} \quad (5.6)$$

By plotting, $-\ln(1 - \chi_m)$ versus time, we can clearly see the reaction order with respect to surface concentration of the fluoro-acrylate moiety. After plotting this, we find that there is approximately a first-order dependence, meaning:

$$-\ln(1 - \chi_m) = kc_{NaOH}^\beta t$$

So, we can similarly check the reaction order with respect to the concentration of sodium hydroxide in solution, by writing:

$$-\frac{\ln(1 - \chi_m)}{t} = kc_{NaOH}^\beta \quad (5.7)$$

Plotting $-\frac{\ln(1 - \chi_m)}{t}$ against concentration of sodium hydroxide will give us a curve through which we can fit a power law to determine reaction order.

5.3.3 Derivation of equations for film composition from FTIR using Beer's law

Solve Beer's law for the molar extinction coefficient

$$\frac{A}{bc} = \epsilon \quad (5.8)$$

Where 'A' is the integral absorbance of IR radiation over some range of wavelengths, 'b' is the pathlength (film thickness), 'c' is the concentration of absorbing species in the polymer film and ϵ is the molar extinction coefficient of the absorbing species.

We assume that the molar extinction coefficient for some given absorbing functionality is the same in the homopolymer and the copolymer. This assumption is valid when there is a linear relationship between the absorption of the peak belonging to some functional group and the concentration of that functional group in the polymer, even in the presence of other components in the polymer. This is the primary assumption of this method. We can write this equality of extinction coefficients for monomer 'i' in the homopolymer and copolymer using subscripts 'h' and 'c' to refer to the homopolymer and copolymer respectively.

$$\frac{A_h}{b_h c_h^i} = \frac{A_c}{b_c c_c^i}$$

Then we can obtain for either component:

$$c_c^i = \frac{A_c b_h}{A_h b_c} c_h^i$$

We would prefer to work with the mole fraction instead of concentration, which can be defined as:

$$x_1 = \frac{n_1}{n_1 + n_2} \quad (5.9)$$

We can also write the definitions of the molar densities. First, we have that the concentration of some species (1) is the number per unit volume. We will consider the homopolymer reference film to be composed of species 1:

$$c_1 = \frac{n_1}{V}$$

The density of the two species in the film, is given by the number of all species present (1 & 2) per unit volume

$$c = \frac{n_1 + n_2}{V}$$

The mole fraction can therefore be re-expressed as:

$$x_1^c = \frac{c_1^c}{c^c} = \frac{A_c b_h c_h^1}{A_h b_c c^c} \quad (5.10)$$

Equation 5.10 is simply a reformulation of Beer's law for the polymer system in question. To find the concentration of functionality in the homopolymer and copolymer films we can begin with the Lorentz-Lorenz equation⁵⁷:

$$M_{r_D} = \frac{n_D^2 - 1}{n_D^2 + 2} \frac{MW}{\rho} \quad (5.11)$$

Where M_{r_D} is the molar refraction of the repeat unit, n_D is the refractive index, MW is the formula weight of the repeat unit, and ρ is the polymer density. For a copolymer film composed of two constituents of unknown composition we use the following relationships between M_{r_D} and MW:

$$M_{r_D} = M_{r_{D_2}} + (M_{r_{D_1}} - M_{r_{D_2}})x_1 \quad (5.12)$$

$$MW = MW_2 + (MW_1 - MW_2)x_1 \quad (5.13)$$

Where subscripts '1' and '2' refer to monomers 1 and 2 respectively. We can substitute (5.12) and (5.13) into (5.11) and solve for the density:

$$\rho = \left(\frac{n_D^2 - 1}{n_D^2 + 2} \right) \frac{MW_2 + (MW_1 - MW_2)x_1}{Mr_{D_2} + (Mr_{D_1} - Mr_{D_2})x_1} \quad (5.14)$$

Writing (5.14) in terms of the concentration instead of density:

$$c = \frac{\left(\frac{n_D^2 - 1}{n_D^2 + 2} \right)}{Mr_{D_2} + (Mr_{D_1} - Mr_{D_2})x_1} \quad (5.15)$$

We can solve this above equation for the mole fraction of component 1 by combining (5.10) and (5.15):

$$x_1^c = \frac{Mr_{D_2}}{Mr_{D_2} \frac{A_h b_c (n_{D,h}^2 + 2)(n_{D,c}^2 - 1)}{A_c b_h (n_{D,h}^2 - 1)(n_{D,c}^2 + 2)} + (Mr_{D_2} - Mr_{D_1})} \quad (5.16)$$

We now have an equation which relates easily measured parameters to mole fraction of copolymer in the polymer.

To progress we need to know the molar refraction of the polymer sub-units. Ideally these would be computed by reference to the bulk homopolymer, but as this data is not available to us, a group contribution method is sufficiently accurate⁵⁸.

$$M_{rD} = \sum_{group} M_{rD}$$

A table of values can be found in Speight (2017)⁵⁸.

5.4. Results and Discussion

5.4.1. iCVD synthesis and FTIR analysis

Thin films of pPFOA, p(PFOA-co-DEGDVE), pHFBA and p(HFBA-co-DEGDVE) were synthesized using the iCVD technique and the conditions listed in table 5.1. Thickness-normalized FTIR transmission spectra were taken of the films (figure 5.3). DEGDVE, PFOA and HFBA monomer spectra were taken using FTIR with attenuated total reflectance (FTIR-ATR), so the absorbance of peaks at higher wavenumber is reduced⁵⁶.

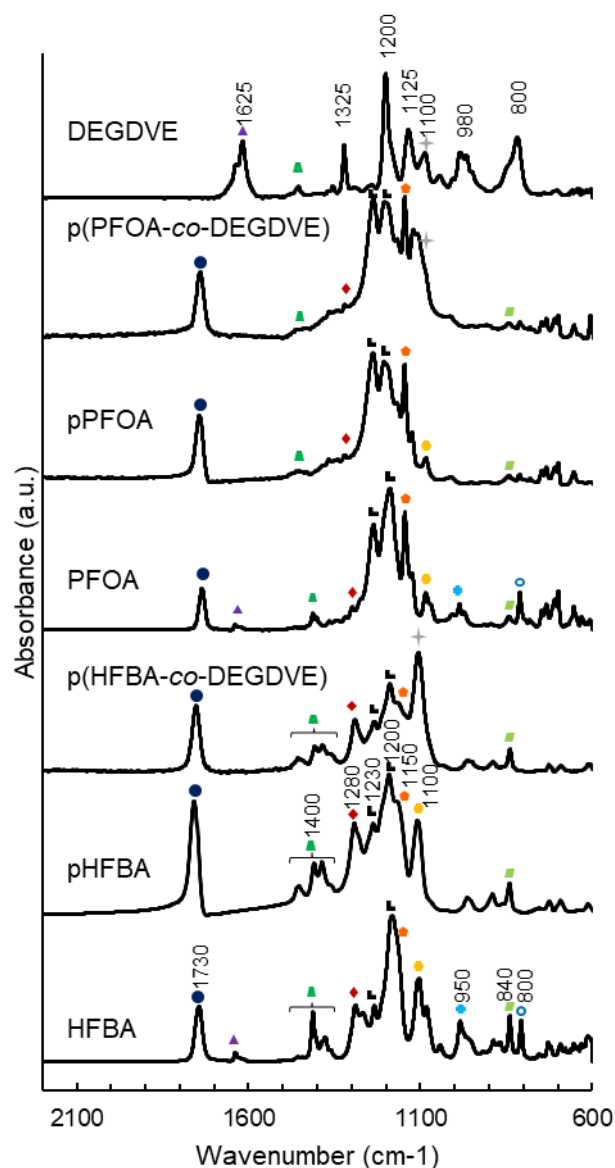


Figure 5.3. FTIR spectra of monomer and polymers. FTIR spectra of DEGDVE monomer, p(PFOA-co-DEGDVE), pPFOA, PFOA monomer, p(HFBA-co-DEGDVE), pHFBA and HFBA monomer. Polymer spectra were collected in transmission mode and were normalized by their film thickness. Monomer spectra were collected using FTIR-ATR, and were normalized to be of the same general scale as they cannot be used for quantification.

Peaks belonging to the acrylic ester functionality (discussed below) can be identified in monomers, homopolymers and copolymers of HFBA and PFOA. In general, the acrylate peaks are larger in the pHFBA spectra than the pPFOA spectra, in part because the number density of the acrylate moiety per unit polymer volume is less for the bulkier PFOA monomer. The peak at $\sim 1730\text{ cm}^{-1}$ (●) is due to C=O stretching in the acrylate. The peak at 1280 cm^{-1} (♦) is due to asymmetric stretching of the acrylate. The peak at 1150 cm^{-1} (◐) is due to the C-O stretching of the acrylate. The peak at 1100 cm^{-1} (◑) is due to either ester asymmetric stretching or the SiO₂ substrate and is difficult to decouple. The peak at 840 cm^{-1} (◒) is due to the C-C bond of the acrylic ester.

Peaks belonging to vinyl groups in both the acrylate and vinyl ether are present in the monomer spectra but are not present in the corresponding polymer spectra. Most notably, the peaks around 1625 cm^{-1} (▲) in monomer spectra are due to C=C stretching, and their absence in polymer spectra indicates full polymerization has occurred for all films. Furthermore, it shows that there are very few, if any, labile vinyl bonds in the crosslinked spectra, indicating that DEGDVE is highly reactive with the acrylic monomers. The peaks around 1400 cm^{-1} (◓) are due to alkane CH₂ scissoring and CH bending. The peak at 950 cm^{-1} (◔) is due to the acrylate C=C. The peak at 800 cm^{-1} (◕) is due to out of plane CH₂ vibration of the vinyl bond.

Peaks belonging to fluorocarbon bond vibrations can be identified in all fluoropolymer spectra. The peaks at 1230 cm^{-1} and 1200 cm^{-1} (◗) are due to the carbon fluorine bond in the fluorocarbon. Peaks around 700 cm^{-1} present in fluorine containing spectra (but not labeled) are due to the C-F deformation broadly.

Peaks in the DEGDVE monomer spectrum are not found prominently in the p(PFOA-co-DEGDVE) or p(HFBA-co-DEGDVE) spectra since the peaks which do not overlap with acrylate

or fluorocarbon peaks are associated with the vinyl ether and disappear on polymerization. The other characteristic DEGDVE peaks appear at around 1100 cm^{-1} in the crosslinked films. The peak at 1325 cm^{-1} is due to $=\text{CH}$ rocking vibration of the vinyl ether. The peak at 1200 cm^{-1} is due to the asymmetric stretching of C-O-C stretching. The peaks at 1125 cm^{-1} and 1100 cm^{-1} (+) are due to stretching of the central ether of DEGDVE. The peak at 980 cm^{-1} is due to vinyl ether wagging and the peak at 800 cm^{-1} is due to vinyl ether symmetric C-O-C stretching.

5.4.2 Composition analysis using FTIR with ellipsometry

Composition of thin films can be determined using FTIR of the “bulk” polymer film. FTIR in the transmission mode is a more effective method of quantification than XPS, since it samples the whole film, and it is not biased by surface non-uniformities. Equation 5.16 gives the mole fraction of fluoro-acrylate (HFBA or PFOA moiety) x_m in the crosslinked film, and is based on the assumption that the Lorenz-Lorentz equation can be used to relate film refractive index to the film density⁵⁷, and that Beer’s law relates the absorption of a peak in FTIR to the concentration of the associated moiety in the film. The Lorentz-Lorentz equation relies on the principle of the molar refraction, which is a measure of the total polarizability of a mole of the substance. Equation 5.16 is used to quantify composition. Densities of the films can be calculated using equation 5.14.

Refractive index and thickness were determined using spectroscopic ellipsometry, and refractive index values were taken at a wavelength of 589 nm, as is standard. Results of the composition analysis are presented in Table 5.2.

Table 5.2. Compositional and density data for homopolymer and crosslinked films

	pHFBA	p(HFBA-co-DEGDVE)	pPFOA	p(PFOA-co-DEGDVE)
A (a.u.)	3.11	0.733	1.39	0.572
b (nm)	459.09±0.10	189.59±0.17	357.45±0.11	153.38±0.175
Mr_{D_m} (cm ³ mol ⁻¹)	33.98	33.98	52.452	52.452
$Mr_{D_{DEGDVE}}$ (cm ³ mol ⁻¹)	-	39.67	-	39.673
n_D	1.394 ± 0.004	1.567±0.046	1.364±0.001	1.392±0.007
ρ (g cm ⁻³)	1.662±0.015	1.71±0.15	1.776±0.004	1.801±0.048
x_m	1	0.456±0.033	1	0.87±0.02

Note that both crosslinked films are denser than their homopolymer counterparts. Without correcting for this, we would underestimate the quantity of crosslinker. Because a group contribution method was used to determine molar refraction, these results should be taken as approximations of the true value, however there is precedent for using refractive index to characterize density of polymer systems⁵⁷. Refractive index of commercially available pHFBA is 1.394, essentially identical to our data from ellipsometry (Table 5.2), however the density is 1.519 g cm⁻³,⁵⁹ meaning that the molar refractivity given by the group contribution method has over-estimated density by 10%. It is reasonable to expect a similar over-estimation for PFOA. Note that due to the deposition kinetics, we were forced to use different crosslinking densities for the two polymers, and we acknowledge this is certainly not ideal for comparison of film properties. We were unable to decrease the crosslinking density in p(HFBA-co-DEGDVE) because reliable DEGDVE flowrates below 0.05 sccm could not be obtained. The remarkably high fraction of

crosslinker is likely due to alternating behavior between the vinyl ether and the HFBA, which is a well-documented tendency in fluoroacrylates⁴. We were unable to increase the crosslinking density in p(PFOA-co-DEGDVE) since a higher partial pressure of DEGDVE would have caused condensation.

5.4.3 Role of Grafting and Crosslinking in Film Stability

The as-deposited pPFOA films were reacted with 1M NaOH at 30°C for 40 min and the pHFBA films were reacted with 0.5 M NaOH at 30°C for 30 min. 30°C was used for all hydrolysis experiments since it was the lowest temperature we could reliably control using the hotplate. The goal of this experiment was solely to illustrate degradation, so a lower concentration of NaOH was used for the hydrolysis of pHFBA films to slow degradation and enable clear characterization. Additionally, different reaction times are reported based on when degradation was most clearly observed. Micrographs of the films show apparent pitting (figure 5.4 a and b), indicating gradual film disintegration during hydrolysis. Because the reaction product of homopolymer pPFOA and pHFBA is poly(acrylic acid), a water soluble polymer⁶⁰, it is unsurprising that films of pPFOA and pHFBA would gradually disintegrate as they are converted to poly(acrylic acid). Based on this hypothesized mechanism of degradation, the only way to prevent degradation is by modifying the film in such a way to make the product insoluble, whereas different strengths of nucleophile or base should have little effect. The film degradation observed in un-crosslinked films made accurate characterization of the surface hydrolysis reaction nearly impossible. Further investigation of the pitting and delamination in figure 5.4 would be necessary to fully understand how these occur and develop over time, however our interest lies instead with the root causes and mitigating measures of degradation. Crosslinking was used to prevent film dissolution. DEGDVE, a common crosslinker in iCVD³¹ was used because it is hydrophilic, enabling us to see the maximum

difference in hydrophilicity following hydrolysis, and is stable in NaOH at moderate temperatures⁵¹.

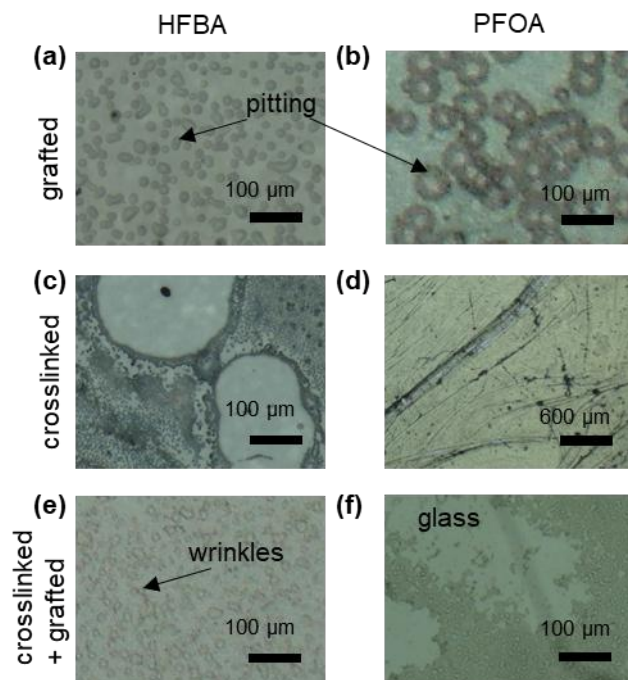


Figure 5.4. Optical micrographs of iCVD films treated with aqueous solutions of NaOH a) pHFBA film grafted with TCVS showing pitting after 30 min in 0.5 M NaOH b) pPFOA film grafted with TCVS showing pitting after 40 min in 1M NaOH c) p(HFBA-co-DEGDVE) film (not grafted) showing delamination after 2 min in 1M NaOH d) p(PFOA-co-DEGDVE) film (not grafted) showing delamination after 2 min in 1M NaOH e) p(HFBA-co-DEGDVE) film grafted with TCVS showing wrinkling after 300 min in 1M NaOH f) p(PFOA-co-DEGDVE) film grafted with TCVS showing degradation after 300 min in 1M NaOH

Ungrafted copolymers, p(PFOA-co-DEGDVE) and p(HFBA-co-DEGDVE) showed serious film delamination (figure 5.4 c and d) after only 2 minutes of exposure to 1M NaOH, indicating that grafting was required to achieve a film capable of withstanding even small amounts

of hydrolysis. Delamination is caused by insufficient Van der Waals interactions between film and substrate and is generally solved by covalently bonding the film to the surface using a silane coupling agent¹⁴. As such, all films used in reaction characterization were grafted to glass coverslips using TCVS.

Grafted and crosslinked p(PFOA-*co*-DEGDVE) and p(HFBA-*co*-DEGDVE) were significantly more stable during hydrolysis in 1M NaOH with both films showing good stability up to 300 minutes. After 300 minutes however, p(HFBA-*co*-DEGDVE) films began to wrinkle (figure 5.4 e). This may be caused by hydrolysis of the siloxane bonds holding it to the surface⁶¹, or it may be caused by film swelling. At 300 min, p(PFOA-*co*-DEGDVE) films were observed to begin to degrade significantly (figure 5.4 f). XPS characterization was performed on less degraded portions of the film, which we know to be continuous based on the lack of silicon peaks. Discontinuous films show silicon peaks from the glass substrate (see figure 5.7). The p(PFOA-*co*-DEGDVE) films contained less crosslinker than p(HFBA-*co*-DEGDVE), so it is unsurprising that they would degrade faster, however the apparent delamination indicates that hydrolysis of the siloxane bonds holding the film to the substrate is occurring, and that better coupling methods must be developed if thin films are to be used in caustic environments for any length of time. Improving the resistance of coupling agents to hydrolysis is a broader problem in the field, with dipodal silanes representing a possible improvement⁶².

The rapid destruction of un-crosslinked and un-grafted fluoro-acrylate thin films indicates that NaOH is an effective method of film removal and could be used to remove hydrolysable films for etching or in the fabrication of Janus materials. These findings also inform the use of fluoro-acrylates more generally, as they indicate that hydrolysis of un-crosslinked fluoro-acrylate bulk polymers is not localized to the top surface, but instead causes degradation of the polymer. While

this is relevant to any application of acrylates in potentially caustic environments, it's especially relevant to fluoro-acrylates like PFOA, which can introduce carcinogenic perfluorinated acid into the environment through hydrolysis. While crosslinking reduces solubility of hydrolyzed films, grafting appears to be necessary to prevent rapid delamination (figure 5.4 f). As such, grafted p(HFBA-co-DEGDVE) and p(PFOA-co-DEGDVE) films were synthesized for reaction kinetics analysis.

5.4.4. Hydrolysis Kinetics

Hydrolysis of crosslinked and grafted p(HFBA-co-DEGDVE) thin films with $45.6 \pm 3.3\%$ HFBA and p(PFOA-co-DEGDVE) thin films with $87 \pm 2\%$ PFOA were performed by floating coated glass slides on NaOH solutions as depicted in figure 5.16 a. XPS was used to characterize the conversion. XPS data only describes the film up to the analysis depth⁶³, which has been estimated using the TPP-2M formula⁵⁵ to be on the order of 10 nm. XPS survey scans of unreacted films were compared with films reacted for 300 minutes in 1M NaOH showing a significant decrease in the prominence of the fluorine 1s peak in PFOA and an almost complete disappearance of the fluorine 1s peak in HFBA samples. This suggests an almost complete loss of fluorine within the XPS analysis depth ($\sim 10\text{nm}$), and an emergence of sodium carboxylate on the surface of the film (the un-protonated reaction product of hydrolysis) (Figure 5.5). As we've already stated, the absence of Si peaks confirms film continuity over the analysis area. All XPS survey scans used for kinetics analysis are also included.

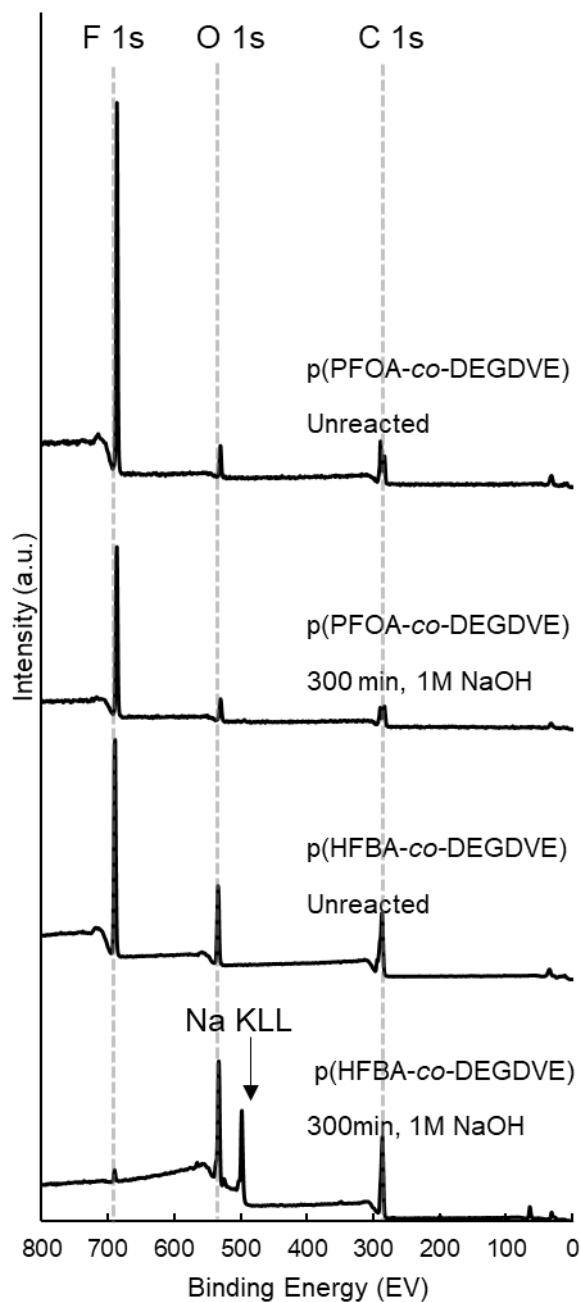


Figure 5.5. XPS data for unreacted p(HFBA-co-DEGDVE) and p(PFOA-co-DEGDVE) thin films and films reacted for 300 min in 1M NaOH. Spectra corresponding to other time points are included here.

Raw XPS data are presented in Figures 5.6-5.11. Data are presented as collected and are not

normalized.

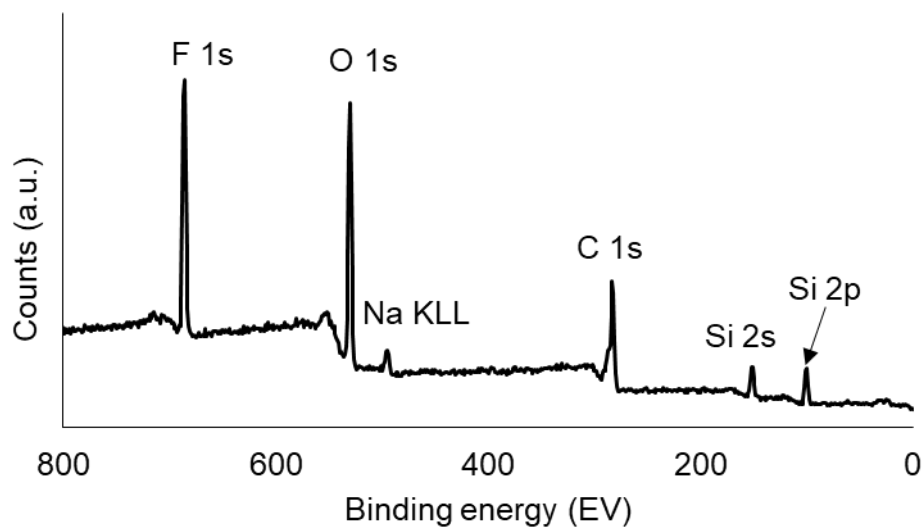


Figure 5.6. XPS scan of pHFBA film showing Si peaks from film degradation after reacting with 0.5M NaOH for 30 minutes.

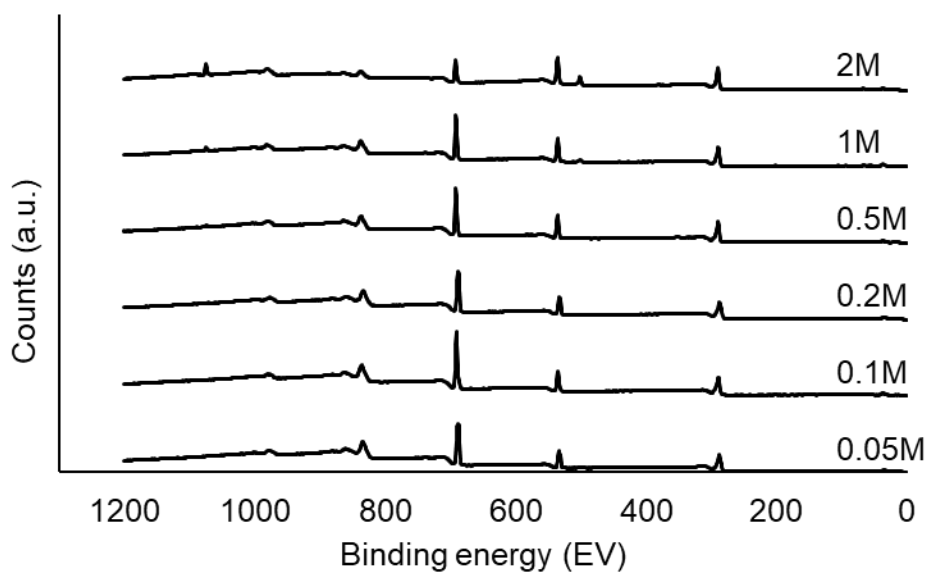


Figure 5.7. XPS data for p(HFBA-co-DEGDVE) in the concentration series. Where NaOH concentration was varied from 0.05-2M, but reacting time and temperature were kept constant at

50 min and 30°C

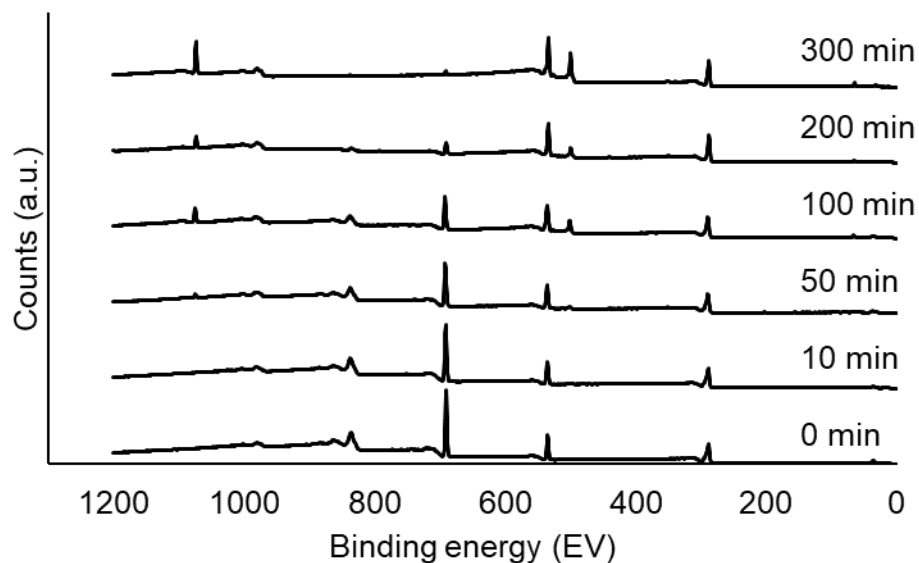


Figure 5.8. XPS data for p(HFBA-co-DEGDVE) time series. Where the reaction time was varied from 0-300min, and concentration and temperature was kept constant at 1M and 30°C.

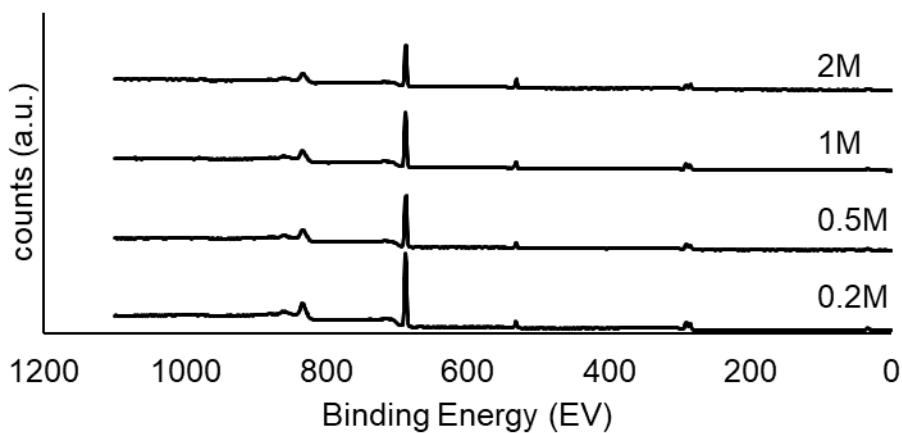


Figure 5.9. XPS data for p(PFOA-co-DEGDVE) concentration series. Where NaOH concentration was varied from 0.2-2M, but reacting time and temperature were kept constant at 180 min and 30°C

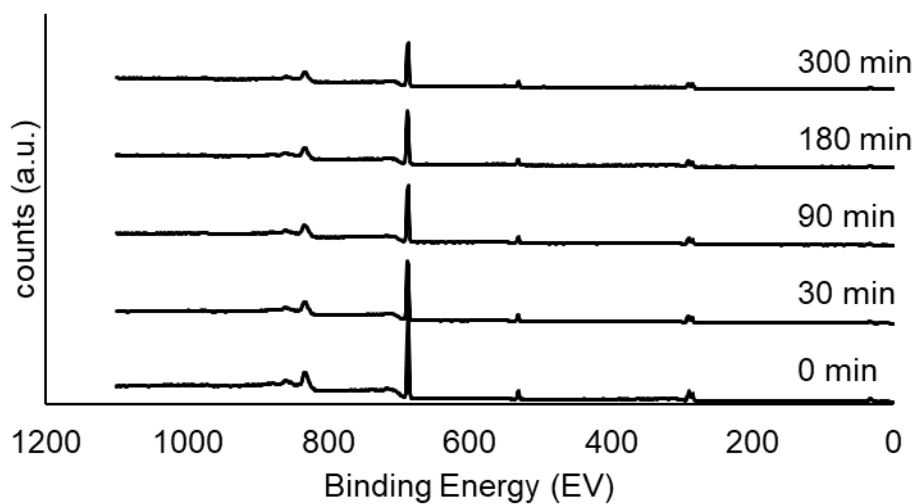


Figure 5.10. XPS data for p(PFOA-co-DEGDVE) time series. Where the reaction time was varied from 0-300min, and concentration and temperature was kept constant at 1M and 30°C.

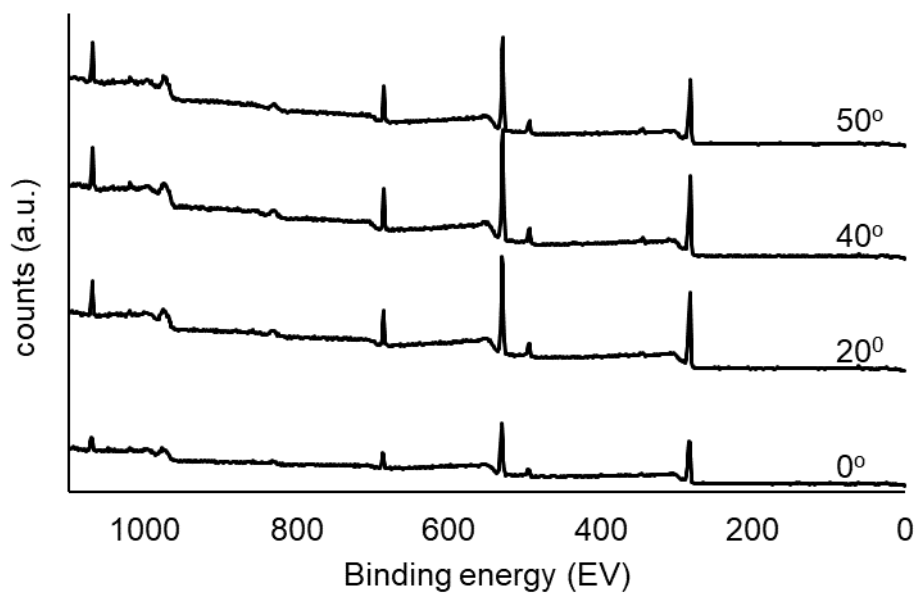


Figure 5.11. Angle resolved XPS data from p(HFBA-co-DEGDVE) reacted for 200min in 1M NaOH

The Dahmkoler number $Da = \frac{kn_m\hat{x}^2}{D}$ describes the relative effects of reaction and diffusion and can be used to characterize the degree to which concentrations measured with XPS are biased by diffusion induced gradients. $\hat{x} = 10$ nm (the analysis depth) and D is the diffusion coefficient of NaOH in the polymer. The Dahmkoler number will be much less than 1 when the system is reaction limited, so XPS characterization is unaffected by diffusion when $D \gg kn_m\hat{x}^2 \sim 10^{-17} m^2 s^{-1}$. Measured diffusion coefficients of water in polymeric systems are generally on the order of $10^{-13} m^2 s^{-1}$,⁶⁴ so we do not expect compositional gradients resulting from reagent diffusion to exist within the XPS analysis depth.

Films of p(HFBA-co-DEGDVE) were reacted in two groups. In series (i), films were reacted in 1M NaOH at 30°C from 5 to 300 minutes to determine kinetics with respect to surface functional groups. In series (ii), films were reacted at 30°C for 50 min at NaOH concentrations ranging from 0.05 M to 2 M, to determine kinetics with respect to NaOH concentration. We implicitly assume constant NaOH concentration over the course of the reaction. Regarding this assumption, there are $\sim 10^{-2}$ mol of -OH ions in the 20 mL of 1M NaOH solution used for reaction, while we estimate $\sim 10^{-7}$ mol of functional groups in the polymer film (based on polymer density). Even so, fresh NaOH was used between samples.

Films of p(PFOA-co-DEGDVE) were also reacted in two groups. Series (i) films were reacted in 1M NaOH at 30°C from 30 min to 300 min to determine kinetics with respect to surface functional groups and Series (ii) films were reacted at 30°C for 180 min in 0.2-2 M NaOH to determine kinetics with respect to NaOH concentration. Surface density of HFBA or PFOA moieties in unreacted and reacted samples were quantified using XPS.

The fractional conversion χ_m , was calculated based on the fluorine atomic ratio (from XPS survey scans) using equation 5.4 (derived in section 5.3), where π_m is the number of fluorine atoms in the fluoro-acrylate monomer (6 for HFBA, 13 for PFOA) and ϕ_m is the number of carbon and fluorine atoms lost when a single fluoro-acrylate moiety is hydrolyzed (10 for HFBA, 21 for PFOA). The surface mole fraction of fluorine was used in lieu of carbon or oxygen to avoid error introduced by adventitious carbon and oxygen. Adventitious carbon and oxygen are common in XPS, while fluorine contamination is not⁶⁵.

We hypothesize a general rate equation of arbitrary reaction order with respect to concentration of NaOH c_{NaOH} and surface number density of either HFBA or PFOA moiety n_m (equation 5.5). Solving equation 5.5 in terms of the fractional conversion of fluoro-acrylate functionality $\chi_m = \frac{n_m^0 - n_m}{n_m^0}$ while keeping the concentration of NaOH constant, gives equation 5.6.

To determine reaction order with respect to fluoro-acrylate moiety, $-\ln(1 - \chi_m)$ is plotted against time (figure 5.12 a), showing an r^2 of 0.99 for HFBA and 0.92 for PFOA for a linear fit. From this data we can conclude that a first order approximation is valid for both hydrolysis of p(PFOA-co-DEGDVE) and p(HFBA-co-DEGDVE) up to a fractional conversion of 0.613 and 0.965 respectively. While a non-integer reaction order less than 1 provides a better fit to HFBA data, mechanistically, as a substitution reaction we expect first order behavior with respect to the concentration of reactive surface moiety. As such, the deviation between the first order and non-integer order models is too small to justify and would be considered overfitting.

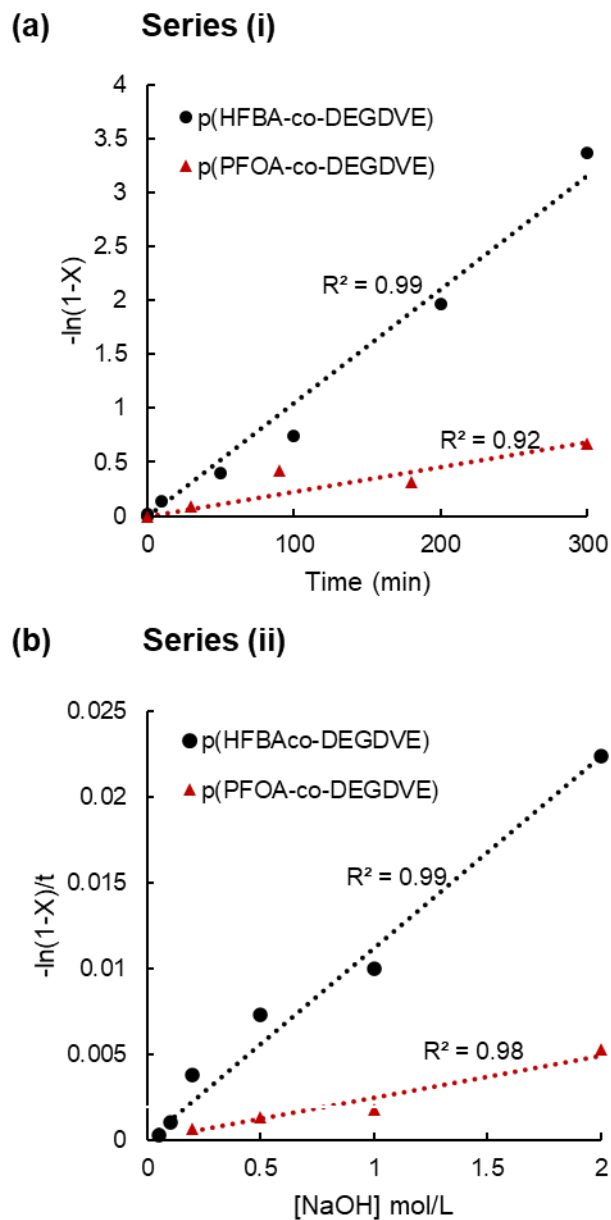


Figure 5.12. Data to determine reaction kinetics. (a) Series (i), $-\ln(1 - X_m)$ versus time for p(HFBA-co-DEGDVE) and p(PFOA-co-DEGDVE) thin films reacted in 1M NaOH showing a 1st order behavior with respect to concentration of acrylate repeat units. (b) Series (ii), $-\ln(1 - X_m)/t$ versus NaOH concentration for p(HFBA-co-DEGDVE) and p(PFOA-co-DEGDVE) thin films showing 1st order behavior with respect to NaOH concentration.

Table 5.3. AR-XPS mole fractions for p(HFBA-*co*-DEGDVE) sample reacted for 200 minutes in 1 M NaOH.

Emission angle (deg)	Analysis depth, λ^1 (nm)	x_C	x_O	x_F	x_{Na}
0	11.4	0.63	0.27	0.057	0.042
20	10.7	0.60	0.28	0.070	0.049
40	8.8	0.58	0.28	0.078	0.058
50	7.4	0.57	0.29	0.077	0.061

It is possible this deviation is due to chain re-orientation within the XPS sampling volume. Fluorinated groups are preferentially excluded from the bulk, even in homo-fluoropolymers⁶⁶, and the presence of carboxylic acid moieties and ethers (present in DEGDVE) likely increases the magnitude of this preferential orientation. As samples are placed in high vacuum for XPS analysis, the carboxylic acid moieties (capable of forming strong hydrogen bonds with each other) exclude the hydrophobic fluorinated side chains to the surface of the film, thereby increasing the concentration of fluorine within the XPS sampling area, and making it appear as though the reaction rate is less. This explanation is supported by AR-XPS, which shows higher density of fluorine at the sample surface (Table 5.3). This effect has been seen elsewhere as well⁶⁶.

As we can see in Table 5.3, the mole fraction of carbon decreases as the angle resolved measurement becomes more surface sensitive (larger angle), and that of oxygen, fluorine and

¹Analysis depth λ of the x-ray used in XPS is reported as $\lambda = 3 * IMFP \cos \theta$ (inelastic mean free path), where the IMFP is calculated from the non-relativistic TPP-2M formula using the QUASES-IMFP-TPP2M Ver. 3.0 software⁵⁵ and θ is the emission angle.

sodium increase, indicating that they are present in a higher proportion at the surface of the material. This is likely due to the aforementioned chain re-orientation, and it indicates that the true fractional conversion of p(HFBA-*co*-DEGDVE) is slightly higher than the measured value.

Taking the reaction to be first order with respect to the fluoro-acrylate moiety we can simplify equation 5.6 by setting $\alpha = 1$. To determine reaction order with respect to NaOH concentration, it's useful to re-express equation 5.6 as equation 5.7. Equation 5.6 shows that for an arbitrary reaction order in NaOH concentration, $-\frac{\ln(1-\chi_m)}{t}$ follows a power law. Figure 5.12 b shows a plot of $-\frac{\ln(1-\chi_m)}{t}$ versus NaOH concentration from series (ii), in which NaOH concentration was varied from 0.05 M to 2 M for HFBA and from 0.2 M to 2 M for PFOA, showing linear behavior with r^2 of 0.99 for HFBA and 0.98 for PFOA indicating the reaction is first order for both polymers. Nucleophilic substitution reactions are categorized into S_N1 and S_N2 types. S_N1 reactions are unimolecular, and depend on the concentration of the nucleophile, while S_N2 reactions are bimolecular and depend on both the concentration of the nucleophile and substrate. Generally, S_N2 reactions are characterized by poor leaving groups, and are generally limited by steric hinderance. Both the PFOA and HFBA side chains are poor leaving groups, so an S_N2 mechanism (first order in both NaOH and fluoroacrylate moiety) is reasonable for our system.

Having determined the reaction order with respect to NaOH concentration and functional group density, equation 5.6 is solved for the rate constant k , and averaged over all reacted samples ($n = 11$ and 8 for HFBA and PFOA respectively) to give $0.011 \pm 0.001 \text{ L mol}^{-1} \text{ s}^{-1}$ for HFBA and $0.0029 \pm 0.0004 \text{ L mol}^{-1} \text{ s}^{-1}$ for PFOA (mean $\pm$ standard error). Base-catalyzed hydrolysis of acrylate and methacrylate monomers in aqueous medium at 30°C have previously been reported to have reaction rates ranging from ~ 0.05 - $0.5 \text{ L mol}^{-1} \text{ s}^{-1}$ ⁶⁷. Partitioning of NaOH into the polymer

matrix explains slower kinetics we observed. At the polymer / water interface, we expect there to be a drop in the concentration of NaOH since it is more easily solvated in the liquid than in the polymer. This effect of partitioning should not change the order of reaction, but it will decrease the concentration of NaOH species available to react within the polymer relative to the bulk solution (where concentration is known), and thereby reduce the measured reaction rate. Decoupling the partition coefficient from the reaction kinetics is out of the scope of this study. Though, it should be noted that reported rate constants are only valid for the crosslinking density studied, and changes in composition will likely change the kinetics. Solvation effects directly impacting the reaction kinetics⁶⁸ may be significant as well, but quantification of these is beyond the scope of this investigation.

The difference in reaction rate between p(PFOA-co-DEGDVE) and p(HFBA-co-DEGDVE) is also unsurprising and is likely due to both steric and inductive effects, since HFBA is less bulky and fluorine atoms are located closer to the ester, thereby more effectively participating in electron withdrawing and making the HFBA functionality a better leaving group than the PFOA.

5.4.5. Dynamic contact angle of hydrolyzed films

Dynamic water contact angles were measured on all p(HFBA-co-DEGDVE) and p(PFOA-co-DEGDVE) samples reacted with NaOH, showing a clear correlation between dynamic contact angle and fractional conversion (Figure 5.13 a and b). Note that dynamic instead of static contact angles are reported because, for high hysteresis surfaces, it becomes likely that measured static contact angles will deviate significantly from the Young's contact angle⁶⁹, so we felt it would be misleading to report static contact angle for these samples. The overlap between data from series (i) and series (ii) mean that two films reacted to the same final conversion using different NaOH

concentration likely have similar surface properties. Above ~30% fractional conversion, water did not recede across the surface of p(HFBA-*co*-DEGDVE), so receding contact angle is reported as 0°. The high hysteresis of both films is likely caused by roughness, viscoelasticity of the film, and chain re-orientation (i.e. surface response to contact with water)⁷⁰. Chemical heterogeneity is cited as a common cause of dynamic contact angle hysteresis, and is commonly modeled using the Cassie equation (equation 5.7), however in practice, the Cassie equation describes the minimum effect of heterogeneity on contact angle⁷¹. Furthermore, the hydrolysis reaction likely changes the mechanical properties and roughness of the film, both of which will contribute to increasing contact angle hysteresis, which is not accounted for in the Cassie equation. The Cassie equation was fitted to dynamic contact angle data of reacted p(HFBA-*co*-DEGDVE) and p(PFOA-*co*-DEGDVE) by performing a linear regression to the data on modified axes (see figure S1). For p(PFOA-*co*-DEGDVE) the r^2 was 0.48 and 0.78 for receding and advancing respectively. For p(HFBA-*co*-DEGDVE) the r^2 was 0.65 and 0.93 for receding and advancing respectively. The dotted lines in figure 5.13 indicate the Cassie equation generated by the aforementioned curve fits of equation 5.17.

$$\cos \theta = f \cos \theta_1 + (1 - f) \cos \theta_2 \quad (5.17)$$

Where f is the area fraction of surface covered with chemistry 1. θ_1 and θ_2 then refer to the contact angle on a surface covered with chemistry 1 and 2 respectively. These were treated as fitting parameters. The fractional conversion χ_m was assumed to be equal to the fractional surface coverage f .

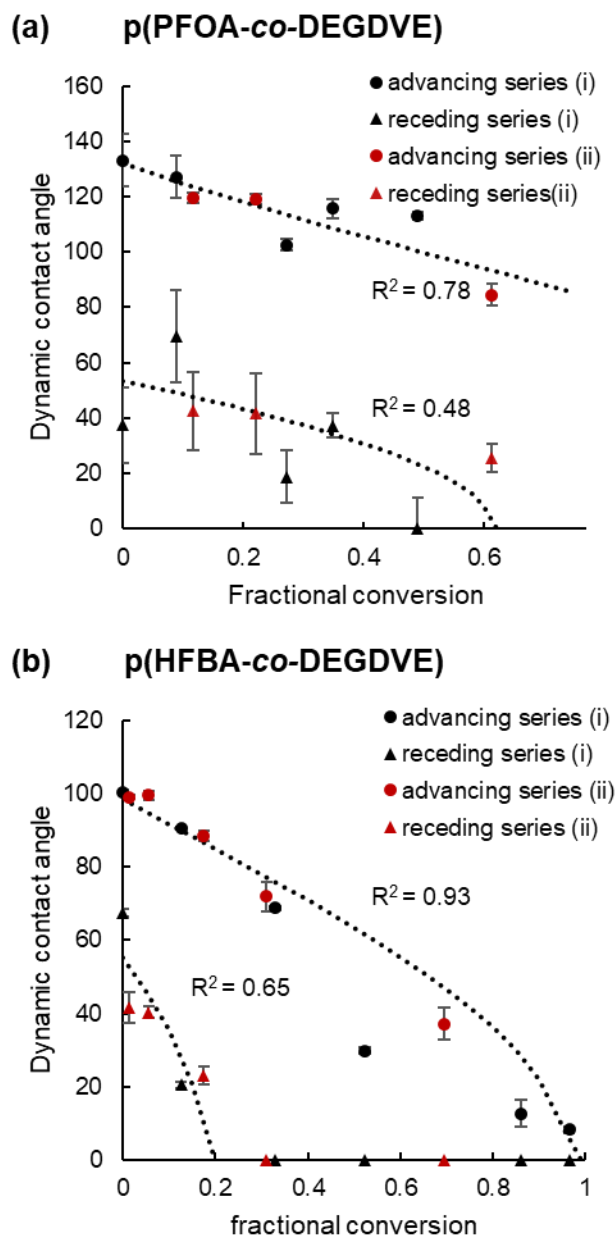


Figure 5.13. Dynamic contact angle data for hydrolyzed p(HFBA-co-DEGDVE) and p(PFOA-co-DEGDVE). Dynamic contact angle of (a) p(PFOA-co-DEGDVE) and (b) p(HFBA-co-DEGDVE) films reacted with aqueous NaOH versus fractional conversion. Series (i) refers to the time series data and series (ii) refers to the concentration series data. The black dotted lines are curve fits of the Cassie equation.

The poor fit for p(PFOA-*co*-DEGDVE) can be explained by random variation between samples, however, there appears to be a systematic overprediction of the advancing contact angle of p(HFBA-*co*-DEGDVE) at high conversion. We believe this overprediction is related to the apparent mobility of the fluorine side chain suggested by AR-XPS. In the high vacuum of XPS analysis, the fluorine side chains will preferentially cluster at the surface. When in contact with water during the dynamic contact angle measurement, they will reorient away from the surface. Re-orientation of fluorine side chains away from the surface would effectively increase the fractional surface coverage of hydrolyzed polymer relative to the measured fractional conversion resulting in the observed over-prediction of the Cassie equation. Correcting for side chain mobility is outside the scope of this investigation.

5.5 Conclusions

Our work has characterized the hydrolysis reaction of p(PFOA-*co*-DEGDVE) and p(HFBA-*co*-DEGDVE) hydrolysis with NaOH, which can enable fabrication of Janus or patterned chemical surface heterogeneity using marginally reactive precursor films by post-synthesis surface reaction. Our kinetic data can be used to quantify the release of perfluorinated alcohols (which degrade to form PFA's) from polymeric sources via hydrolysis. Such quantification is necessary to understand the impact of using polymers which can release these potentially toxic long chain PFA's in the environment.⁷ Ellipsometric and FTIR data were used to determine thin film composition using a novel method which combines the Lorentz-Lorenz equation with Beer's law, and demonstrates that when homopolymer and copolymer films have significant density differences, only using FTIR to characterize polymer composition causes errors.

By observing degradation of pHFBA and pPFOA, we show that modification of surface chemistry using the hydrolysis method requires both crosslinking and grafting. We have

determined the rate equation for p(HFBA-co-DEGDVE) and p(PFOA-co-DEGDVE) hydrolysis with NaOH, showing both reactions to be well described by an S_N2 type model which is first order in both the concentration of NaOH and the concentration of reactive moieties on the surface and 2nd order overall. They have rate constants at 30°C of $0.011 \pm 0.001 \text{ L mol}^{-1} \text{ s}^{-1}$ for HFBA and $0.0029 \pm 0.0004 \text{ L mol}^{-1} \text{ s}^{-1}$ for PFOA. The more rapid reaction of HFBA can be explained by steric and inductive effects. AR-XPS performed on samples at high conversion shows densification of fluorine at the surface of the film caused by migration of the hydrophobic fluorinated side chains to the surface of the film once it is exposed to air.

Dynamic contact angle performed on the hydrolyzed samples shows that the wetting properties are strongly correlated with the fractional conversion. Furthermore, the absence of a significant change in contact angle hysteresis indicates that hydrolysis results in a homogeneous distribution of reacted moieties (under the assumption that roughness does not change significantly during hydrolysis). This indicates that partial hydrolysis of fluorinated films like those studied here may yield high quality amphiphilic surfaces, which are useful for antifouling applications⁸.

In future research, hydrolysis followed by ammonolysis could enable the fabrication of diverse surface chemistries, including zwitterionic films^{52,72}. Because S_N2 reaction kinetics are often determined by sterics, we expect that the kinetics of nucleophilic amine substitutions onto hydrolyzed surfaces would be significantly more rapid than on the native fluoroacrylate. We would suggest further investigation into whether film removal of acrylates by NaOH can be used effectively as an etching method in film patterning or efficient fabrication of Janus materials. We would also suggest further investigation of the role that changing crosslinking density has in determining reaction kinetics.

5.6 References

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CHAPTER 6:
CONCLUSIONS AND SUMMARY

6.1. Conclusions

This thesis contains fundamental research seeking to better understand the underlying processes important for applied research in iCVD implementation, process innovations and post-deposition modification. Four investigations of fundamental phenomena occurring in the iCVD process are undertaken in Chapters 2 through 5.

Chapter 2 investigates a previously unreported defect which occurs in iCVD. The potential for defects poses a risk for the use of iCVD in precision applications such as optical thin films, where smooth and uniform films are needed. Defects are shown to occur due to the buildup of short-chain oligomeric material during normal reactor operation which can contaminate depositions. The oligomeric material crystallizes into aggregates before the deposition begins and these aggregates locally inhibit deposition. As normal deposition proceeds on the surrounding aggregate free surface, holes form. These holes correspond to the size and shape of the aggregates, reaching millimeter scale. We show that high stage temperature inhibits defect formation. If deposition conditions which might normally result in defects are required (such as low stage temperature) they can be accessed by changing deposition conditions immediately after starting the deposition.

Chapters 3 and 4 develop a model of the vapor phase in iCVD. Chapter 3 examines the physical aspects of mixing and heat transfer, while Chapter 4 focuses on the reactive vapor phase and the thermal decomposition of TBPO. Together, Chapters 3 and 4 support innovations in reactor design and process modifications, providing a foundational framework for future advancements in iCVD modelling.

The study of mixing in Chapter 3 centers on the RTD. To establish an ideal mixing condition in iCVD, the RTD for 1 sscm flow of argon in IPA is measured using the Pirani pressure gauge, confirming the well-mixed assumption. Multiphysics simulations further elucidate reactor mixing, demonstrating the reactor is well-mixed below a critical Peclet number of ~ 0.3 . Not only does this establish that the iCVD vapor phase is most closely described as a well-mixed system, which has remained poorly understood until now, but provides a general method to characterize deviations from the ideal state using the Peclet number.

Measurements of diffusive heat transfer from the filament array in argon and IPA vapor suggest that heat transfer is pressure-dependent for Knudsen numbers greater than ~ 0.4 . This establishes an ideal diffusive heat transfer dominated state for the iCVD reactor and provides a straightforward method to quantify deviations using a heat transfer model for the filament array. To describe the effect of the filament temperature on batch processes in iCVD, the effective temperature is defined such that the ideal gas law holds for the reactor. A model is developed for the effective temperature and is fitted against data, showing excellent agreement with the heat transfer studies performed in the continuous mode.

Chapter 4 investigates the reactive vapor phase, specifically the thermal decomposition of di-tert-butyl peroxide (TBPO). Consistent with existing work on the thermal decomposition of TBPO at higher pressures and uniform temperature, primary decomposition limits TBPO decomposition in iCVD. This challenges existing perspectives by implying that there is only a trace amount of free radicals in the vapor phase, and that the stoichiometry of TBPO decomposition has large effects on partial pressures of monomer. In continuous flow-through operation of the iCVD reactor, we find that the well-mixed assumption validated in Chapter 3 accurately predicts the flow rate, temperature and pressure dependence of TBPO decomposition. Further studies

investigating TBPO decomposition in the presence of monomer (CHMA) do not suggest that TBPO or any of its degradation products reacts with CHMA.

Finally, the model of the vapor phase developed here predicts deposition using an ideal linear free radical polymerization model. The apparent activation energy of deposition rate indicates that β -scission occurs relatively far from the filament array. If the deposition rate predicted by our model is plotted against filament temperature on Arrhenius axes, there is a notable decrease in the apparent activation energy at high filament temperature. This decrease has been observed in past work and is generally attributed to diffusion limitation. Notably, the phenomenon in our model arises when the rate of TBPO decomposition is the same order of magnitude as the total flow of TBPO into the reactor. This finding emphasizes that a model of the iCVD vapor phase is needed to accurately model iCVD kinetics.

Chapter 5 investigates the base-catalyzed hydrolysis of two fluoroacrylic polymers with sodium hydroxide. This reaction is relevant to innovations in the post-deposition modification of polymers, since it pushes the limits of the nucleophilic substitution of esters, showing where the method breaks down, and the limiting factors in its potential applications. We find that the decomposition process requires the film to be covalently grafted to the substrate. Additionally, the degradation of the grafting agent over time indicates a need for more robust grafting procedures. The kinetics for hydrolysis of both polymers was determined, and both were found to undergo S_N2 substitution reactions with OH^- ions. This work may enable estimation of perfluorinated acid release from fluoroacrylic sources and is a proof of concept for post-deposition modification of marginally reactive surfaces.

6.2 Future Work

In Chapter 1 of this thesis, the distinction between fundamental and applied research in the field of iCVD was discussed. Applied research was broken down into three broad categories of implementation of iCVD, innovations to the iCVD process and innovations in post-deposition modification. We propose that most well-posed fundamental research will support or enable further progress in one of these domains. Each of the chapters of this thesis represents such fundamental research. As such, the goal of each chapter is to provide a foundational perspective upon which future work can be built.

In Chapter 2, the investigation of defects in iCVD demonstrates a potential vulnerability for applications of iCVD which require defect-free films. While this work raises fundamental questions about polymer crystallization and potentially represents a new way of synthesizing polymer monolayers, its most direct application in the iCVD literature is more grounded. While the methods of preventing defects in this work were found to be effective, they are not optimized for efficiency. Design of reactors to mitigate the buildup of oligomers and to streamline the “conditioning-film” method we’ve presented will be necessary for precision applications of iCVD where small defects can be catastrophic.

In Chapters 3 and 4, we sought to address a gap in our current understanding of the iCVD vapor phase. However, the work in Chapter 3 and 4 requires significant follow-up and validation. The use of the Pirani gauge as a composition sensor is promising, however the measurements it provides will be significantly more trustworthy if performed in concert with alternative composition sensors like in-situ FTIR or in-situ mass spectrometry. More generally, Chapters 3 and 4 underscore the need for standardization in iCVD research. As an example of this, the thermal

decomposition of TBPO highlights the un-reliability of thermocouples for measuring the temperature of the filament in iCVD. Resistance based temperature measurement would enable true filament temperature to be determined without inconsistency between experiments and research groups where the thermocouple may be attached in slightly different ways, causing potentially large errors. Finally, we hope that the work in these chapters inspires further innovations which we have not conceived of.

In Chapter 5 we investigated post-deposition modification. While most research in this area is purely applied, this has resulted in poorly understood limitations to the procedure. While the hydrolysis of fluoroacrylates has potential applications in the synthesis of Janus membranes and patterned surfaces, our work highlights the potential for film degradation during post-deposition modification. As such, the primary takeaway from our work in Chapter 5 was the need for improved grafting agents to facilitate conversion of passivating surfaces.

APPENDIX A.

Solution of Laplace's equation for an infinite line of point sources

Diffusive heat transfer in a gas follows:

$$\mathbf{j}_q = \kappa \nabla T$$

The units of $\mathbf{j}_q$ are W m^{-2} , and the total heat transferred from the array per unit length of filament is q_{fil} , having units of W m^{-1} . At steady-state, conservation of energy requires:

$$\nabla \kappa \nabla T = 0$$

Because $\kappa \sim \sqrt{T}$ in gases, the equation governing heat transfer can be written as Laplace's equation,

$$\nabla^2 \phi = 0$$

(A.1)

where the potential ϕ is defined from equation (A.2).

$$\phi = -\frac{2}{3} \kappa' T^{\frac{3}{2}}$$

(A.2)

And kinetic theory of gases gives equation (A.3).

$$\kappa' = \frac{\kappa}{\sqrt{T}} = \frac{f}{3d^2} \sqrt{\frac{k_b^2}{\pi^3 m}}$$

(A.3)

To solve equation (A.1) in the iCVD geometry it is helpful to consider that spacing between the filaments is small compared to the distance to the stage, and they can therefore be approximated as an infinite line of equally spaced point sources of equal magnitude in 2D. This is if we ignore edge effects in the reactor.

The Cauchy-Reimann equations show that the real (and imaginary) part of any complex function is a solution of Laplace's equation. Using this fact, any 2D solution of Laplace's equation expressed in terms of circular harmonics, or Green's functions (point sources, sinks, dipoles, quadrupoles etc.), consists of finding a complex function with the correct singularities in the

complex domain. If a complex function is found with the appropriate singularities, then it is the solution.

To obtain a solution of equation (A.1) for an infinite line of point sources therefore, it is first necessary to consider a single point source, which we obtain from the complex natural logarithm. Using the polar representation for the complex plane, $z = re^{i\theta}$, it can be shown that the real part of the complex natural log, corresponds to the natural log of the absolute value of its argument.

$$Re(\ln(z)) = \ln|z| = \ln(r) \quad (A.4)$$

From this, we can see already that $Re(\ln(z))$ corresponds to the solution of Laplace's equation with a point source at the origin. Since we're interested in a line of point sources, we're only interested in logarithmic singularities.

Extending equation (A.4), we can see that the real part of the natural log of a complex function corresponds to the natural log of the absolute value of the function.

$$Re(\ln(f(z))) = \ln|f(z)|$$

The natural log function goes to infinity at zero and infinity, but nowhere else, therefore singular values of $\ln|f(z)|$ are places where $|f(z)|$ are zero or infinity.

Because $\ln(x)$ goes to negative infinity at zero, and infinity at infinity, zeros of $|f(z)|$ correspond to point sinks, and poles of $|f(z)|$ correspond to point sources, or vice versa for $-\ln(x)$. Places where the limiting value of $|f(z)|$ is both zero and infinity corresponds to higher order harmonics. For instance, $|\exp(1/z)|$ corresponds to a dipole at the origin.

Finding the solution of Laplace's equation for a line of equally-spaced point sources of equal magnitude therefore consists of finding a periodic function with a line of equally spaced

zeros. The only zeros of $\sin(z)$ are located at $n\pi$, so we expect it matches the solution to our problem.

$$\text{Re}(\ln(\sin(z))) = \ln |\sin(z)|$$

Note that in the limit approaching zero, $\ln |\sin(z)|$ behaves like $\ln(r)$, which we know to be the solution for a point source.

$$\lim_{z \rightarrow 0} \ln |\sin(z)| = \ln(|z|) = \ln(r)$$

Because $\ln(\sin(z))$ is periodic, this means that indeed $\ln |\sin(z)|$ corresponds to an infinite line of equally spaced point sources of equal magnitude.

It can also be shown that $\ln|\sin(z)|$ has appropriate behavior for large y , by considering the following limit:

$$\ln|\sin(x + iy)| = \ln \left| \frac{e^{i(x+iy)} - e^{-i(x+iy)}}{2i} \right|$$

$$\lim_{y \rightarrow \infty} \ln|\sin(z)| = \ln \left| -\frac{e^y}{2i} \right|$$

$$\lim_{y \rightarrow \infty} \ln|\sin(z)| = \ln \left| i \frac{e^y}{2} \right| = y - \ln(2)$$

This is the correct behavior for large values of y , so we consider that the solution of Laplace's equation for an infinite line of equally spaced point sources of equal magnitude is given by equation (A.5).

$$\phi \sim \text{Re}(\ln(\sin(z)))$$

(A.5)

To make this a solution to a temperature field, we need to add in the homogenous linear temperature profile from the stage to the lid. We will assume the filament is halfway between the stage and the lid, and that they are maintained at the same temperature, making the homogenous

solution constant. We also need to introduce a proportionality constant which will be related to the total flow of heat from the filament. We also need to convert the complex numbers to cartesian coordinates, so we will substitute $z = x + iy$, and re-scale them so that the periodicity of $\sin(z)$ matches the spacing of the filament (L):

$$\phi = c_1 + c_2 \operatorname{Re} \left(\ln \left(\sin \left(\frac{x\pi}{L} + \frac{iy\pi}{L} \right) \right) \right)$$

Evaluate $\sin(z)$ as:

$$\phi = c_1 + c_2 \operatorname{Re} \left(\ln \left(\sin \left(\frac{x\pi}{L} \right) \cosh \left(\frac{y\pi}{L} \right) + i \cos \left(\frac{x\pi}{L} \right) \sinh \left(\frac{y\pi}{L} \right) \right) \right)$$

The real part of $\ln(z)$ is equal to $\ln(|z|)$:

$$\phi = c_1 + c_2 \ln \left(\sqrt{\sin^2 \left(\frac{x\pi}{L} \right) \cosh^2 \left(\frac{y\pi}{L} \right) + \cos^2 \left(\frac{x\pi}{L} \right) \sinh^2 \left(\frac{y\pi}{L} \right)} \right)$$

Next, we need to solve for c_1 and c_2 . Noting that $j_e = \nabla \phi$, we can obtain c_2 by taking the derivative with respect to y at $x = 0$, far from the filament:

$$\phi = c_1 + c_2 \ln \left(\sinh \left(\frac{y\pi}{L} \right) \right)$$

$$\frac{\partial \phi}{\partial y} * 2L = q_{fil} = 2\pi c_2 \text{ for large } y$$

Solving for c_2 .

$$\phi = c_1 + \frac{q_{fil}}{2\pi} \ln \left(\sqrt{\sin^2 \left(\frac{x\pi}{L} \right) \cosh^2 \left(\frac{y\pi}{L} \right) + \cos^2 \left(\frac{x\pi}{L} \right) \sinh^2 \left(\frac{y\pi}{L} \right)} \right)$$

Now, we know that at $x = 0$ and $y = R$, $\phi = \phi_R$, and note that because R_{fil} is small compared to L , $\sinh \left(\frac{R_{fil}\pi}{L} \right) \sim \frac{R_{fil}\pi}{L}$:

$$\phi_R = c_1 + \frac{q_{fil}}{2\pi} \ln \left(\sinh \left(\frac{R_{fil}\pi}{L} \right) \right) \sim c_1 + \frac{j_e}{2\pi} \ln \left(\frac{R_{fil}\pi}{L} \right)$$

Solving for c_1 and plugging back in, we get equation (3.2) from the main text:

$$\phi = \phi_R + q_{fil} \frac{\ln \left(\frac{L}{R_{fil}\pi} \sqrt{\sin^2 \left(\frac{x\pi}{L} \right) \cosh^2 \left(\frac{y\pi}{L} \right) + \cos^2 \left(\frac{x\pi}{L} \right) \sinh^2 \left(\frac{y\pi}{L} \right)} \right)}{2\pi} \quad (\text{A.6})$$

Next, we want to use this temperature field to obtain a shape factor for the system to account for the system geometry in the overall heat transfer problem. To do so, we first consider that at y -values close to the stage, there is very little variation in equation (A.6) with respect to x , therefore the temperature at these y -values is approximately equal to the temperature when $x = 0$. Using this logic, the difference in potential between the filament and the stage can be obtained:

$$\frac{(\phi_{stage} - \phi_R)2\pi}{q_{fil}} = \ln \left(\frac{L \sinh \left(\frac{H\pi}{L} \right)}{R_{fil}\pi} \right) \quad (\text{A.7})$$

Where equation (A.7) describes the resistance to heat transfer due to conduction in the gas phase.